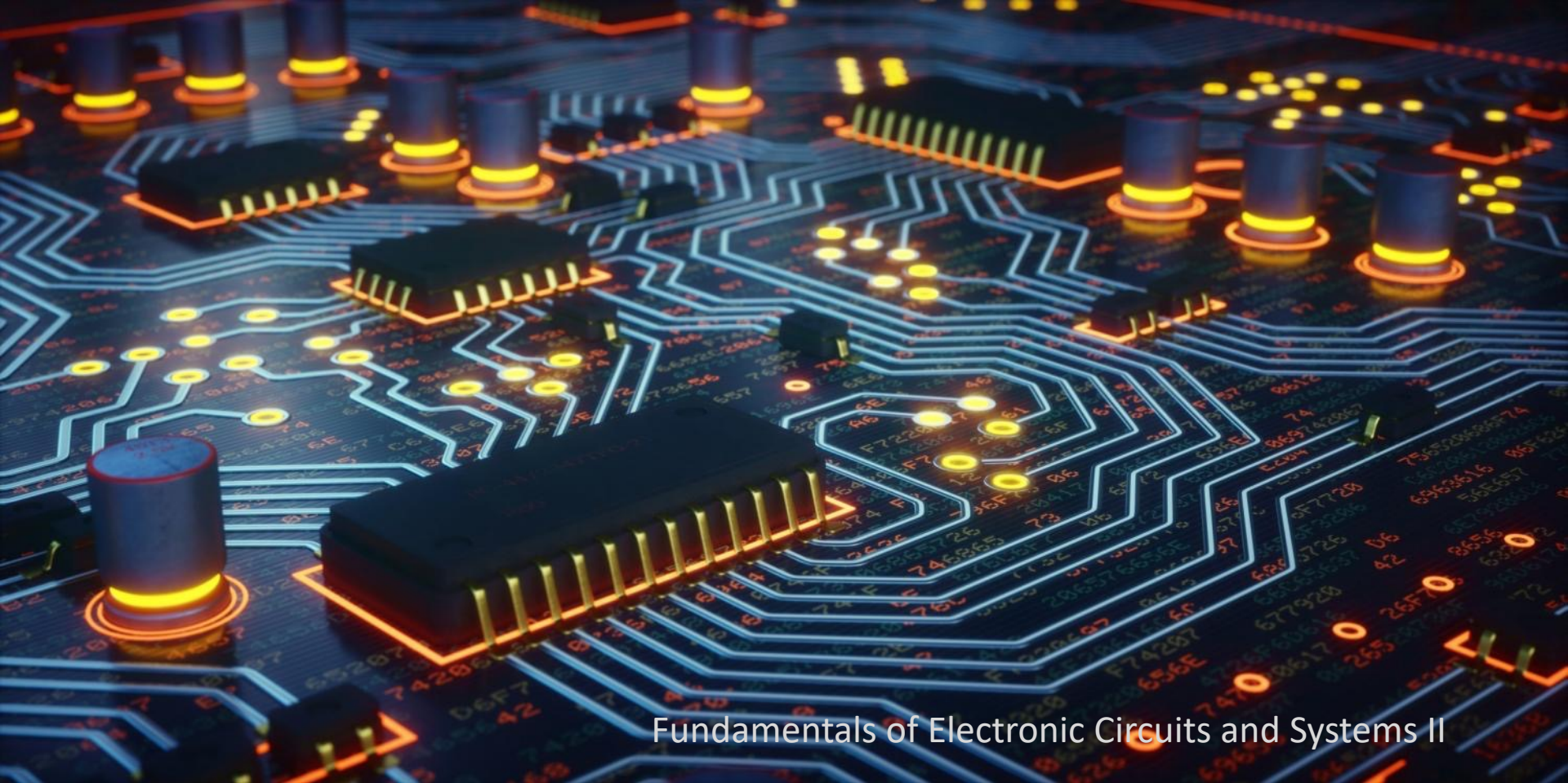




Fundamentals of Electronic Circuits and Systems II

MOS Field-Effect Transistors

Milin Zhang

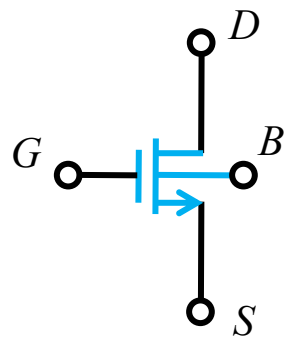
Dept of EE, Tsinghua University

Outline

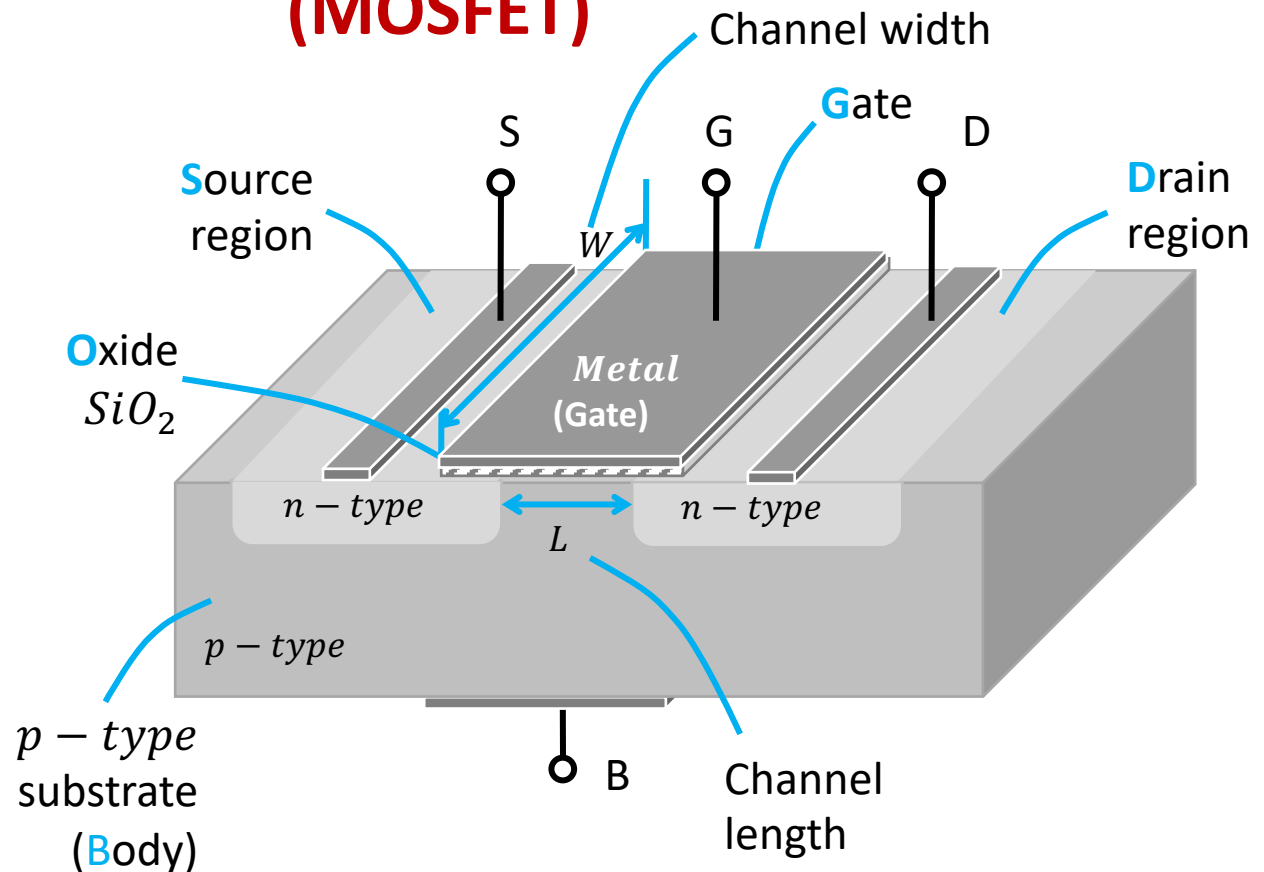
- Introduction to MOSFET
 - WHAT does it look like
 - HOW does it work
- The characteristic curves
- Circuit analysis techniques with MOSFET
- MOSFET v.s. BJT
- More examples of useful modules

Structure of MOSFET

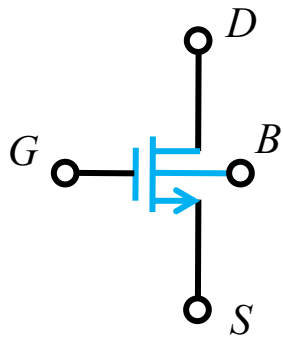
Metal-Oxide-Semiconductor Field-Effect Transistors (MOSFET)



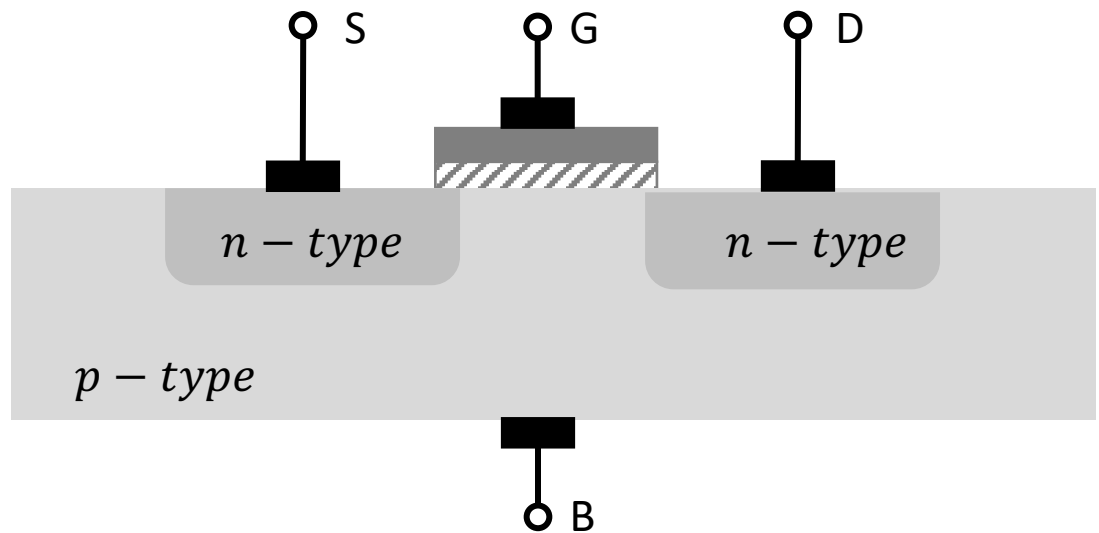
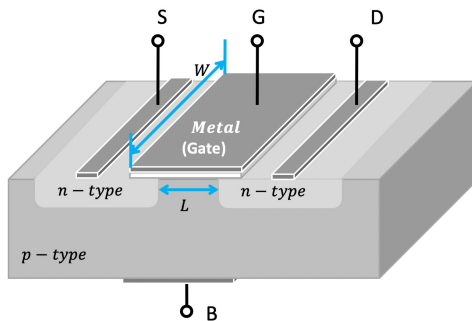
Enhancement-type NMOS transistor



Structure of MOSFET

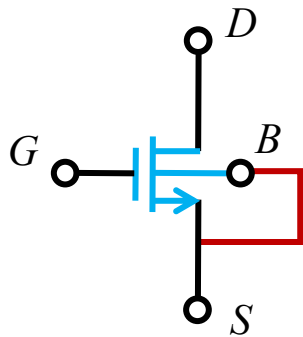


**Enhancement-type
NMOS transistor**



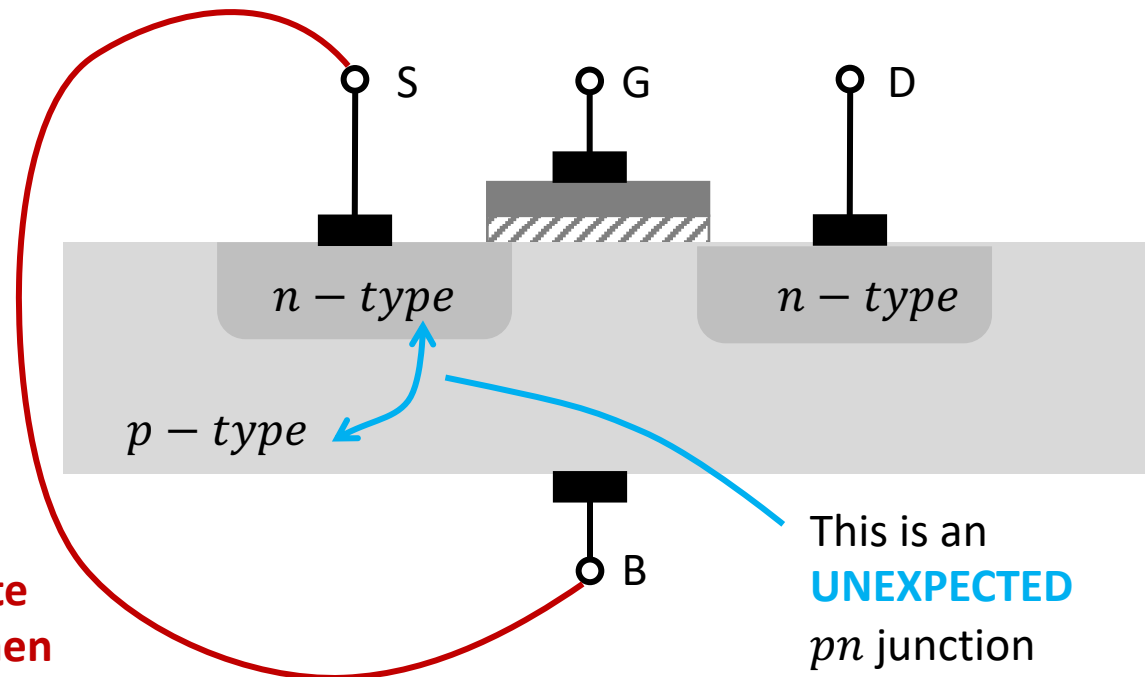
Cross section of an enhancement-type NMOS transistor

Structure of MOSFET



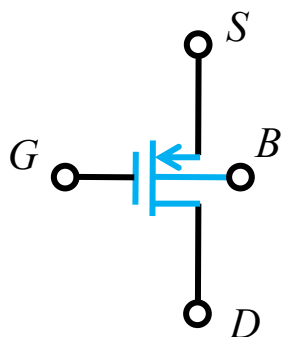
Enhancement-type
NMOS transistor

Body effect: the substrate acts as a second gate when the pn junction between the base and the source is forward biased

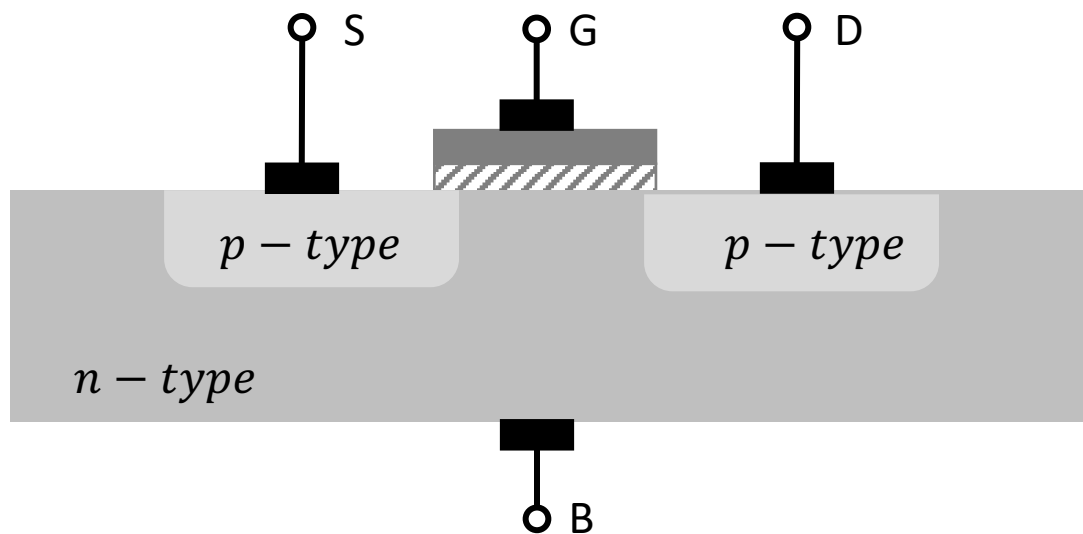


**Enable a zero bias
(cutoff mode)**

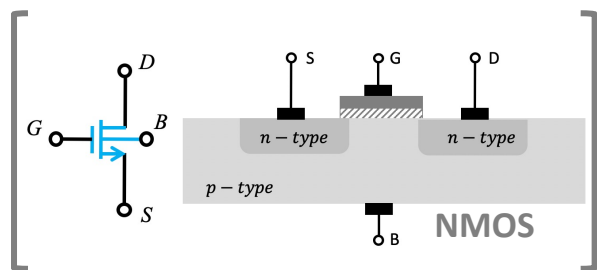
Structure of MOSFET



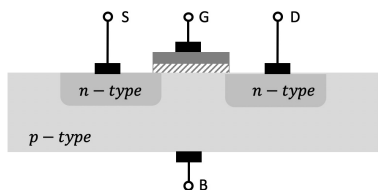
**Enhancement-type
PMOS transistor**



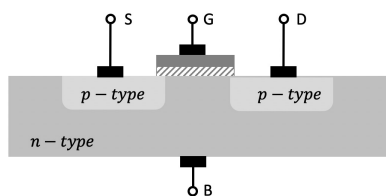
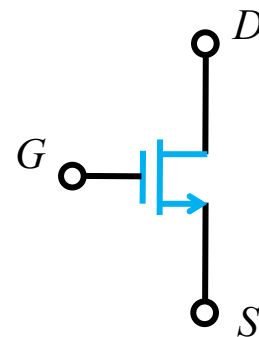
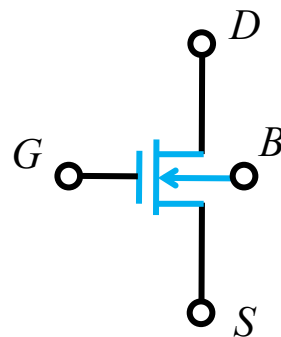
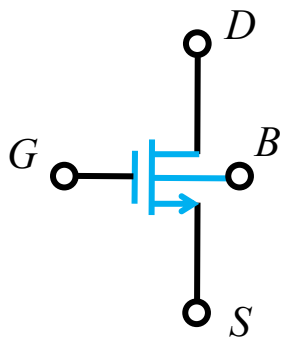
Cross section of an enhancement-type PMOS transistor



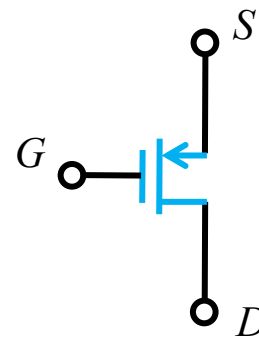
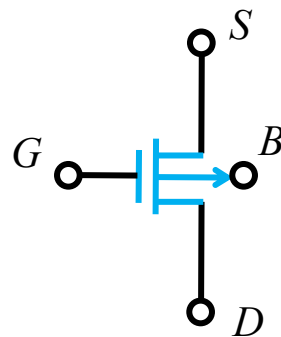
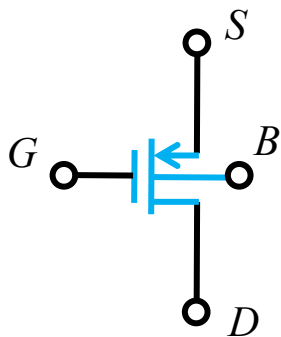
Circuit symbols of MOSFET



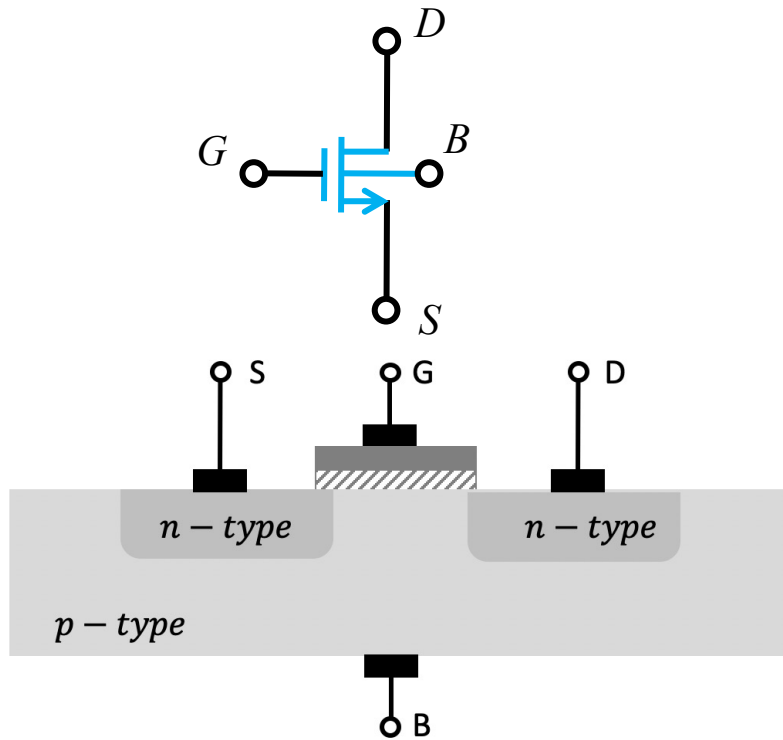
Enhancement-type
NMOS transistor



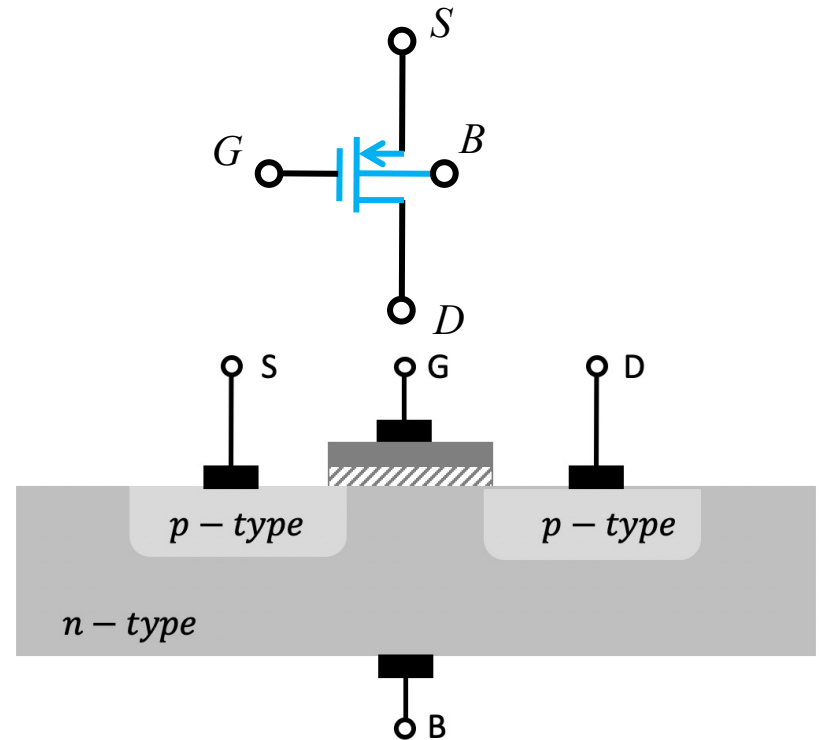
Enhancement-type
PMOS transistor



NMOS v.s. PMOS



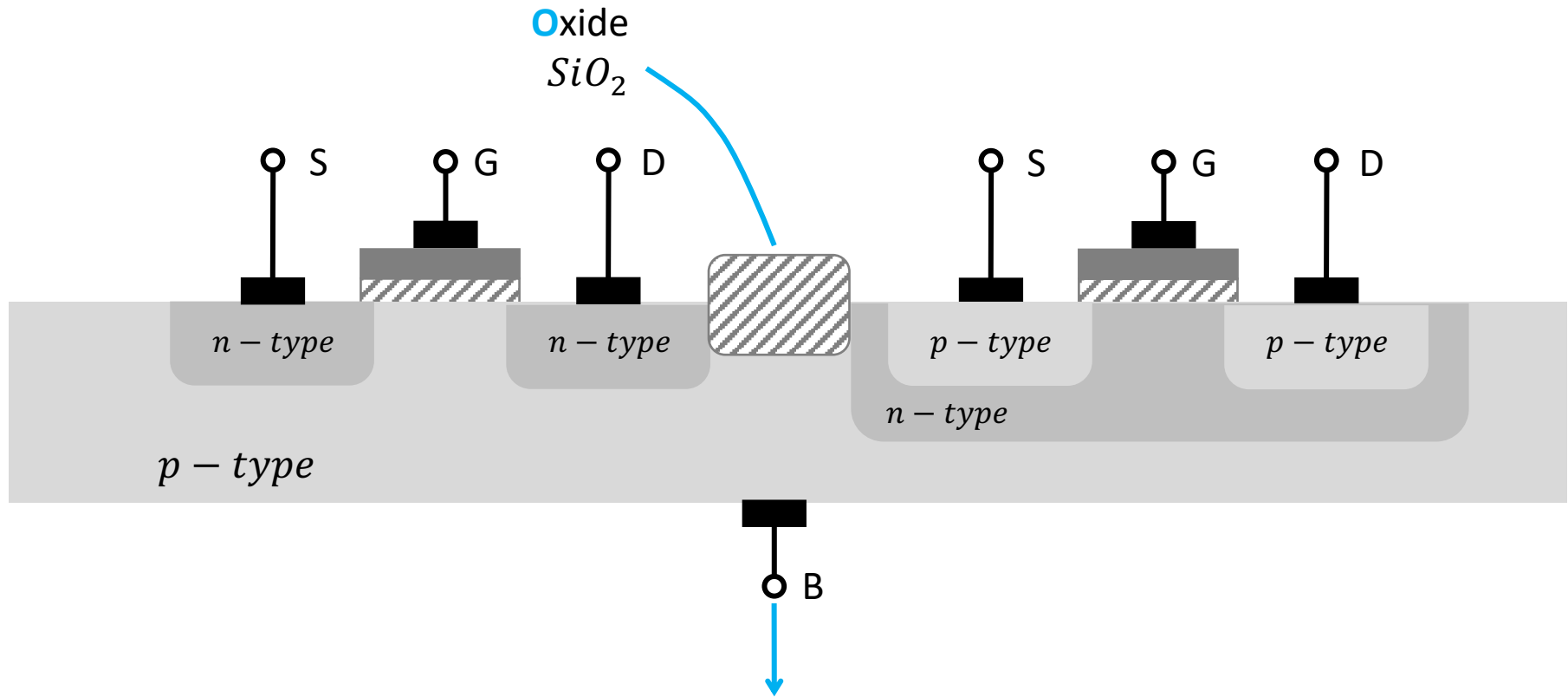
**Enhancement-type
NMOS transistor**



**Enhancement-type
PMOS transistor**

How to integrate NMOS & PMOS onto a same substrate?

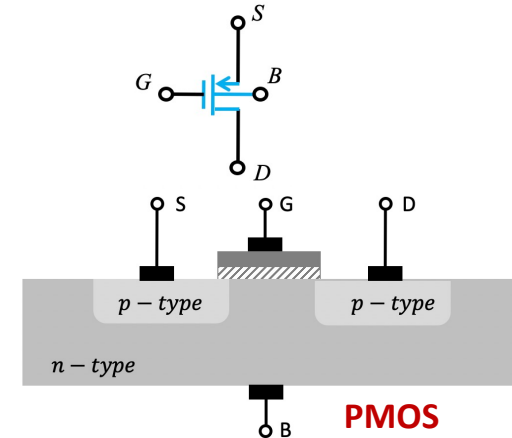
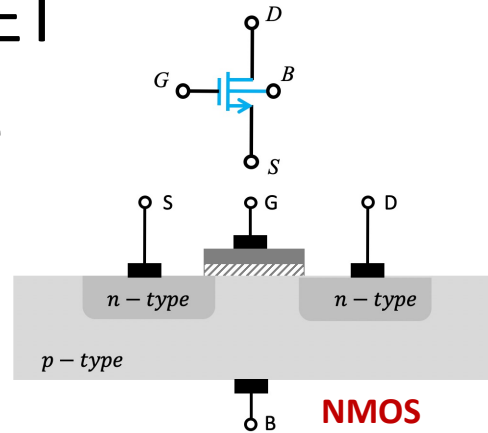
Complementary MOS (CMOS)



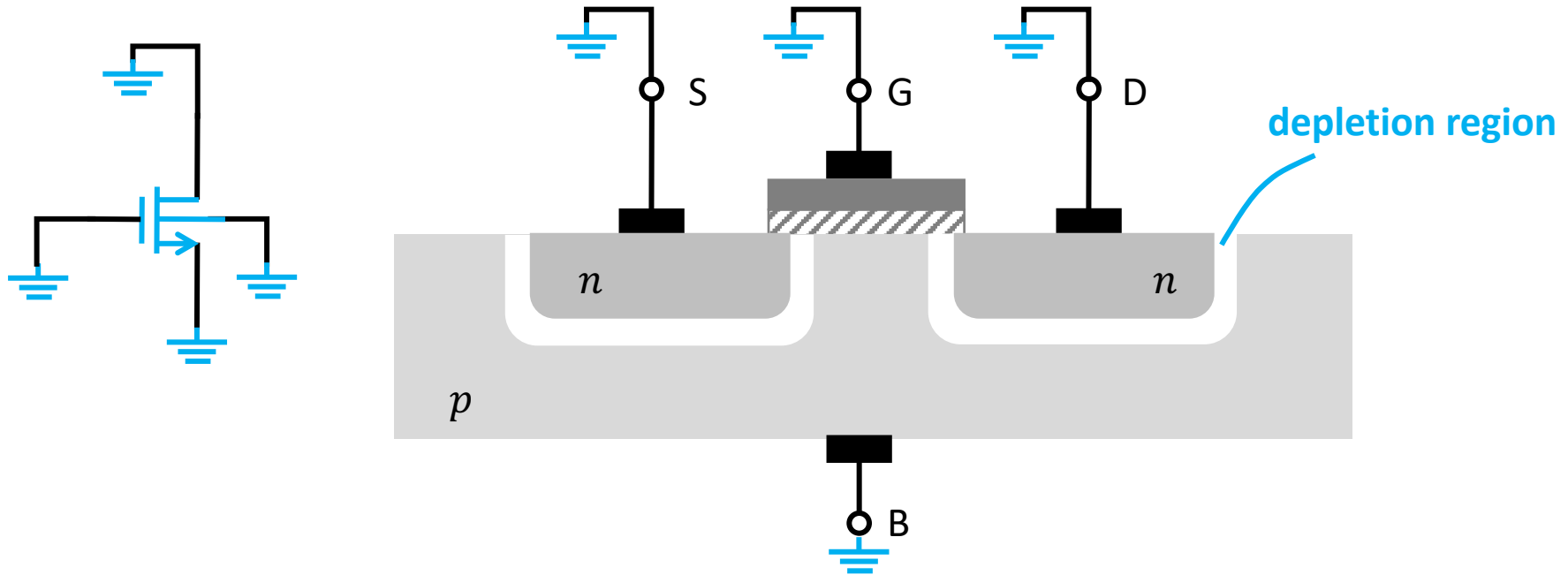
Connected to the most negative power supply to avoid body-effect

Outline

- Introduction to MOSFET
 - WHAT does it look like
 - **HOW does it work**

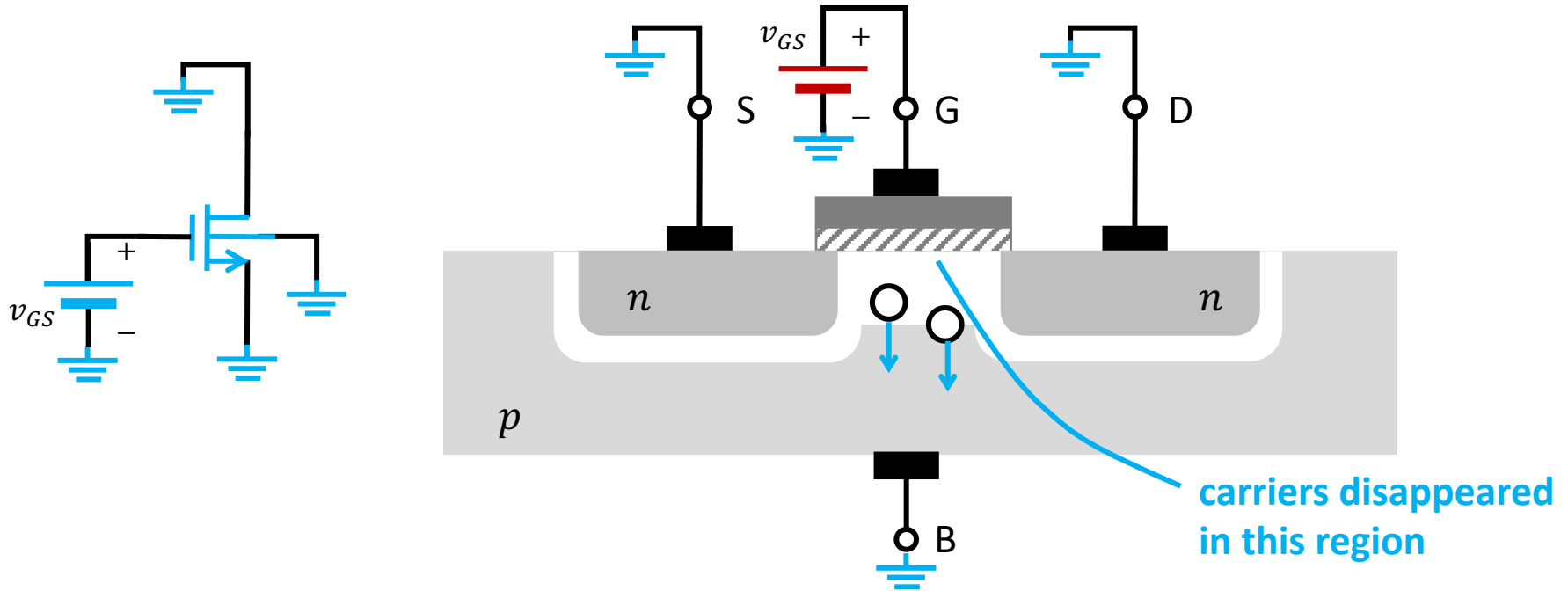


How Does It Work



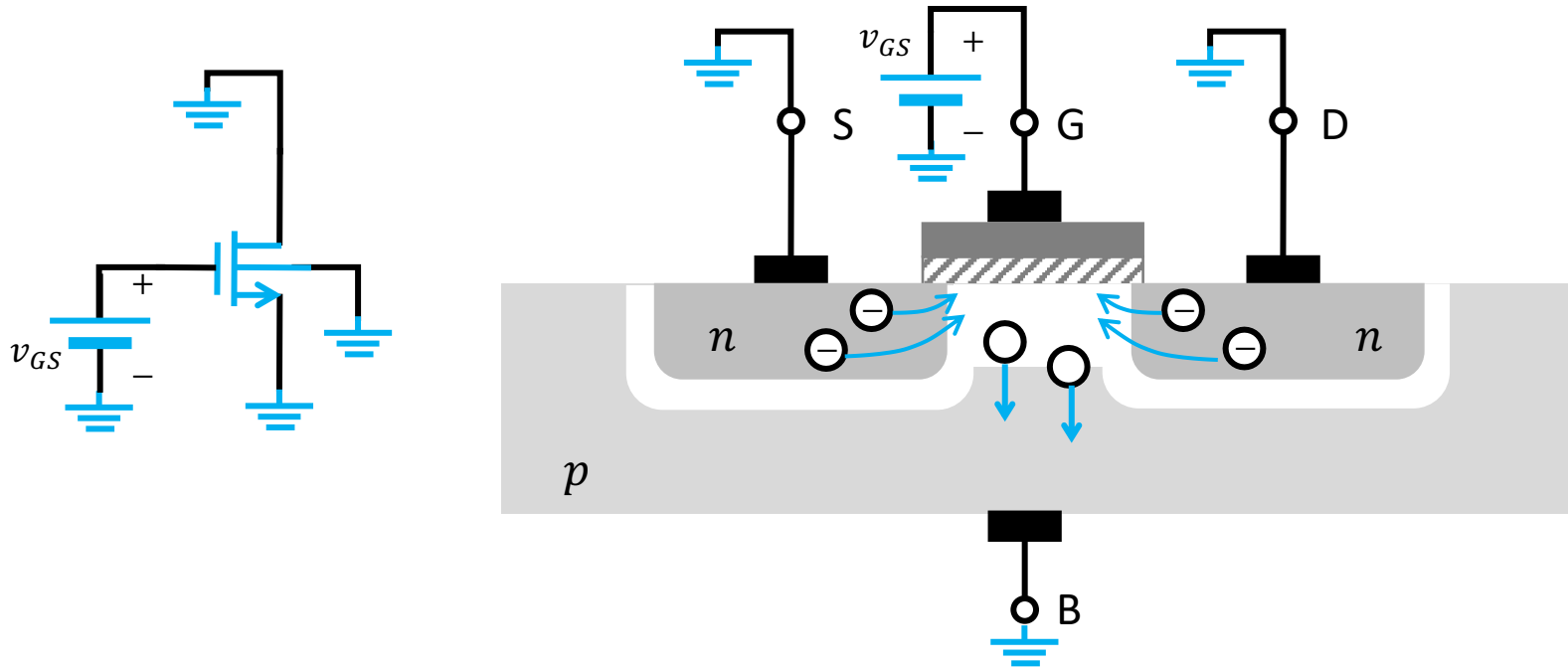
- Depletion regions are generated around both the Source and the Drain

How Does It Work



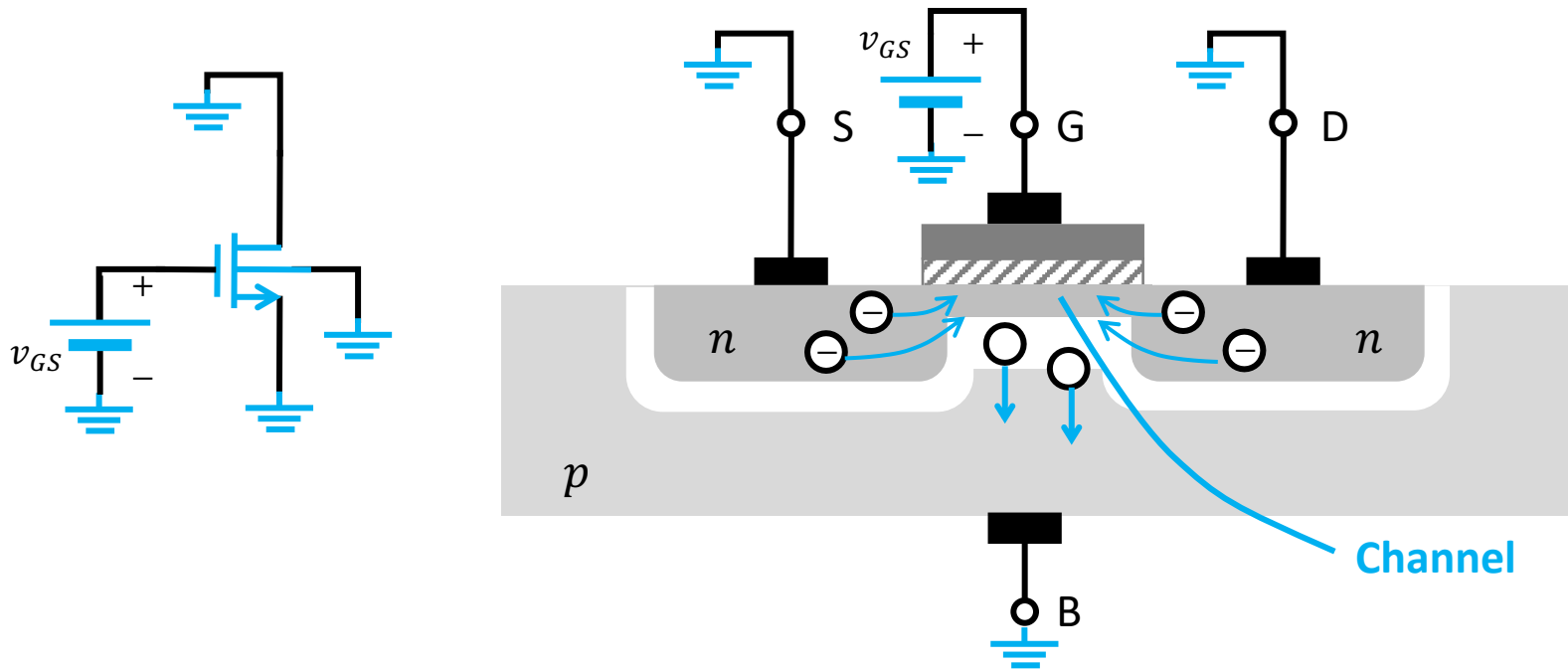
- Apply a voltage v_{GS} to the gate
- The **holes** are pushed downward into the substrate

How Does It Work



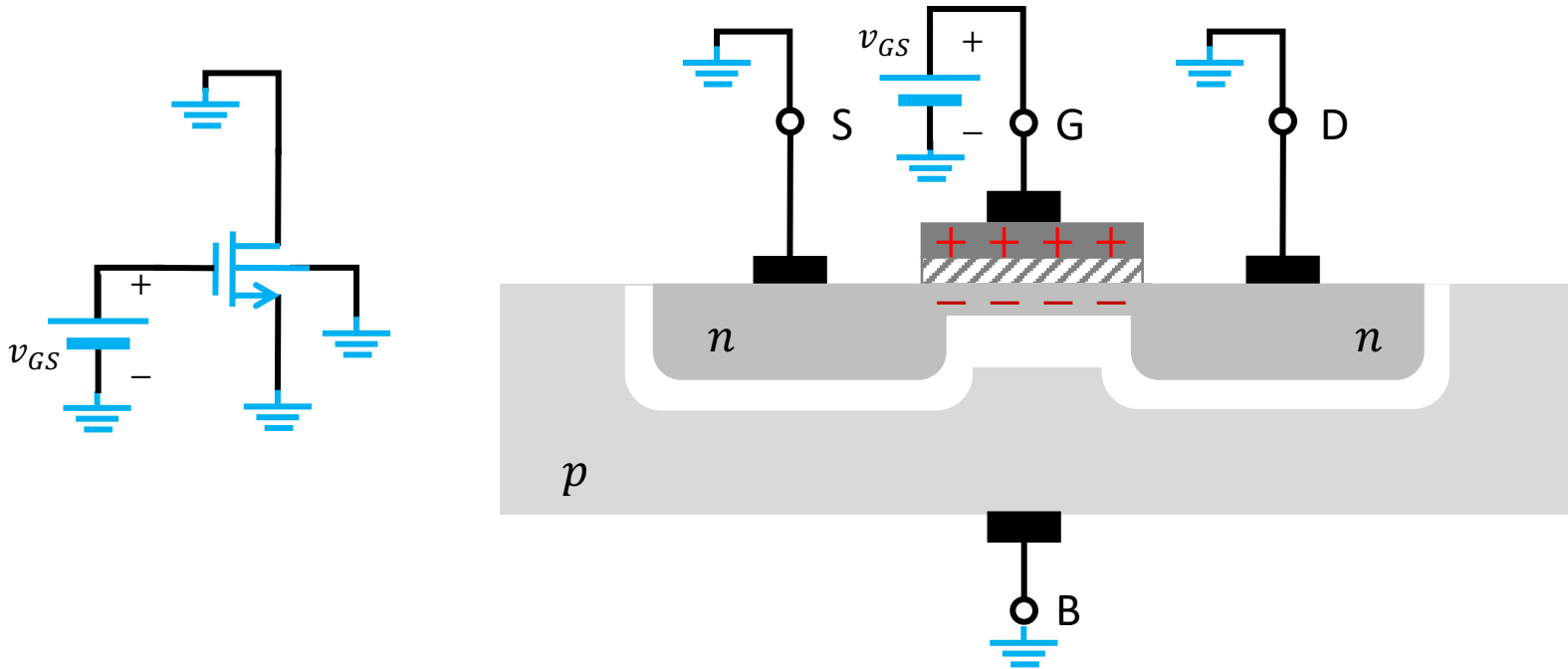
- The **electrons** from source and drain are attracted into the "carrier-disappeared" region

How Does It Work



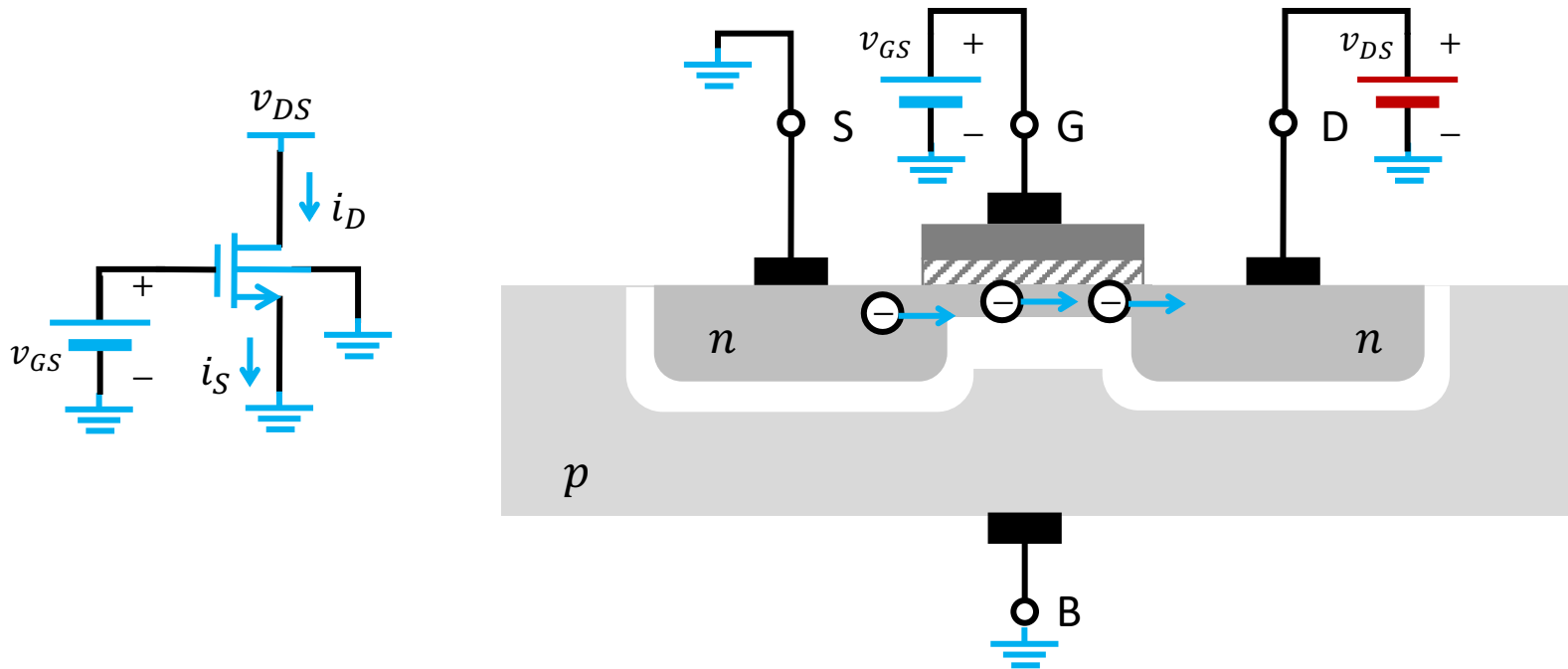
- The induced electrons form a n type **channel**
- this type of MOSFET is called **$n - channel$ MOSTFET**

How Does It Work



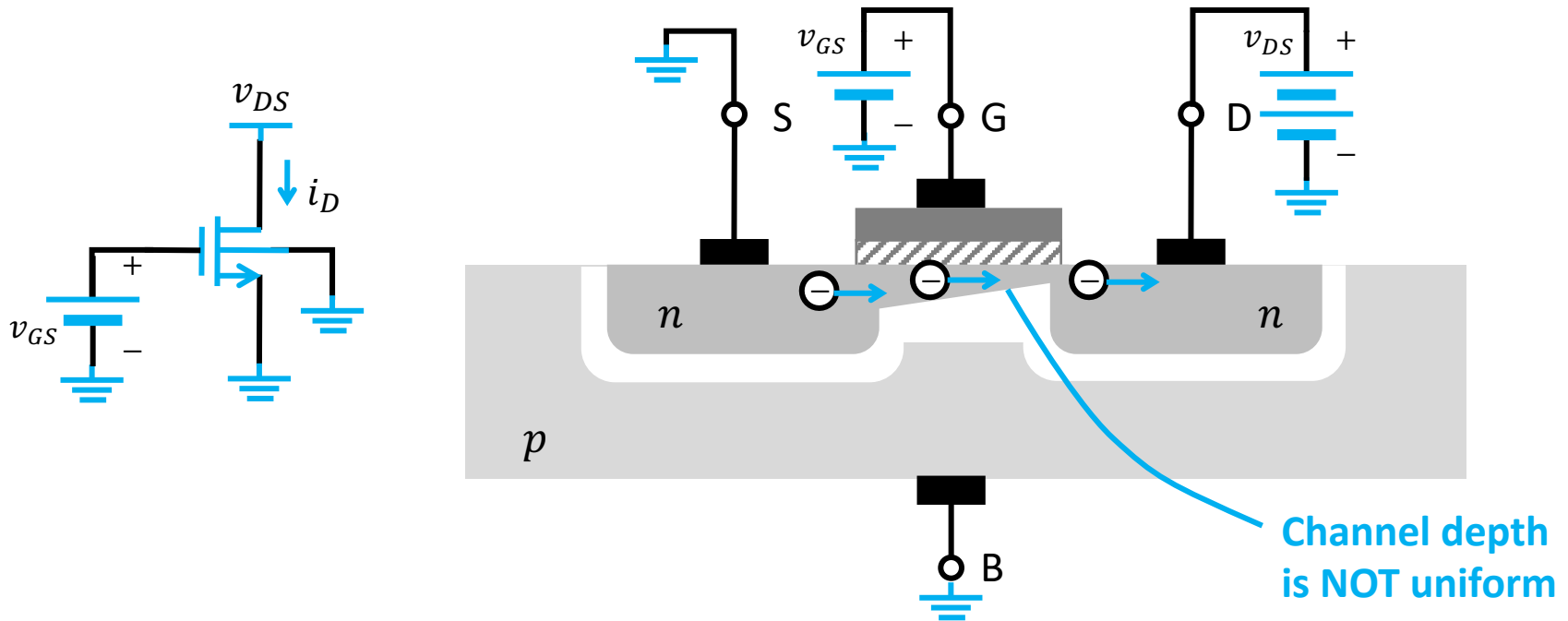
- The **gate** and the **channel region** form a parallel-plate capacitor C_{ox} , with the **oxide layer** acting as the capacitor dielectric

How Does It Work



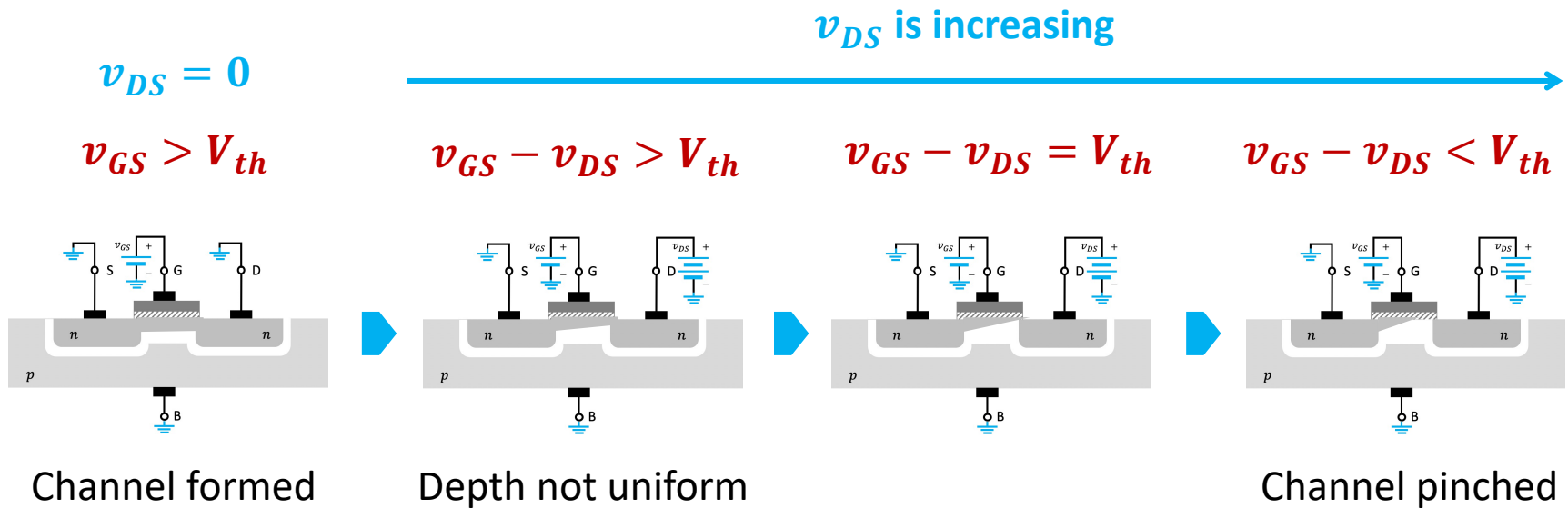
- Apply a small voltage v_{DS} to the gate
- When v_{DS} is higher than the threshold voltage v_{th} , A current i_D flows through the channel
- Currents through the drain and source are the same $i_D = i_S$
- No current @ Gate $i_G = 0$

How Does It Work



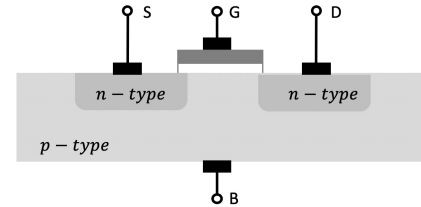
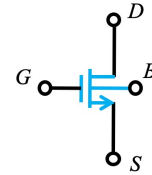
- When the voltage v_{DS} is increased
- The voltage difference $v_{GS} \gg v_{GD} = v_{GS} - v_{DS}$
- Channel is **NOT uniform depth**

How Does It Work



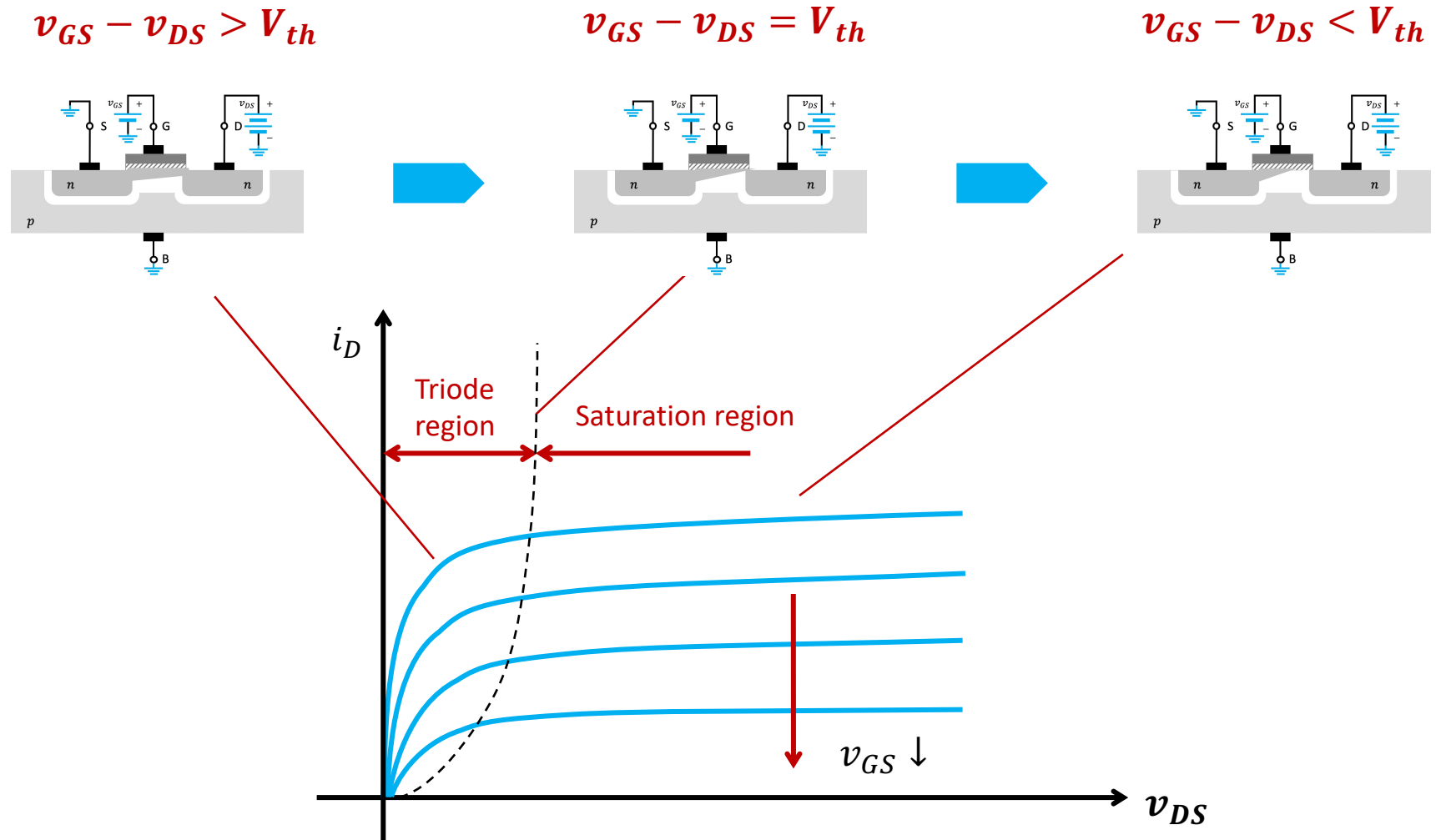
Outline

- Introduction to MOSFET
 - WHAT does it look like
 - HOW does it work
- The characteristic curves



$i_D - v_{DS}$ characteristics

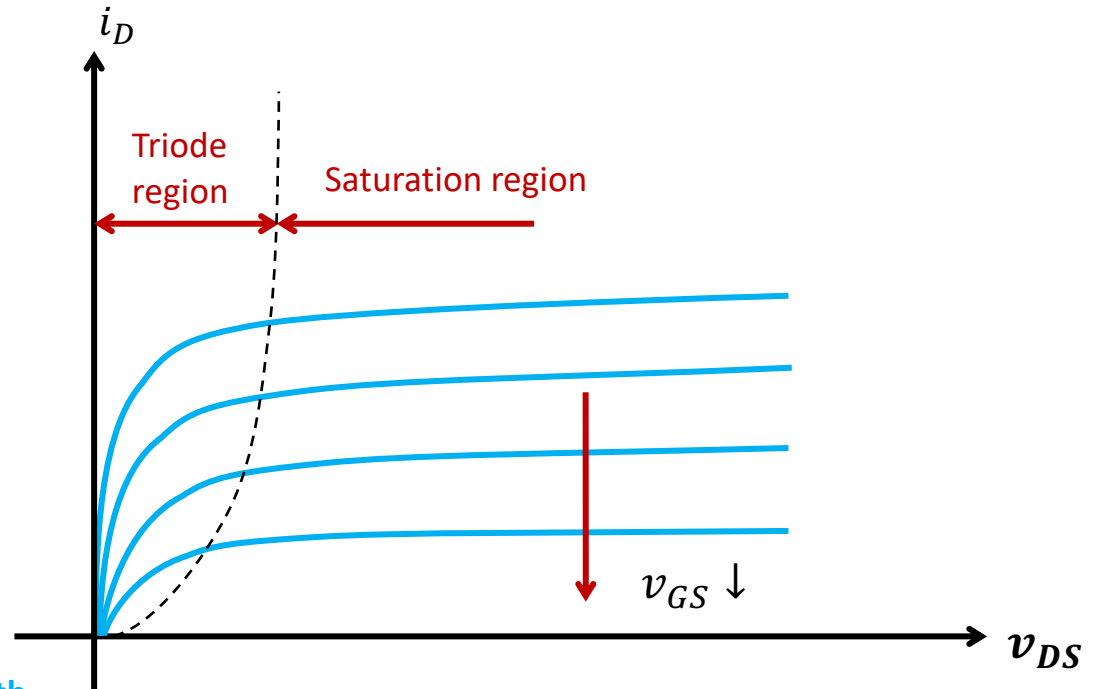
v_{DS} is increasing



$i_D - v_{DS}$ characteristics

@ Triode region

$$\begin{cases} v_{GS} > V_{th} \\ v_{GS} - v_{DS} > V_{th} \end{cases}$$



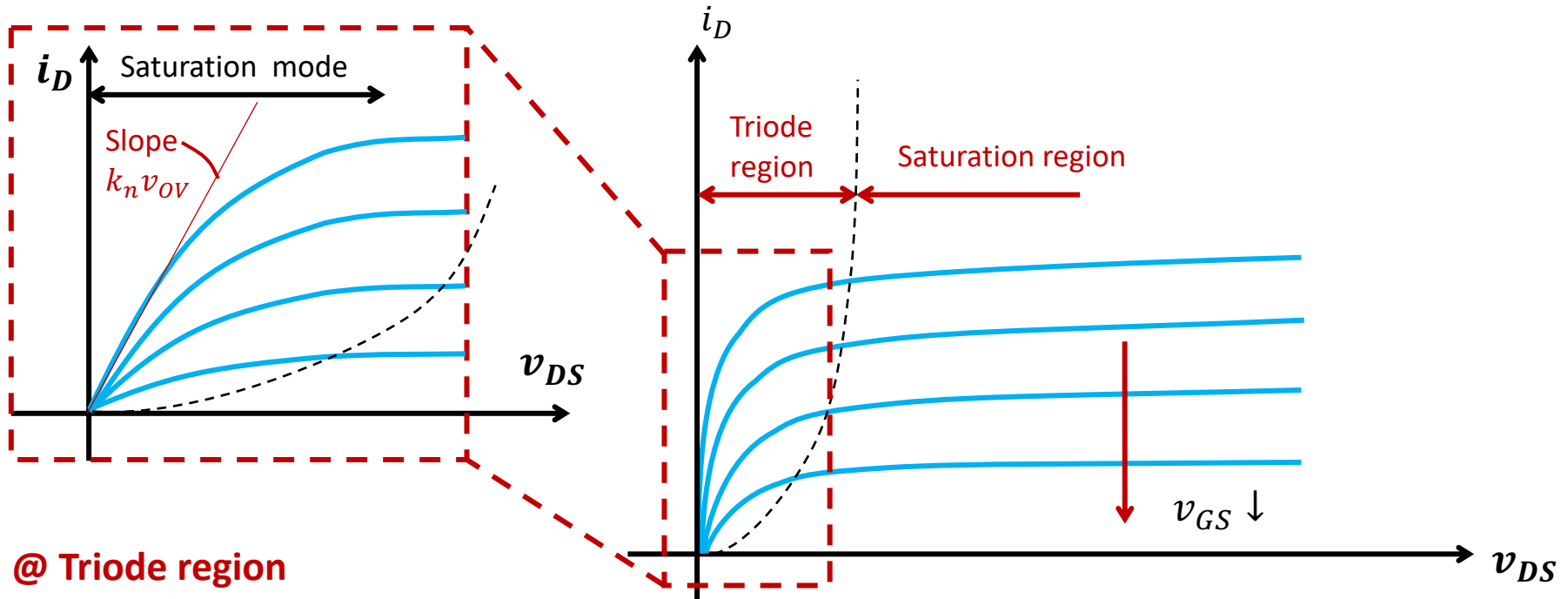
$$i_D = \underbrace{(\mu_n C_{ox}) \left(\frac{W}{L} \right)}_{\text{electron mobility, oxide capacitance, channel width, channel length}} \left[\underbrace{(v_{GS} - V_{th})}_{\text{threshold voltage}} v_{DS} - \frac{1}{2} v_{DS}^2 \right]$$

Define aspect ratio $= \frac{W}{L}$

Define overdrive voltage $v_{OV} = v_{GS} - V_{th}$

Define MOSFET transconductance parameter $k_n = (\mu_n C_{ox}) \left(\frac{W}{L} \right) = k'_n \left(\frac{W}{L} \right)$

$i_D - v_{DS}$ characteristics



@ Triode region

$$i_D = k_n \left[(v_{GS} - V_{th})v_{DS} - \frac{1}{2}v_{DS}^2 \right]$$

↓ If $v_{DS} \rightarrow 0$

$$\approx k_n (v_{GS} - V_{th})v_{DS}$$

Define $r_{DS} = \frac{1}{k_n (v_{GS} - V_{th})} = \frac{1}{k_n v_{OV}}$

**MOSFET behaves as
a linear resistance r_{DS}
when v_{DS} kept small**

$i_D - v_{DS}$ characteristics

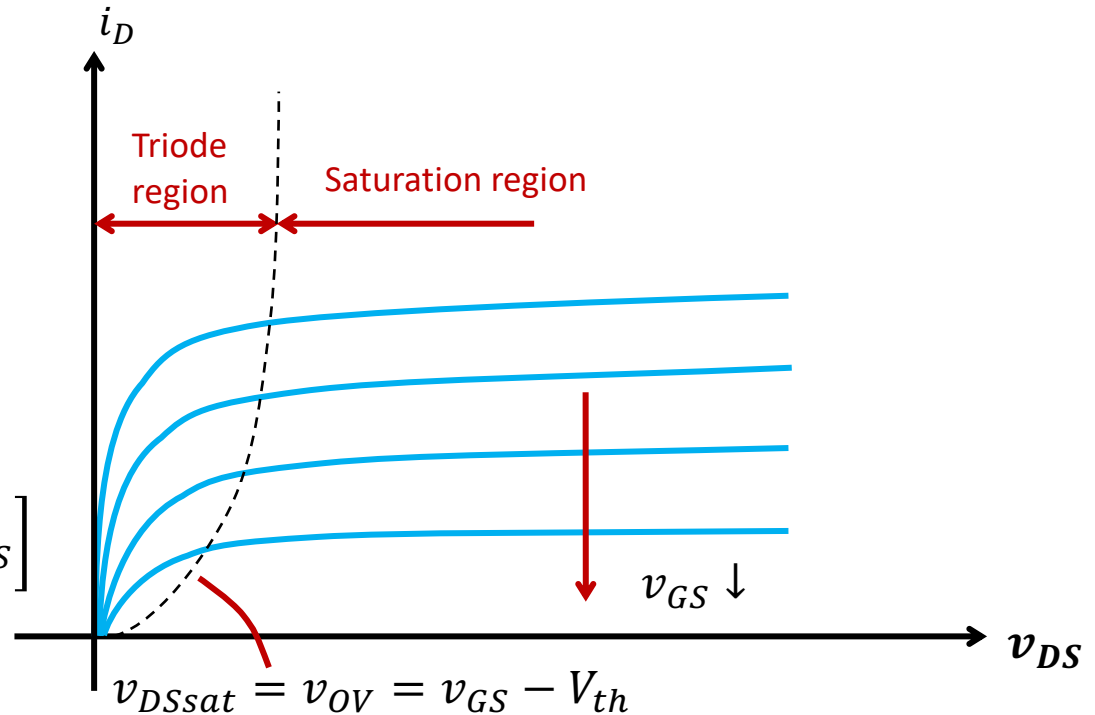
@ boundary between regions

$$\begin{cases} v_{GS} > V_{th} \\ v_{GS} - v_{DS} = V_{th} \end{cases}$$

$$i_D = k_n \left[(v_{GS} - V_{th})v_{DS} - \frac{1}{2}v_{DS}^2 \right]$$

$$v_{DSsat} = v_{OV}$$

$$= \frac{1}{2}k_n(v_{GS} - V_{th})^2$$



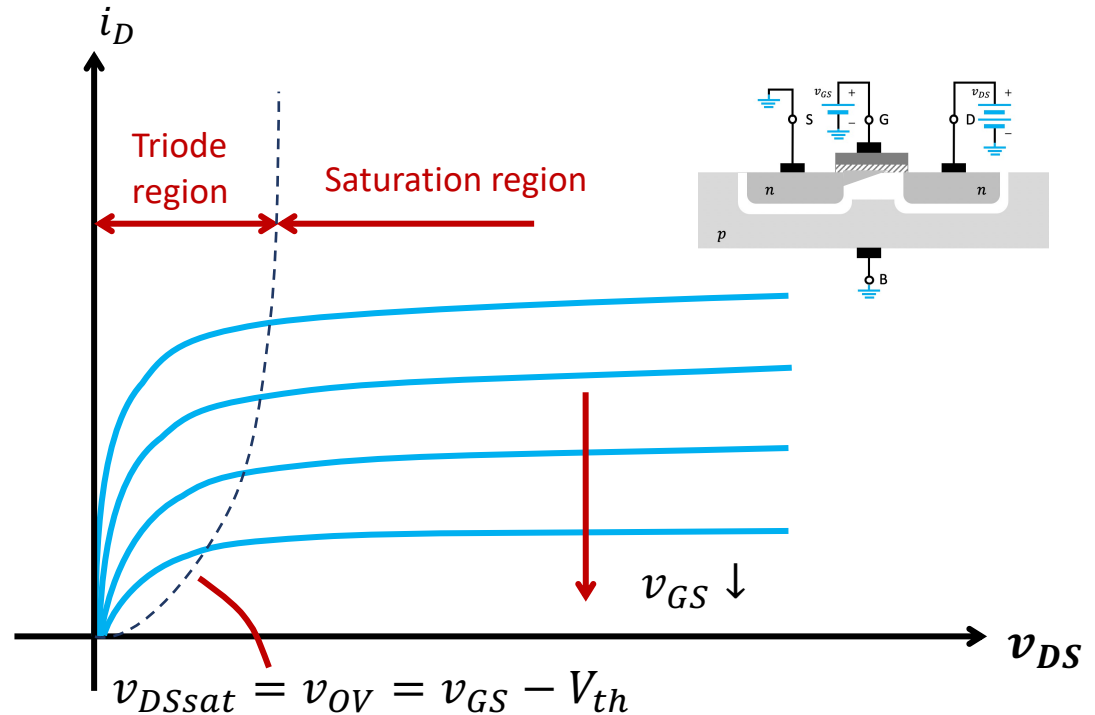
$i_D - v_{DS}$ characteristics

@ Saturation region

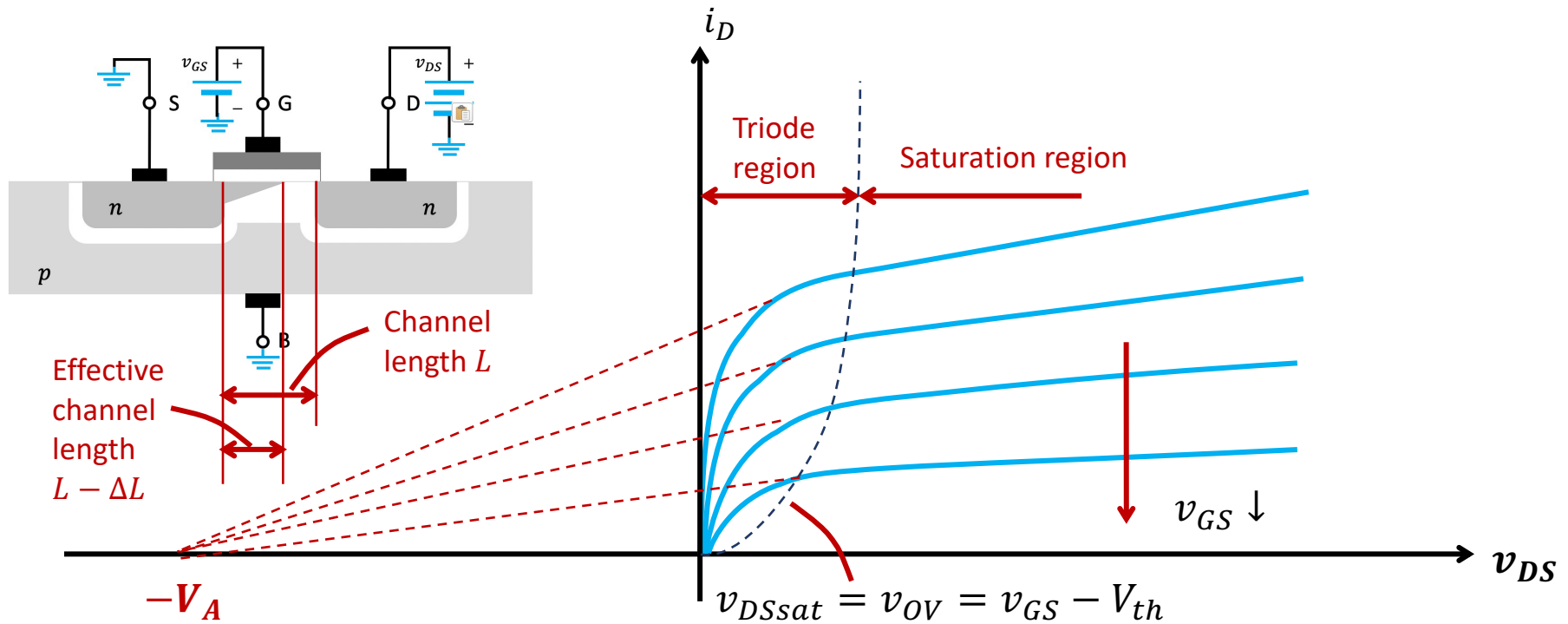
$$\begin{cases} v_{GS} > V_{th} \\ v_{GS} - v_{DS} < V_{th} \end{cases}$$

$$i_D = \frac{1}{2} k_n (v_{GS} - V_{th})^2$$

Current saturates since
the channel is pinched off



$i_D - v_{DS}$ characteristics

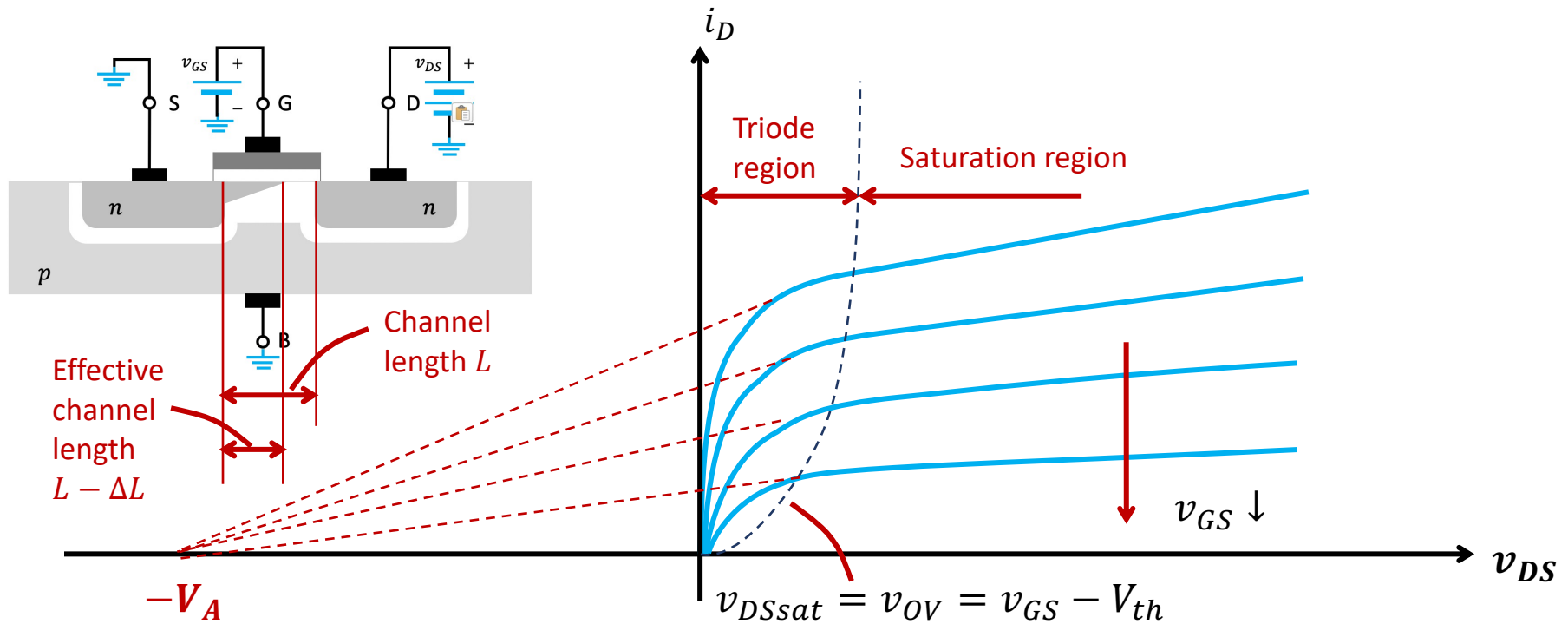


$$i_D = \frac{1}{2} (\mu_n C_{ox}) \left(\frac{W}{L - \Delta L} \right) (v_{GS} - V_{th})^2 = \frac{1}{2} (\mu_n C_{ox}) \left(\frac{W}{L} \right) \left(\frac{1}{1 - \frac{\Delta L}{L}} \right) (v_{GS} - V_{th})^2$$

$$\approx \frac{1}{2} (\mu_n C_{ox}) \left(\frac{W}{L} \right) \left(1 + \frac{\Delta L}{L} \right) (v_{GS} - V_{th})^2$$

i_D increases

$i_D - v_{DS}$ characteristics



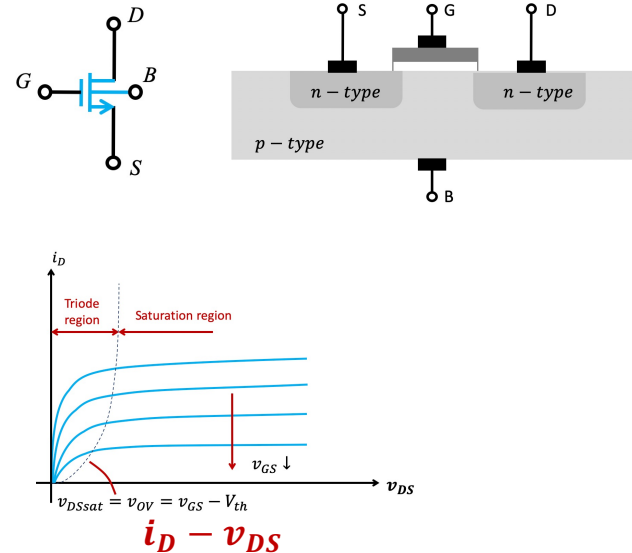
Early Voltage

$$i_D = \frac{1}{2} k_n (v_{GS} - V_{th})^2 \left(1 + \frac{v_{DS}}{V_A} \right)$$

**Channel-length modulation
(THE EARLY EFFECT)**

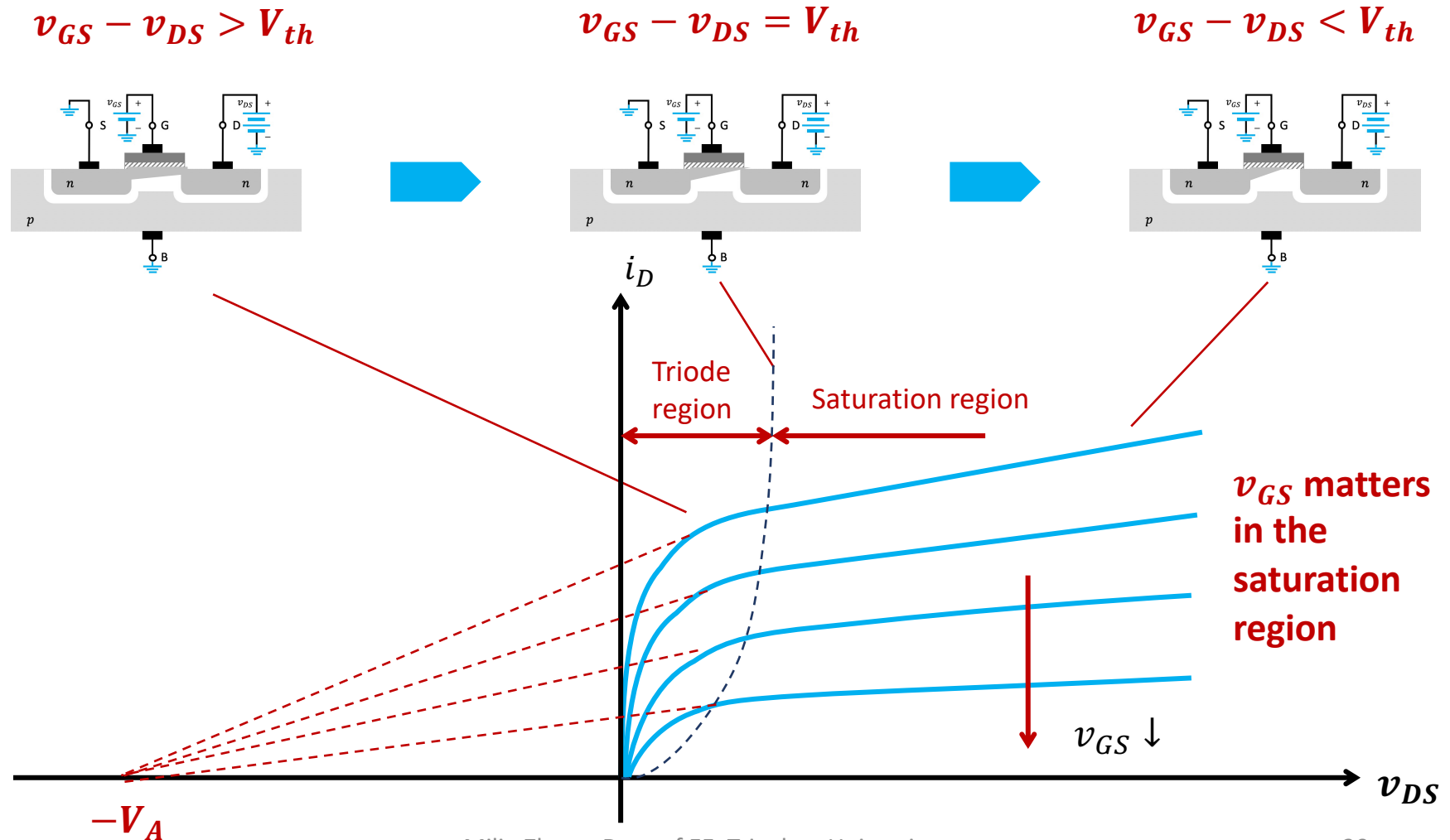
Outline

- Introduction to MOSFET
 - WHAT does it look like
 - HOW does it work
- The characteristic curves
 - $i_D - v_{DS}$ characteristics
 - $i_D - v_{GS}$ characteristics



Recall: $i_D - v_{DS}$ characteristics

v_{DS} is increasing



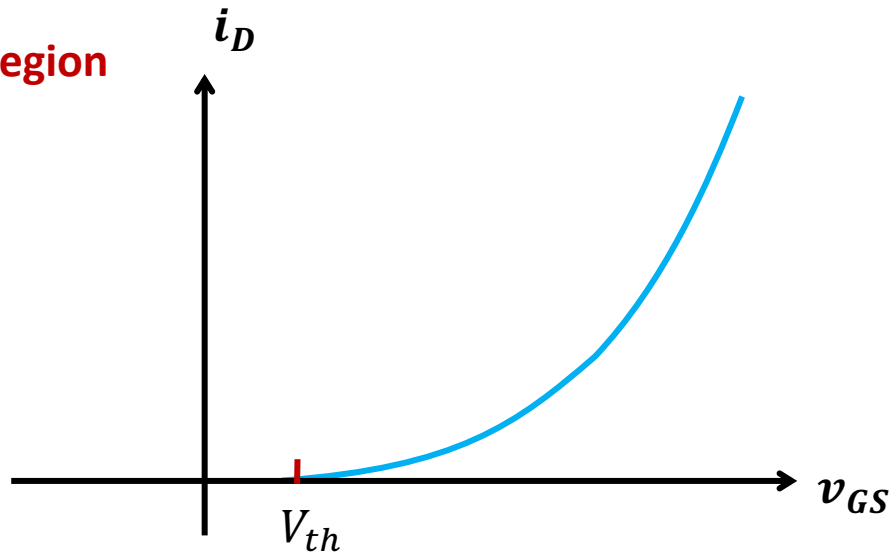
$i_D - v_{GS}$ characteristics

v_{GS} matters in the saturation region

@ Saturation region

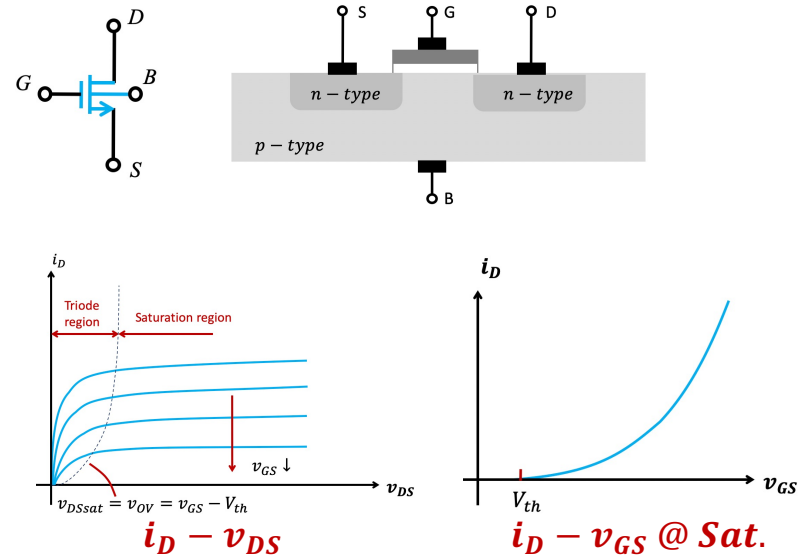
$$\left\{ \begin{array}{l} v_{GS} > V_{th} \\ v_{GS} - v_{DS} < V_{th} \end{array} \right.$$

$$i_D = \frac{1}{2} k_n (v_{GS} - V_{th})^2$$

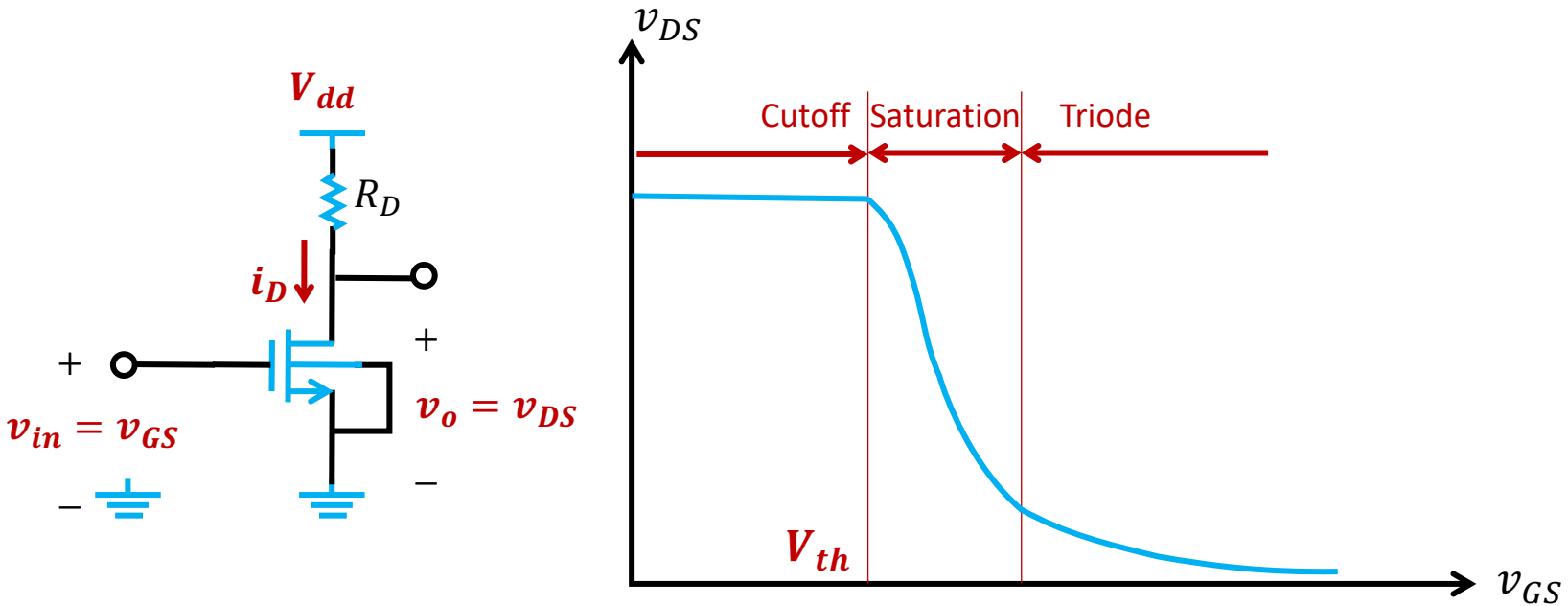


Outline

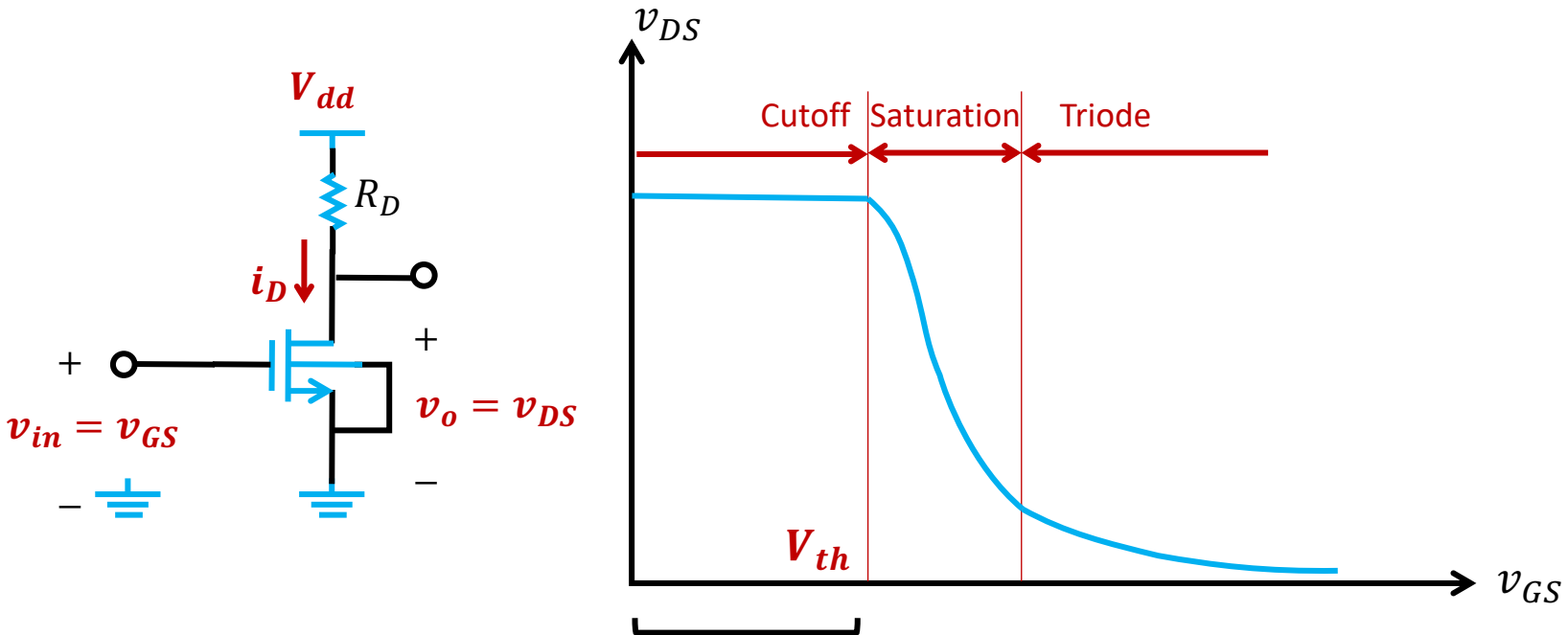
- Introduction to MOSFET
 - WHAT does it look like
 - HOW does it work
- The characteristic curves
 - $i_D - v_{DS}$ characteristics
 - $i_D - v_{GS}$ characteristics
 - **The transfer characteristic**



The Transfer Characteristic



The Transfer Characteristic

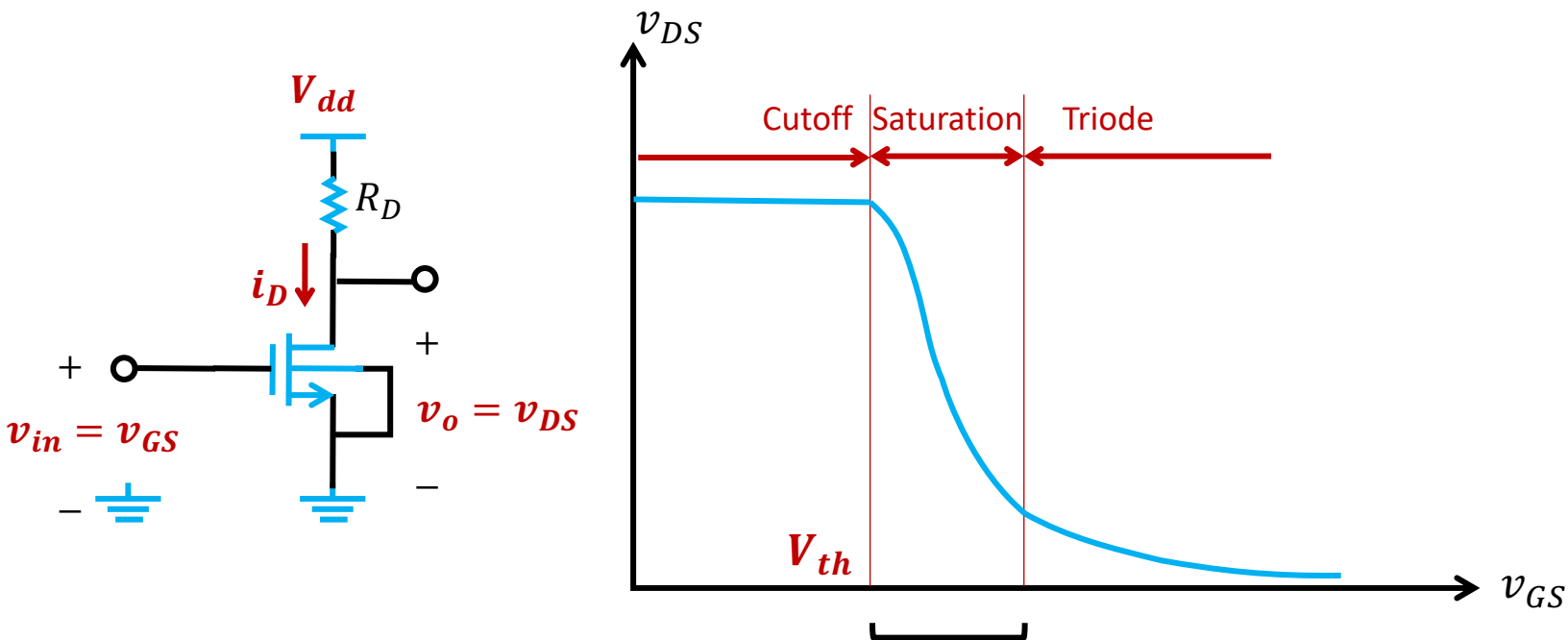


$$v_{in} < V_{th}$$

$$i_D = 0$$

$$v_o = V_{dd} - i_D R_D = V_{dd}$$

The Transfer Characteristic



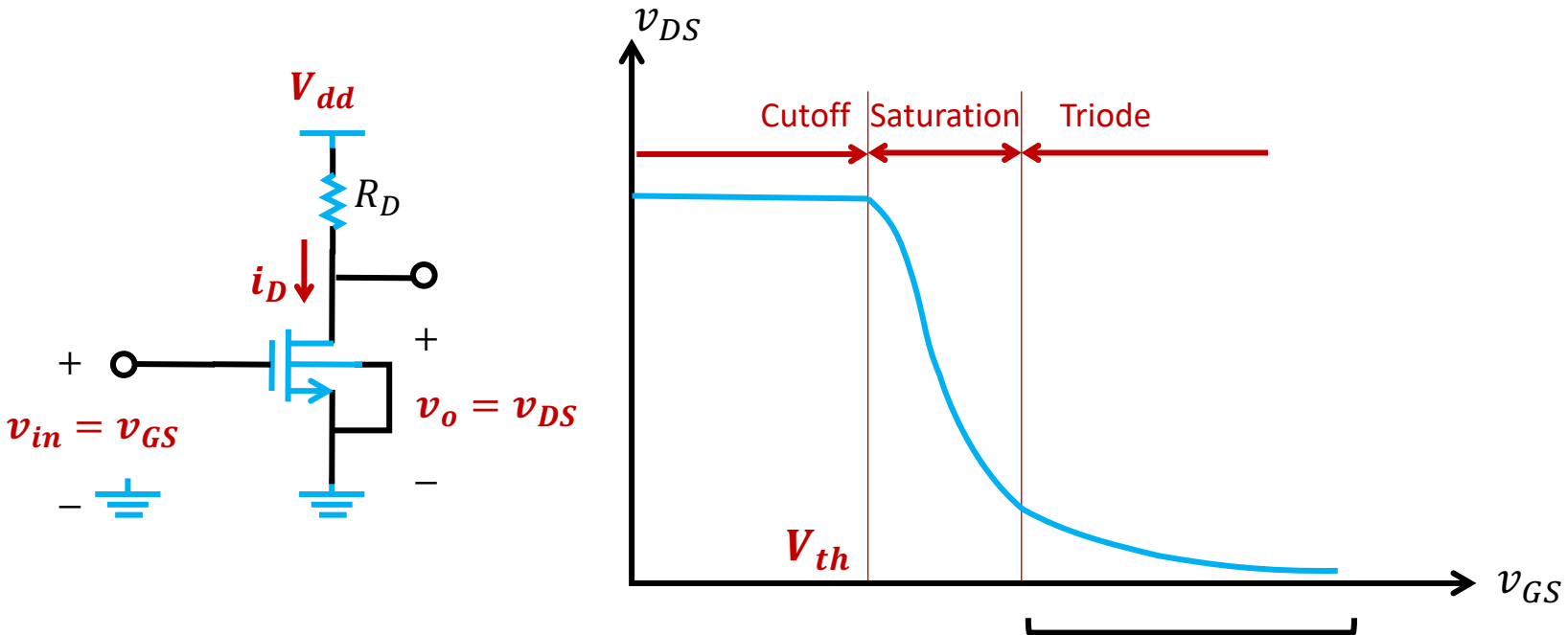
$$v_{in} > V_{th}$$

$$v_{GS} - v_{DS} < V_{th}$$

$$i_D = \frac{1}{2} k_n (v_{GS} - V_{th})^2$$

$$v_o = V_{dd} - i_D R_D = V_{dd} - \frac{1}{2} k_n (v_{GS} - V_{th})^2 R_D$$

The Transfer Characteristic



$$v_{in} > V_{th}$$

$$v_{GS} - v_{DS} > V_{th}$$

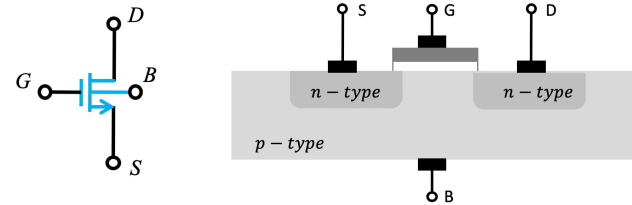
$$i_D = k_n \left[(v_{GS} - V_{th})v_{DS} - \frac{1}{2}v_{DS}^2 \right]$$

$$v_o = V_{dd} - i_D R_D$$

Outline

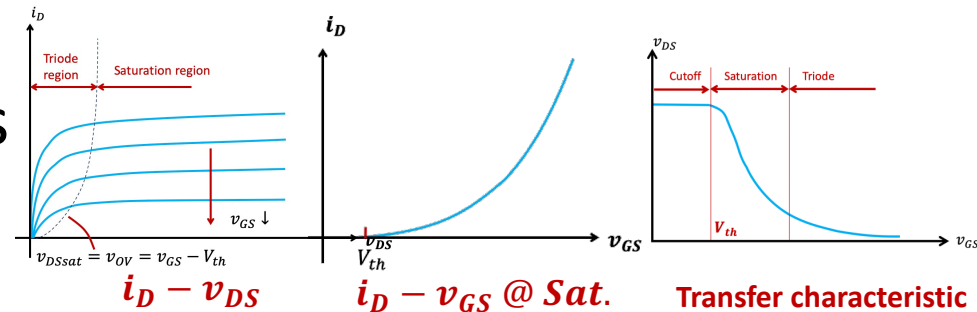
■ Introduction to MOSFET

- WHAT does it look like
- HOW does it work



■ The characteristic curves

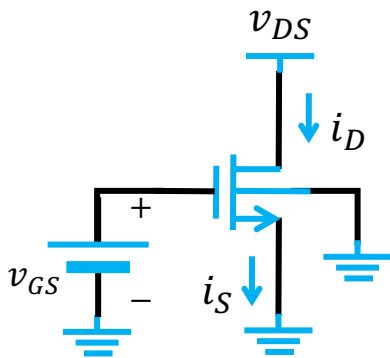
- $i_D - v_{DS}$ characteristics
- $i_D - v_{GS}$ characteristics
- The transfer characteristic



■ Circuit analysis techniques with MOSFET

- **DC analysis techniques**

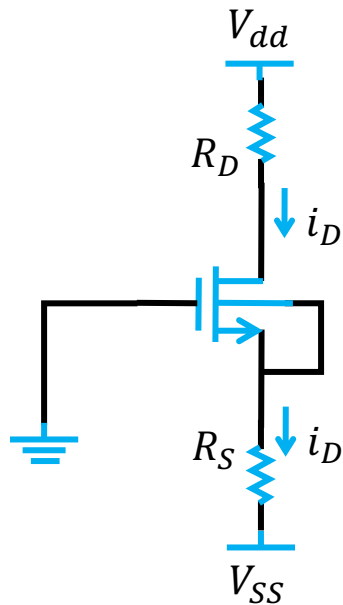
Characteristics Summary of MOSFET



	Triode region	Saturation region
when	$\begin{cases} v_{GS} > V_{th} \\ v_{GS} - v_{DS} > V_{th} \end{cases}$	$\begin{cases} v_{GS} > V_{th} \\ v_{GS} - v_{DS} < V_{th} \end{cases}$
i_D	$k_n \left[(v_{GS} - V_{th})v_{DS} - \frac{1}{2}v_{DS}^2 \right]$	$i_D = \frac{1}{2}k_n(v_{GS} - V_{th})^2$
i_S	$i_S = i_D$	$i_S = i_D$

Example 1

QUESTION: Find out the resistance of R_D and R_S to bias the transistor at $I_D = 0.4mA$ and $V_D = 0.5V$ with $V_{th} = 0.7V$, $V_{dd} = 2.5V$, $V_{SS} = -2.5V$, $W = 32\mu m$, $L = 1\mu m$, $\mu_n C_{ox} = 100\mu A/V^2$. Neglect the channel-length modulation effect.



- Since $V_D = 0.5V$, $V_G = 0V$
 $v_{GS} - v_{DS} = -0.5V < V_{th}$
- Assume the transistor is biased in saturation region
- According to KVL & transistor characteristics

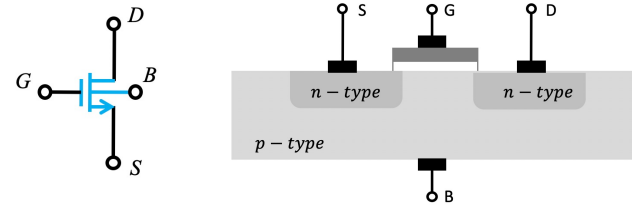
$$\left\{ \begin{array}{l} V_D = V_{dd} - i_D R_D \\ V_{SS} + i_D R_S + V_{GS} = 0 \\ i_D = \frac{1}{2} k_n (v_{GS} - V_{th})^2 \end{array} \right. \rightarrow \left\{ \begin{array}{l} R_D = 5k\Omega \\ R_S = 3.25k\Omega \\ V_{GS} = 1.2V \end{array} \right.$$

Check assumption

Outline

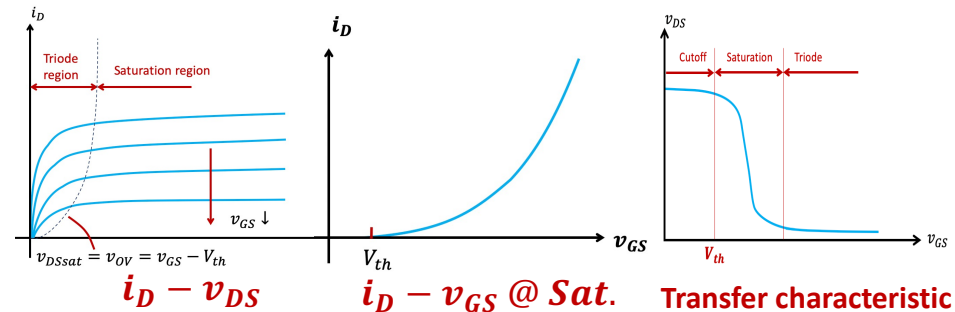
■ Introduction to MOSFET

- WHAT does it look like
- HOW does it work



■ The characteristic curves

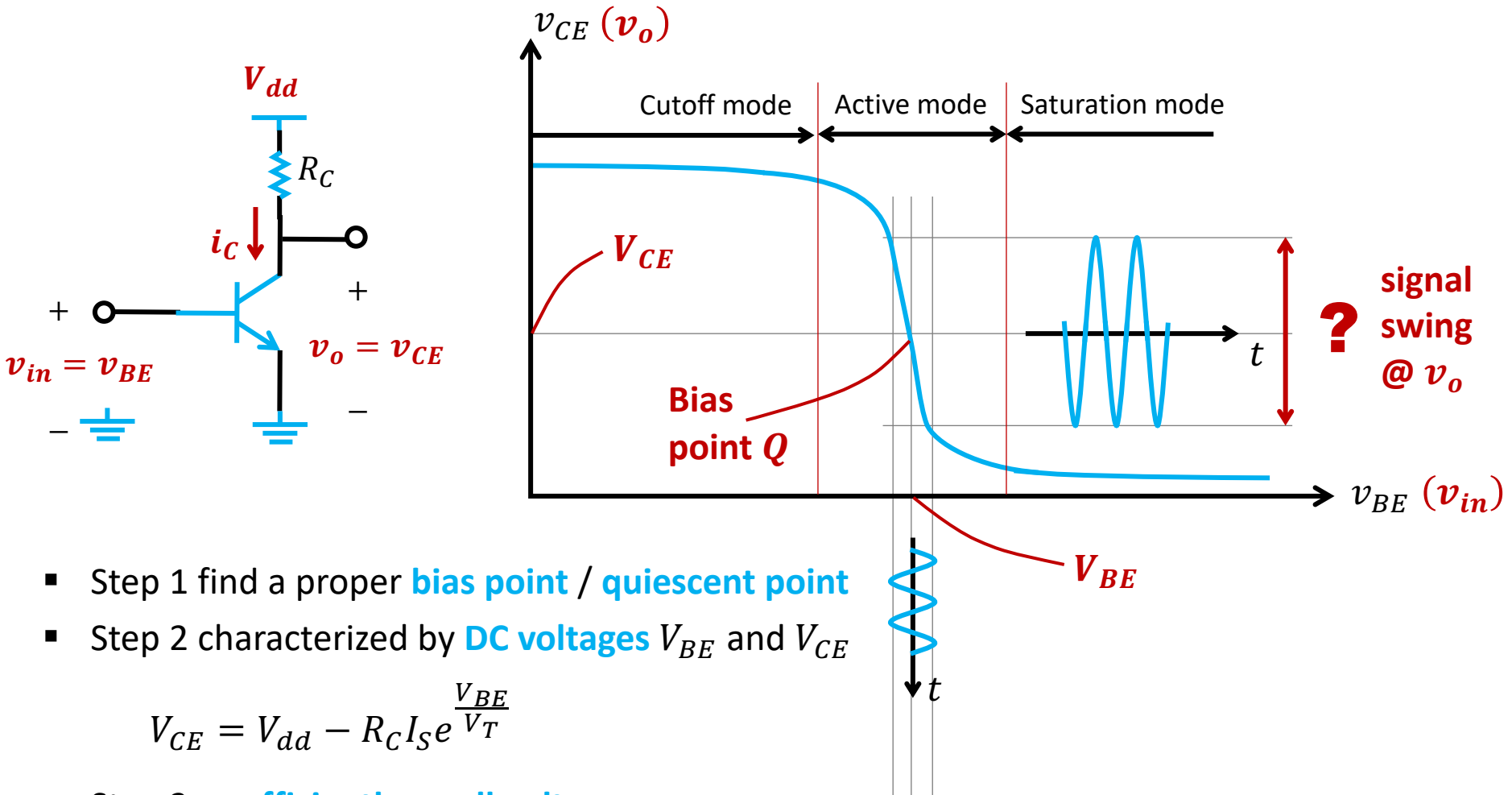
- $i_D - v_{DS}$ characteristics
- $i_D - v_{GS}$ characteristics
- The transfer characteristic



■ Circuit analysis techniques with MOSFET

- DC analysis techniques
- **AC analysis techniques**

Recall: Signal Amplification of BJT

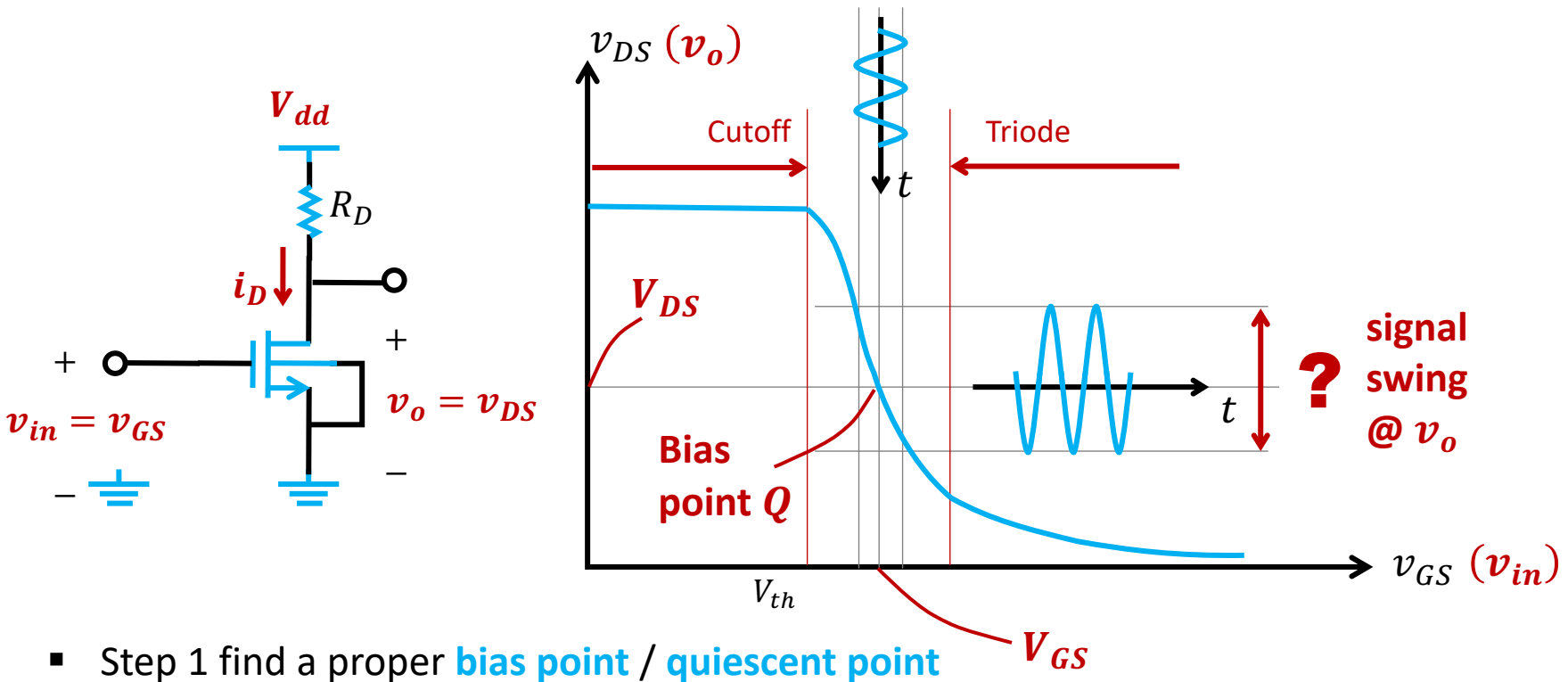


- Step 1 find a proper **bias point / quiescent point**
- Step 2 characterized by **DC voltages** V_{BE} and V_{CE}

$$V_{CE} = V_{dd} - R_C I_S e^{\frac{V_{BE}}{V_T}}$$

- Step 3 a **sufficiently small voltage** v_{BE} superimposed on V_{BE}

Signal Amplification of MOSFET

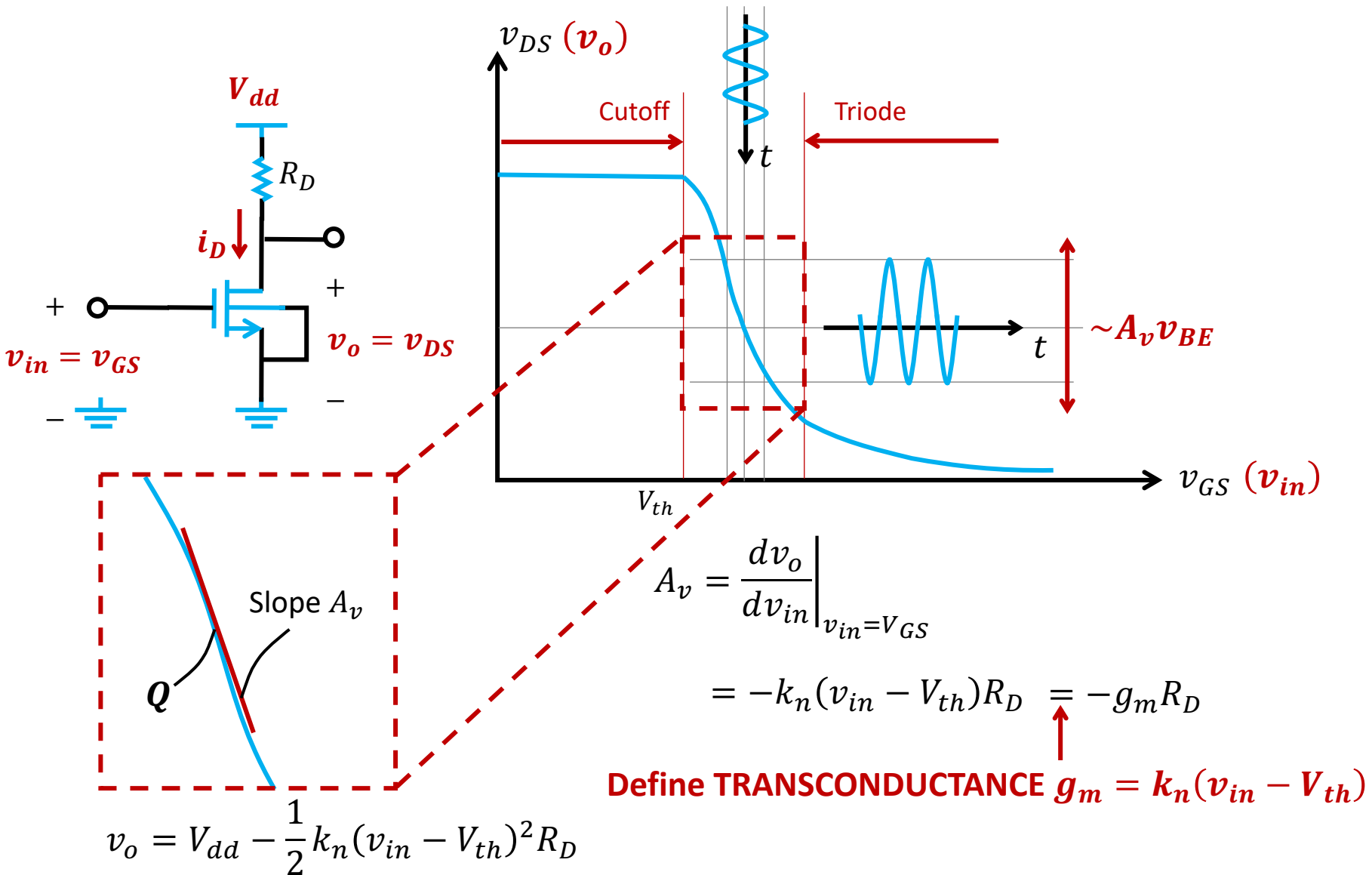


- Step 1 find a proper **bias point / quiescent point**
- Step 2 characterized by **DC voltages** V_{GS} and V_{DS}

$$V_{DS} = V_{dd} - R_D I_D = V_{dd} - \frac{1}{2} k_n (v_{GS} - V_{th})^2 R_D$$

- Step 3 a **sufficiently small voltage** v_{GS} superimposed on V_{GS}

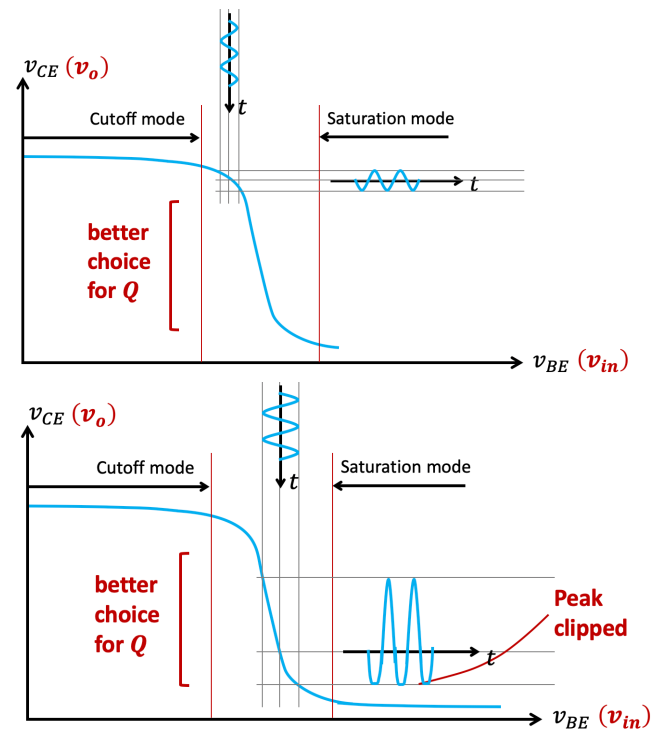
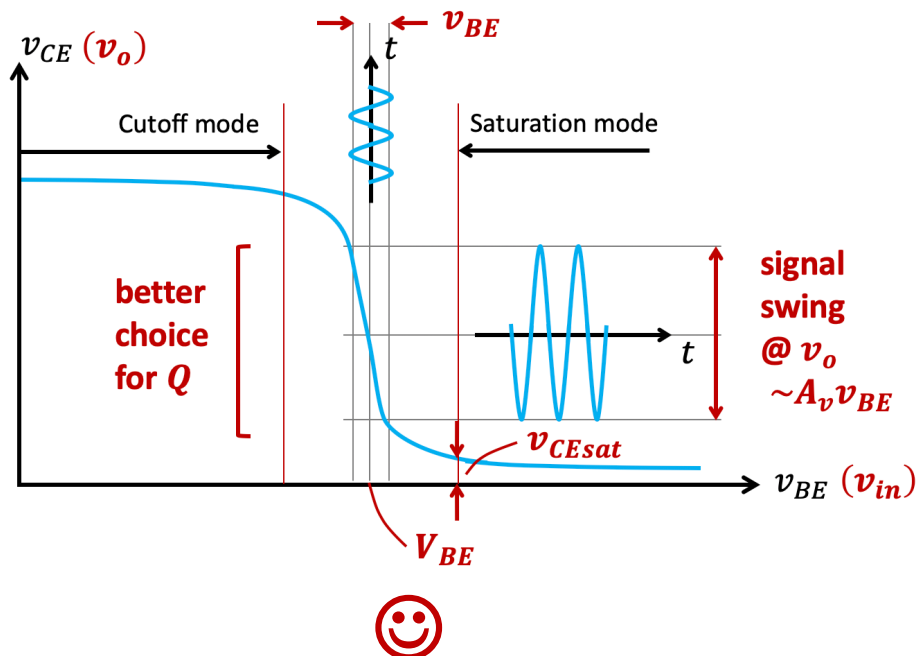
Signal Amplification of MOSFET



Recall: Locate the Bias Point for BJT

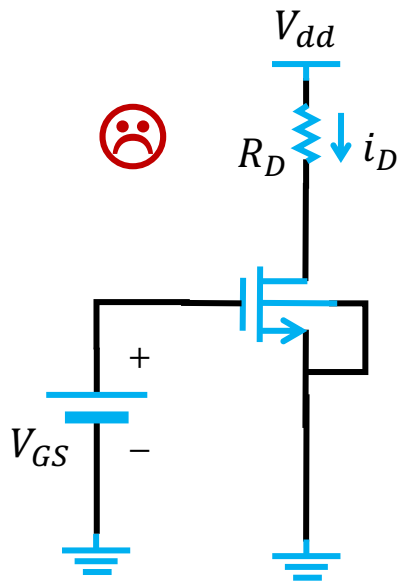
WHY? Key to small signal amplification

- Q is determined by V_{BE}
- Q determines the **GAIN** A_v
- Q has effects on the **OUTPUT SIGNAL SWING**

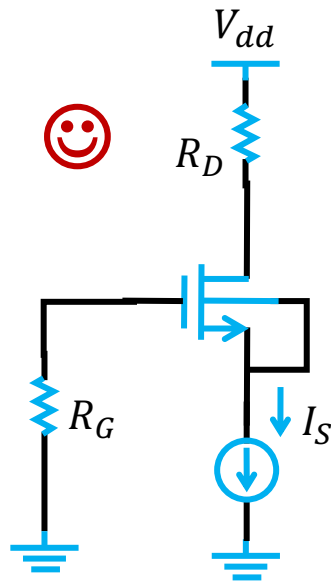


Locate the Bias Point for MOSFET

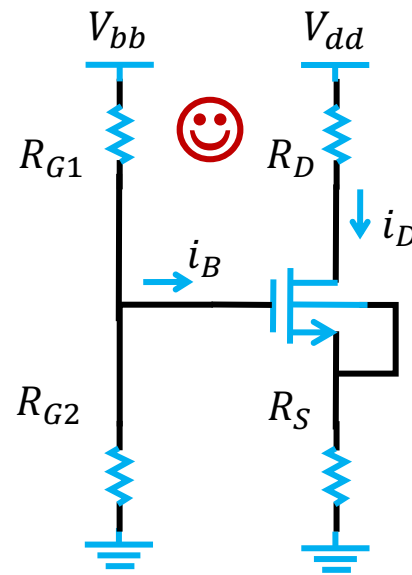
- Method 1
By fixing V_{GS}



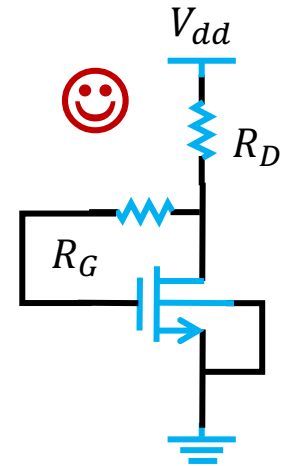
- Method 2
By fixing I_S



- Method 3
By fixing V_G &
Connecting a R_S



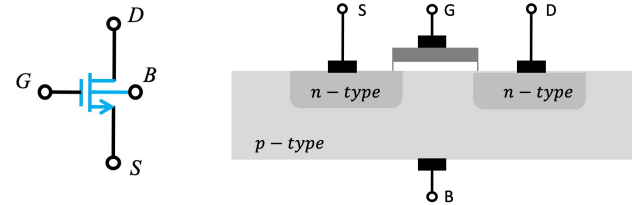
- Method 4 –
Using Drain-to-Gate
feedback resistor



Outline

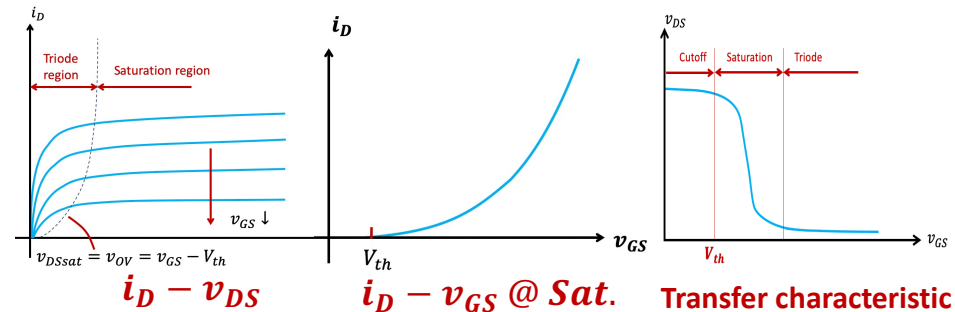
■ Introduction to MOSFET

- WHAT does it look like
- HOW does it work



■ The characteristic curves

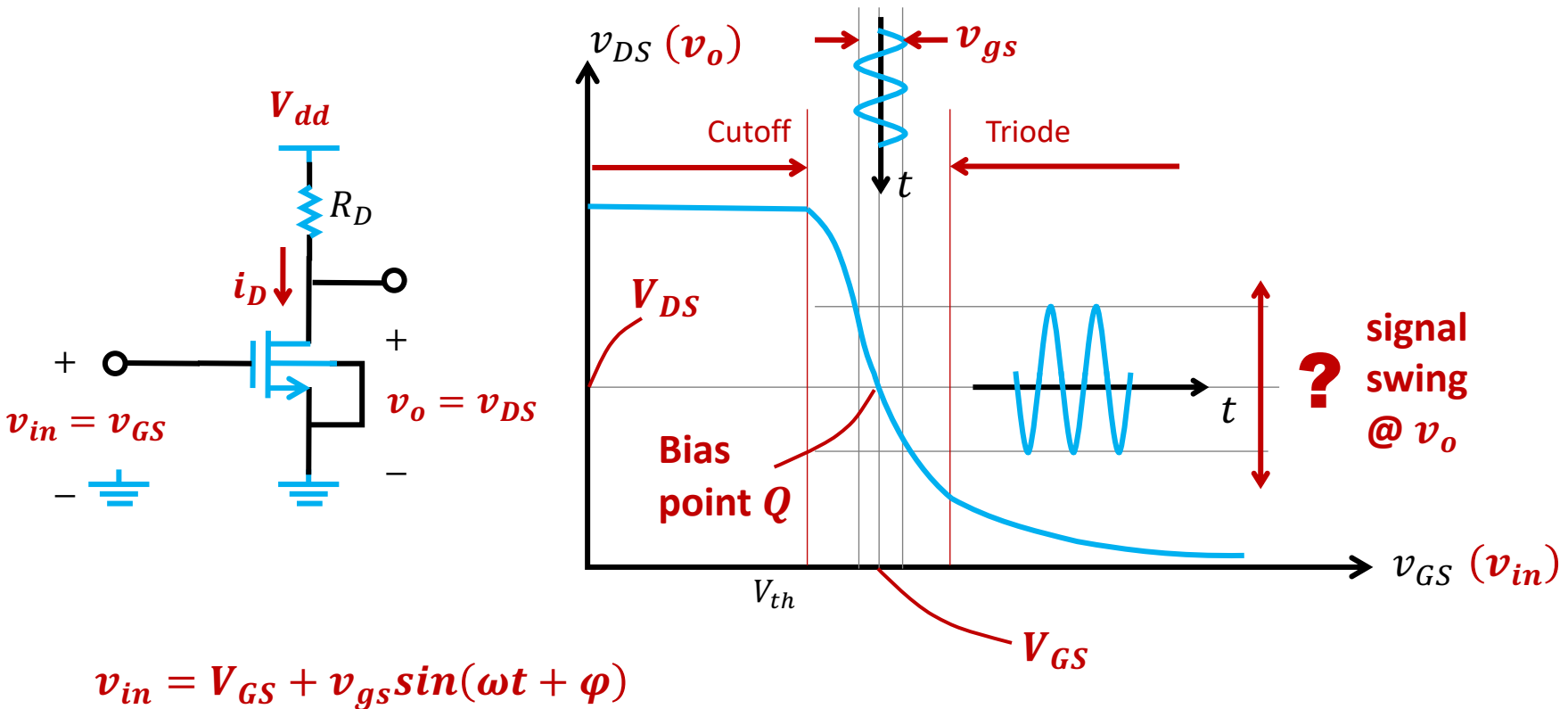
- $i_D - v_{DS}$ characteristics
- $i_D - v_{GS}$ characteristics
- The transfer characteristic



■ Circuit analysis techniques with MOSFET

- DC analysis techniques
- AC analysis techniques
 - Locate the bias point
 - **Small-signal operation & model**

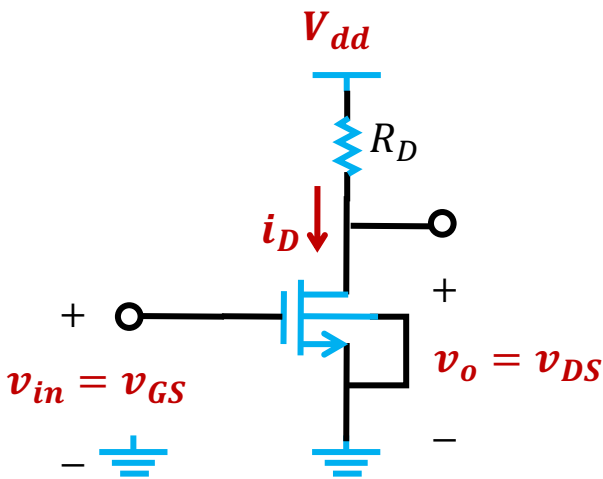
Signal Amplification of MOSFET



$$v_{in} = V_{GS} + v_{gs} \sin(\omega t + \varphi)$$

- The transistor is biased @ $v_{GS} = V_{GS}$
- A small AC signal is applied to input

Signal Amplification of MOSFET



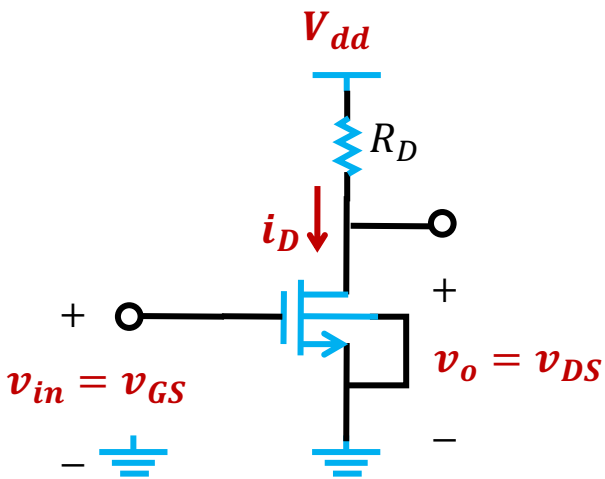
**Recall: Define
TRANSCONDUCTANCE**
 $g_m = k_n(v_{in} - V_{th})$

$$v_{in} = V_{GS} + v_{gs}\sin(\omega t + \varphi) = V_{GS} + v_{in,AC}$$

- In saturation region, according to the transistor characteristic

$$\begin{aligned} i_D &= \frac{1}{2}k_n(V_{GS} + v_{in,AC} - V_{th})^2 \\ &= \frac{1}{2}k_n(V_{GS} - V_{th})^2 + k_n(V_{GS} - V_{th})v_{in,AC} + \frac{1}{2}k_nv_{in,AC}^2 \\ &\quad \xrightarrow{\text{If } (V_{GS} - V_{th})v_{in,AC} \gg \frac{1}{2}v_{in,AC}^2} \\ &\approx \underbrace{I_D}_{I_D} + \underbrace{k_n(V_{GS} - V_{th})v_{in,AC}}_{g_m} \end{aligned}$$

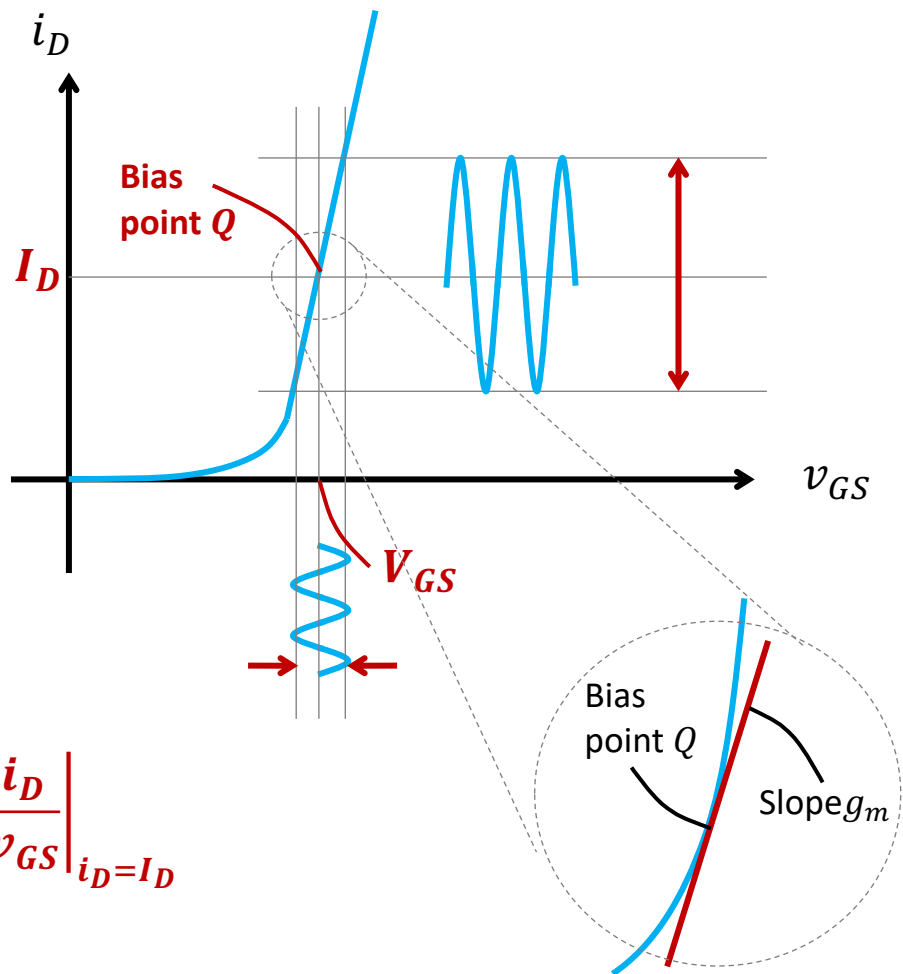
Transconductance g_m



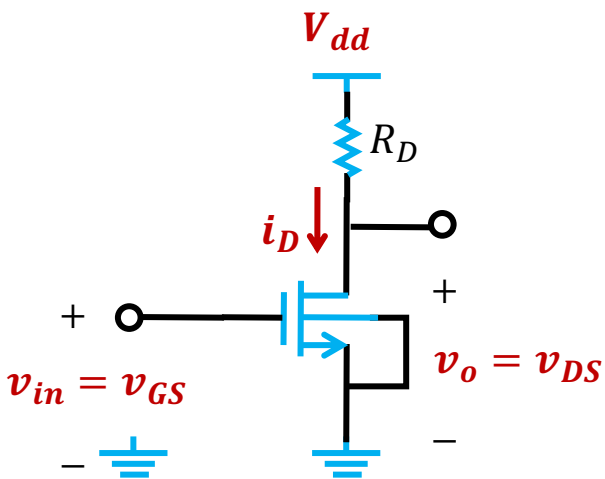
$$v_{in} = V_{GS} + v_{in,AC}$$

$$i_D = I_D + g_m v_{in,AC}$$

$$\text{TRANSCONDUCTANCE } g_m = \left. \frac{\partial i_D}{\partial v_{GS}} \right|_{i_D = I_D}$$



Voltage Gain A_v



- According to KVL

$$\begin{aligned}
 v_o &= V_{dd} - i_D R_D = V_{dd} - \left(i_D \Big|_{DC} + i_D \Big|_{AC} \right) R_D \\
 &= \underbrace{V_{dd} - I_D R_D}_{V_{DS}} - \underbrace{i_D \Big|_{AC} R_D}_{g_m v_{in,AC}}
 \end{aligned}$$

- Define voltage gain

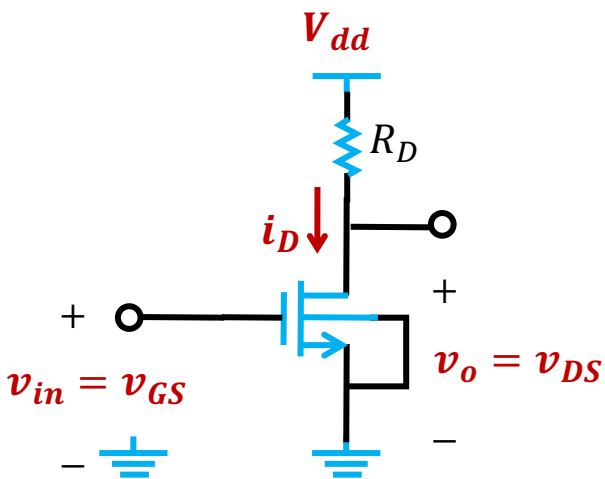
$$v_{in} = V_{GS} + v_{in,AC}$$

$$i_D = I_D + g_m v_{in,AC}$$

$$g_m = \left. \frac{\partial i_D}{\partial v_{GS}} \right|_{i_D = I_D}$$

$$A_v = \frac{v_o \Big|_{AC}}{v_{in} \Big|_{AC}} = \frac{-g_m v_{in,AC} R_D}{v_{in,AC}} = -g_m R_D$$

Summary: Small-Signal Operation

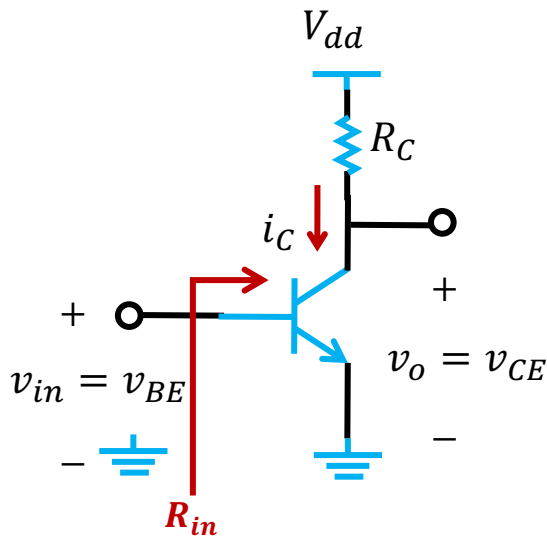


$$v_{in} = V_{GS} + v_{gs}\sin(\omega t + \varphi) = V_{GS} + v_{in,AC}$$

Voltage gain $A_v = \frac{v_o|_{AC}}{v_{in}|_{AC}} = -g_m R_D$

Transconductance $g_m = \left. \frac{\partial i_D}{\partial v_{GS}} \right|_{i_D=I_D} = k_n (V_{GS} - V_{th})$

Recall: BJT Small-Signal Operation



$$v_{in} = V_{BE} + v_{be}\sin(\omega t + \varphi) = V_{BE} + v_{AC}$$

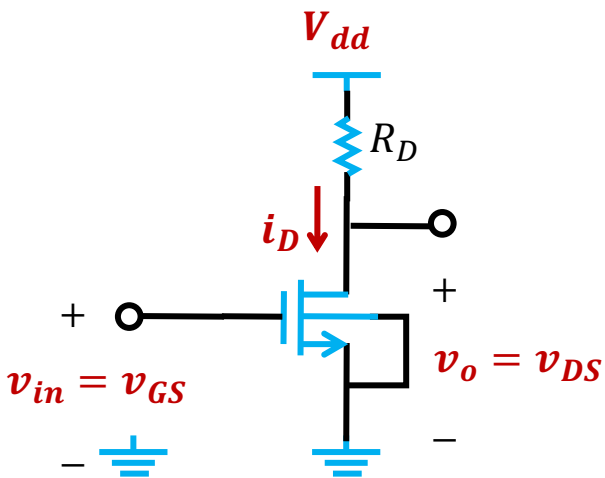
Voltage gain $A_v = \frac{v_o|_{AC}}{v_{in}|_{AC}} = -g_m R_C$

Transconductance $g_m = \left. \frac{\partial i_C}{\partial v_{BE}} \right|_{i_C=I_C} = \frac{I_C}{V_T}$

Input resistance @ Base $R_{in} = \frac{V_T}{I_B} = \frac{\beta}{g_m}$

Why we don't discuss input resistance for MOSFET?

Summary: Small-Signal Operation



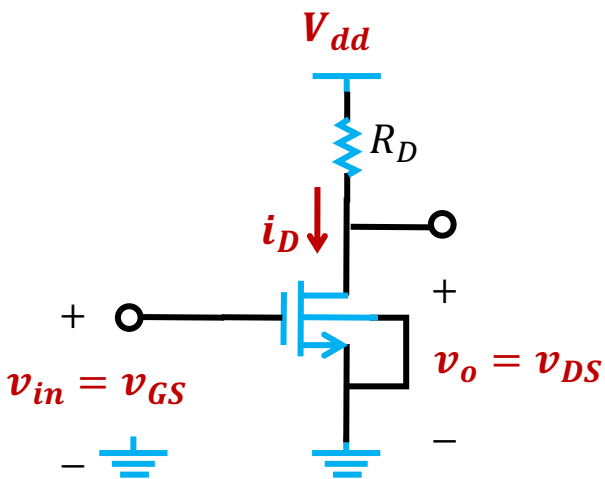
$$v_{in} = V_{GS} + v_{gs}\sin(\omega t + \varphi) = V_{GS} + v_{in,AC}$$

Voltage gain $A_v = \frac{v_o|_{AC}}{v_{in}|_{AC}} = -g_m R_D$

Transconductance $g_m = \left. \frac{\partial i_D}{\partial v_{GS}} \right|_{i_D=I_D} = k_n(V_{GS} - V_{th})$

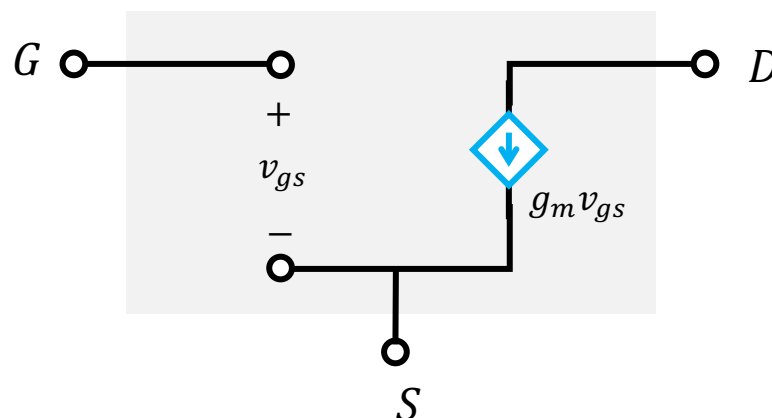
Input resistance @ Base $R_{in} = \infty$ Since $i_G = 0$

Small-Signal Model

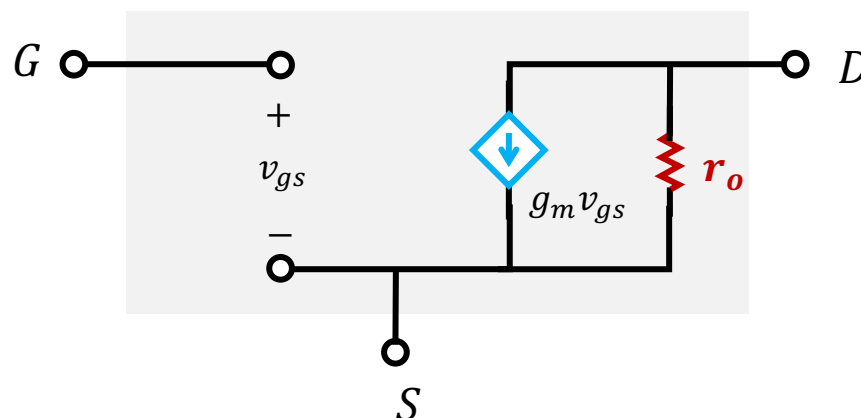


$$\left[\begin{array}{l} v_{in} = V_{GS} + v_{gs}\sin(\omega t + \varphi) = V_{GS} + v_{in,AC} \\ \text{Voltage gain } A_v = \frac{v_o|_{AC}}{v_{in}|_{AC}} = -g_m R_D \\ \text{Transconductance } g_m = \left. \frac{\partial i_D}{\partial v_{GS}} \right|_{i_D = I_D} = k_n (V_{GS} - V_{th}) \end{array} \right]$$

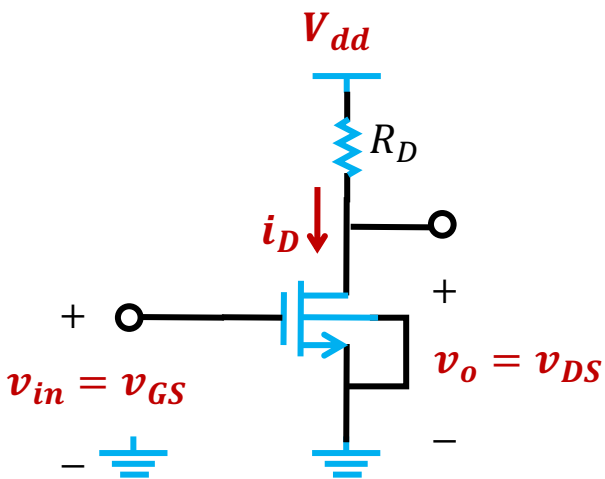
SIMPLIFIED HYBRID- π MODEL



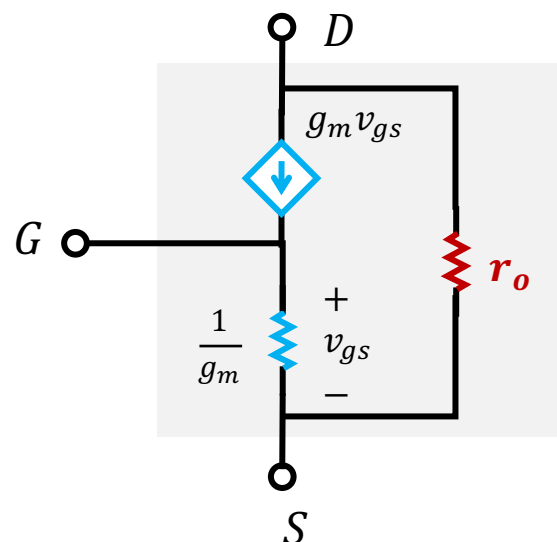
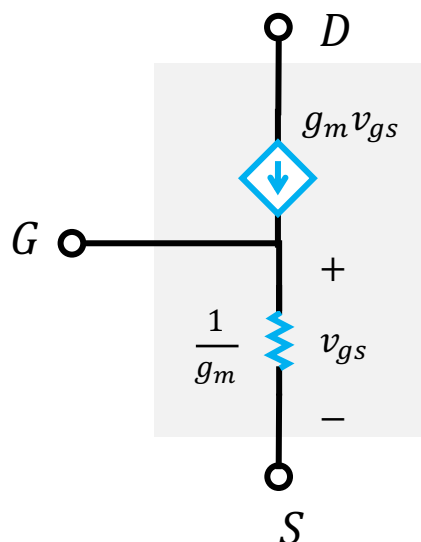
IF take channel-length modulation into account



Small-Signal Model

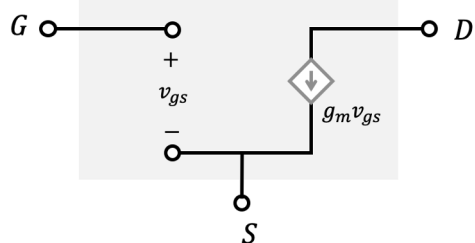


SIMPLIFIED T MODEL



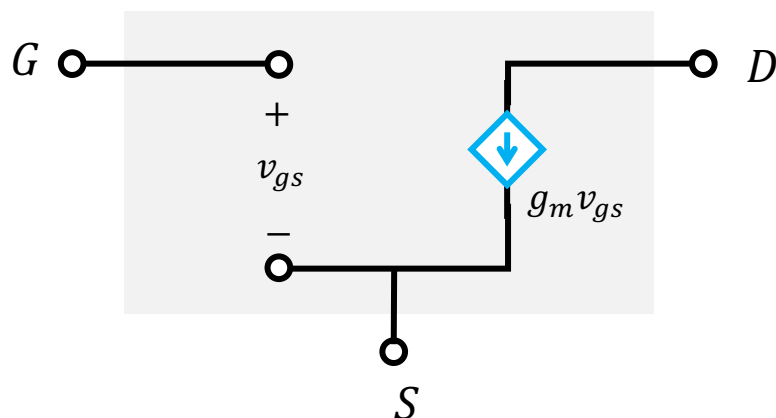
IF take channel-length modulation into account

SIMPLIFIED HYBRID- π MODEL

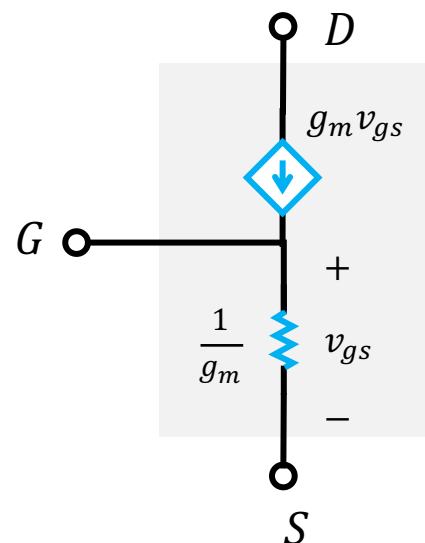


Small-Signal Model

SIMPLIFIED HYBRID- π MODEL



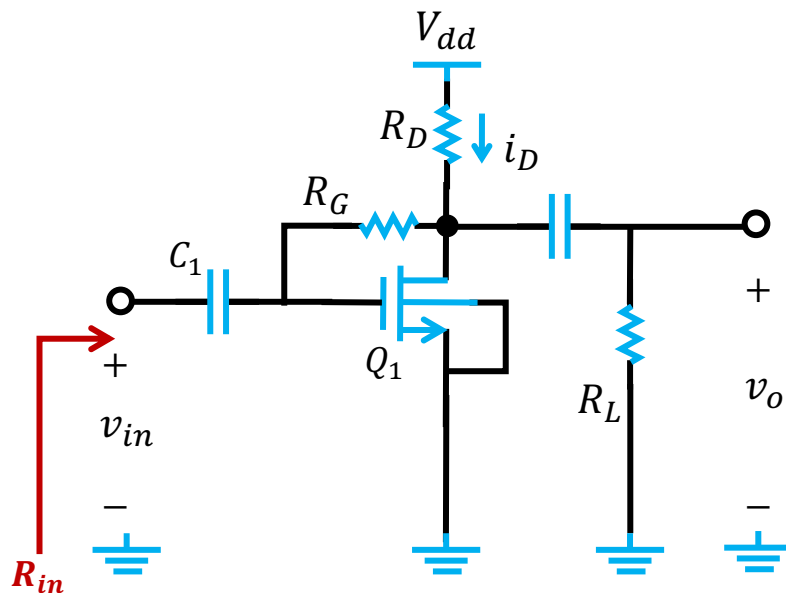
SIMPLIFIED T MODEL



HYBRID- π MODEL and T MODEL are equivalent

Example 2

QUESTION: Find out the small-signal voltage gain A_v , input resistance R_{in} and the largest allowable input signal with $V_{th} = 1.5V$, $V_{dd} = 15V$, $V_A = 50V$, $k_n = 0.25\mu A/V^2$.



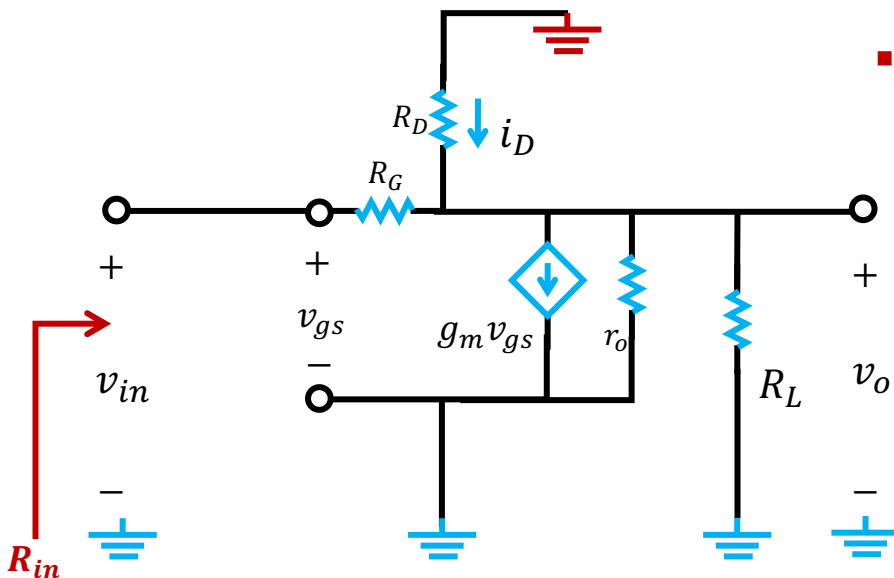
- **Step 1: perform DC analysis**
- Let's **ASSUME** Q_1 is biased in **Saturation mode**
- According to KVL & transistor characteristics

$$\begin{cases} v_D = V_{dd} - I_D R_D \\ I_D = \frac{1}{2} k_n (v_{GS} - V_{th})^2 \\ I_G = 0 \end{cases} \Rightarrow \begin{cases} I_D = 1.06mA \\ v_D = 4.4V \\ v_{GS} = v_D \end{cases}$$

Check assumption

Example 2

QUESTION: Find out the small-signal voltage gain A_v , input resistance R_{in} and the largest allowable input signal with $V_{th} = 1.5V$, $V_{dd} = 15V$, $V_A = 50V$, $k_n = 0.25\mu A/V^2$.



- **Step 2: perform AC analysis**

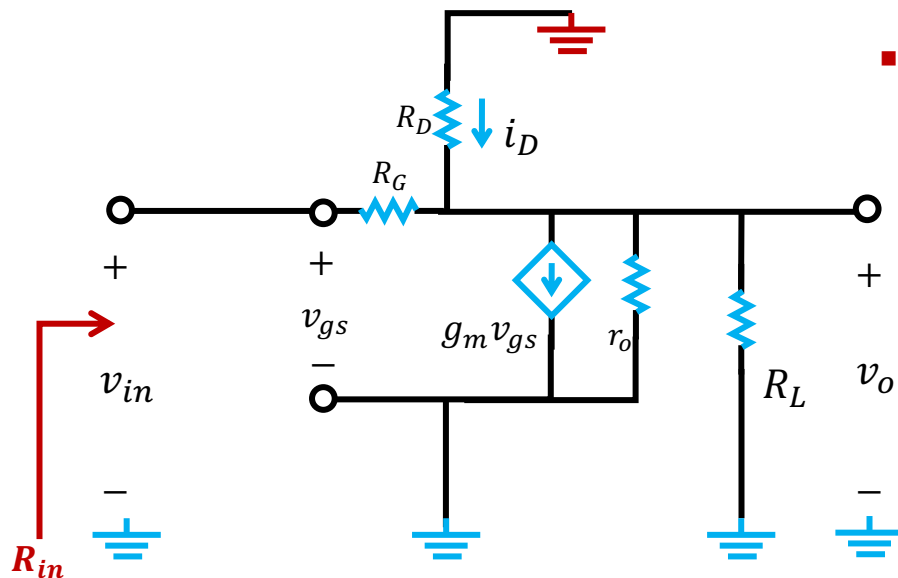
- **Step 2.1: replace the transistor with the small-signal model**
- **Step 2.2: turn off DC sources**
 - SHORT all voltage sources
 - OPEN all current sources
- **Step 2.3: Calculate small-signal model parameters**

$$g_m = k_n (v_{GS} - V_{th}) = 0.725 \text{ mA/V}$$

$$r_o = \frac{V_A}{I_D} = 47 \text{ k}\Omega$$

Example 2

QUESTION: Find out the small-signal voltage gain A_v , input resistance R_{in} and the largest allowable input signal with $V_{th} = 1.5V$, $V_{dd} = 15V$, $V_A = 50V$, $k_n = 0.25\mu A/V^2$.



- **Step 2: perform AC analysis**

- **Step 2.4: Analyze resulting circuit**

$$v_o \approx -g_m v_{gs} (R_D || R_L || r_o)$$

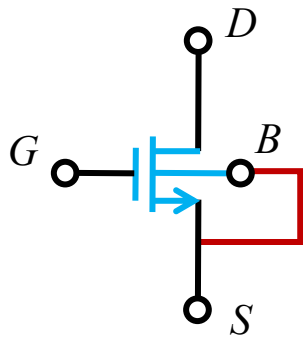
$$= -g_m v_{in} (R_D || R_L || r_o)$$

The voltage gain

$$A_v = \frac{v_o}{v_{in}} = -g_m (R_D || R_L || r_o)$$

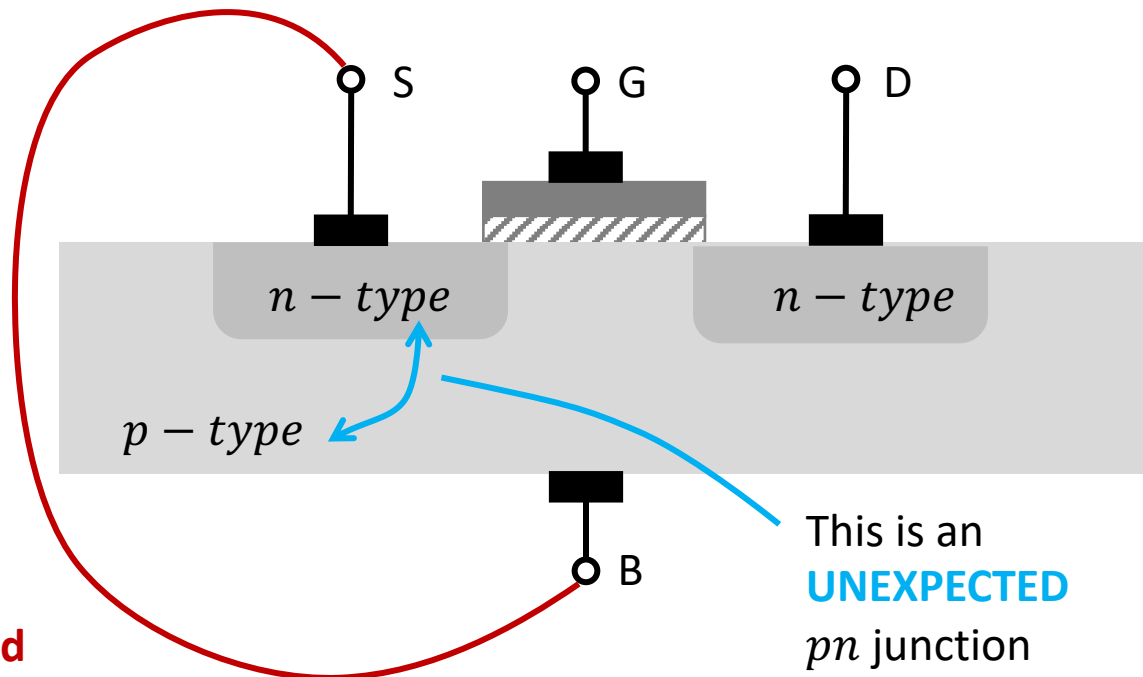
$$= -3.3$$

Recall: Body Effect of MOSFET



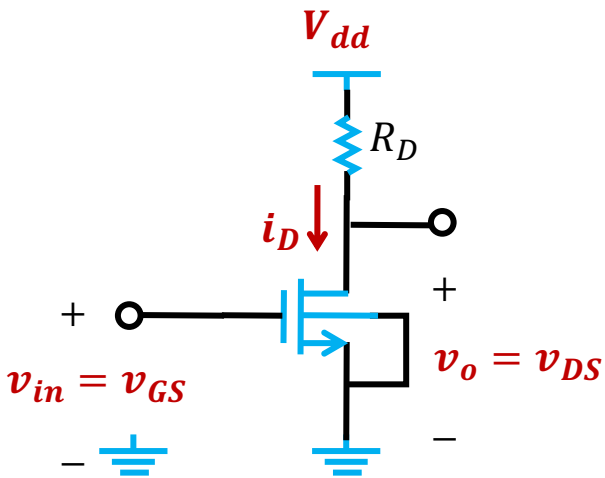
Enhancement-type
NMOS transistor

BODY EFFECT: the substrate acts as a second gate when the *pn* junction between the base and the source is forward biased

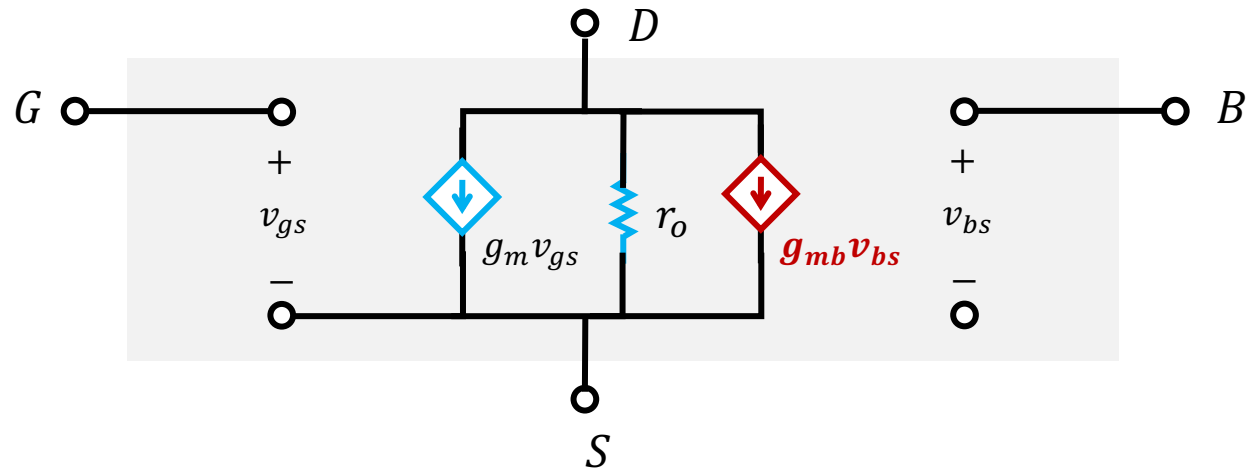


**Enable a zero bias
(cutoff mode)**

Small-Signal Model



IF take body effect into account



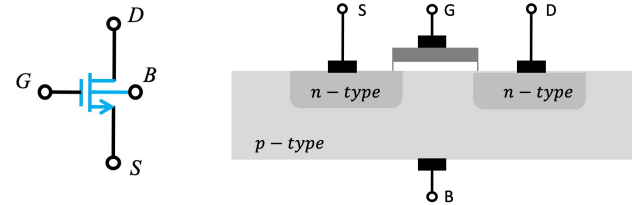
Define BODY TRANSCONDUCTANCE

$$g_{mb} = \left. \frac{\partial i_D}{\partial v_{BS}} \right|_{v_{GS}=\text{constant}, v_{DS}=\text{constant}}$$

Outline

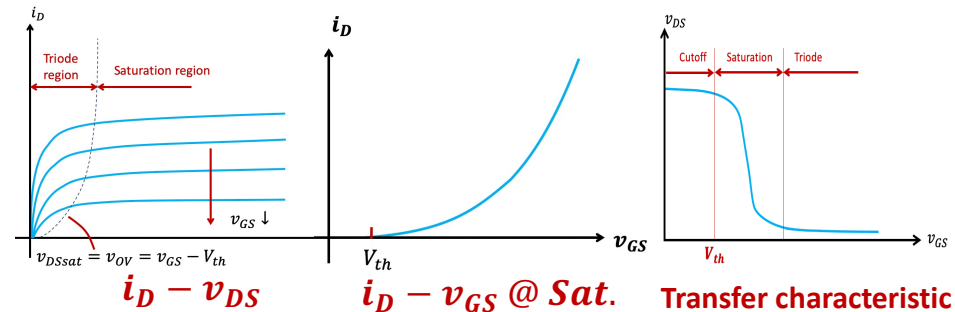
■ Introduction to MOSFET

- WHAT does it look like
- HOW does it work



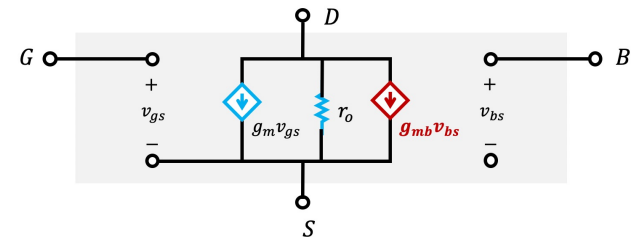
■ The characteristic curves

- $i_D - v_{DS}$ characteristics
- $i_D - v_{GS}$ characteristics
- The transfer characteristic

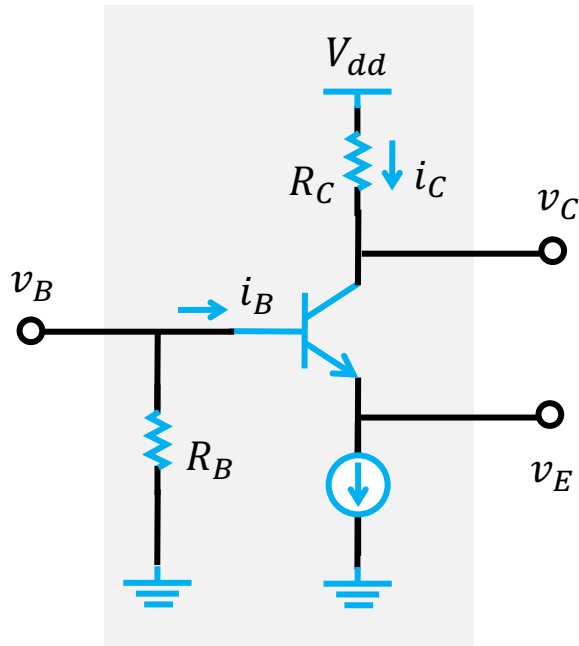


■ Circuit analysis techniques with MOSFET

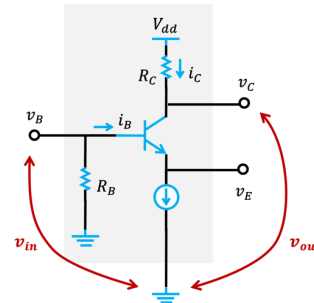
- DC analysis techniques
- AC analysis techniques
 - Locate the bias point
 - Small-signal operation & model
 - **Basic MOSFET Amplifier Configurations**



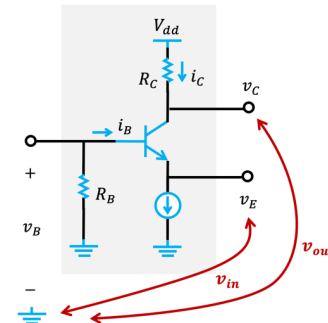
Recall: Basic BJT Amp. Configurations



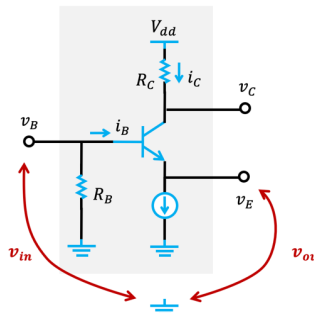
- Assume the transistor is biased in active mode
- There are three ports: **Base**, **Collector** and **Emitter**
- An amplifier requires two voltages: v_{in} and v_o



Common-Emitter (CE) Amplifier

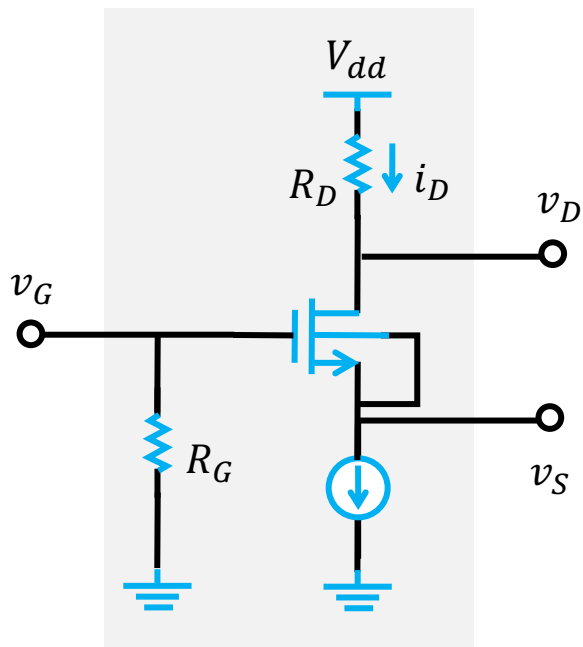


Common-Base (CB) Amplifier

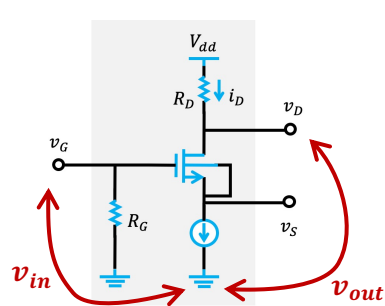


Common-Collector (CC) Amplifier

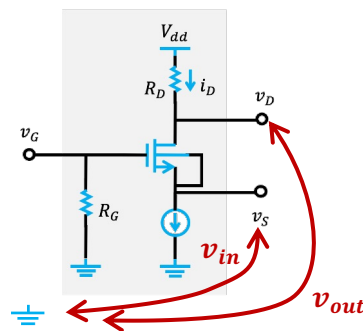
Basic MOSFET Amp. Configurations



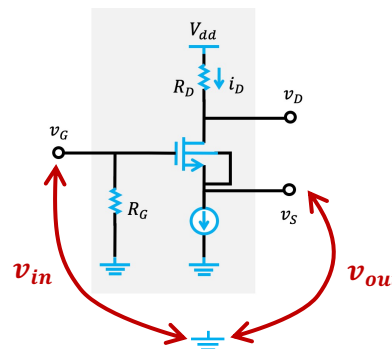
- Assume the transistor is biased in saturation mode
- There are three ports: **Gate**, **Source** and **Drain**
- An amplifier requires two voltages: v_{in} and v_o



Common-Source (CS) Amplifier



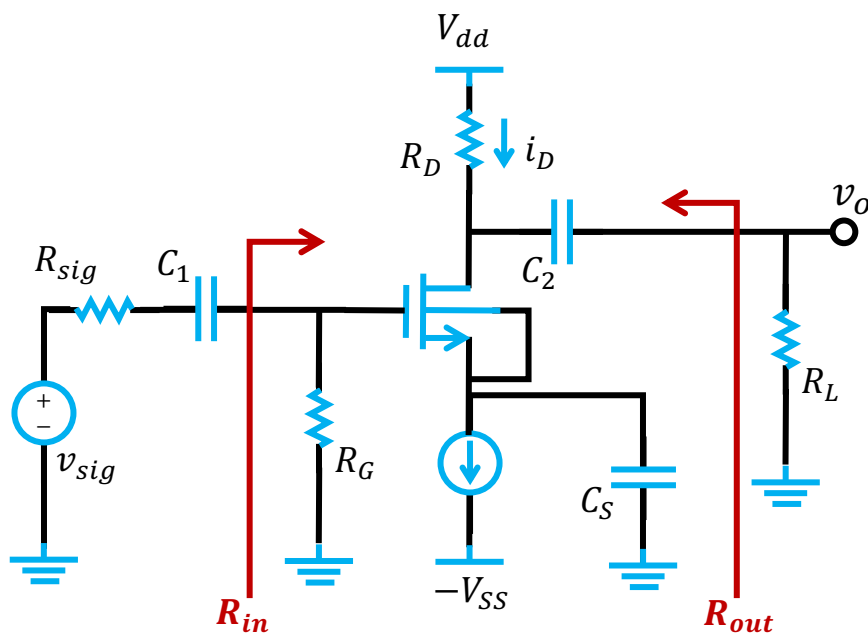
Common-Gate (CG) Amplifier



Common-Drain (CD) Amplifier

Example 3: CS Amplifier

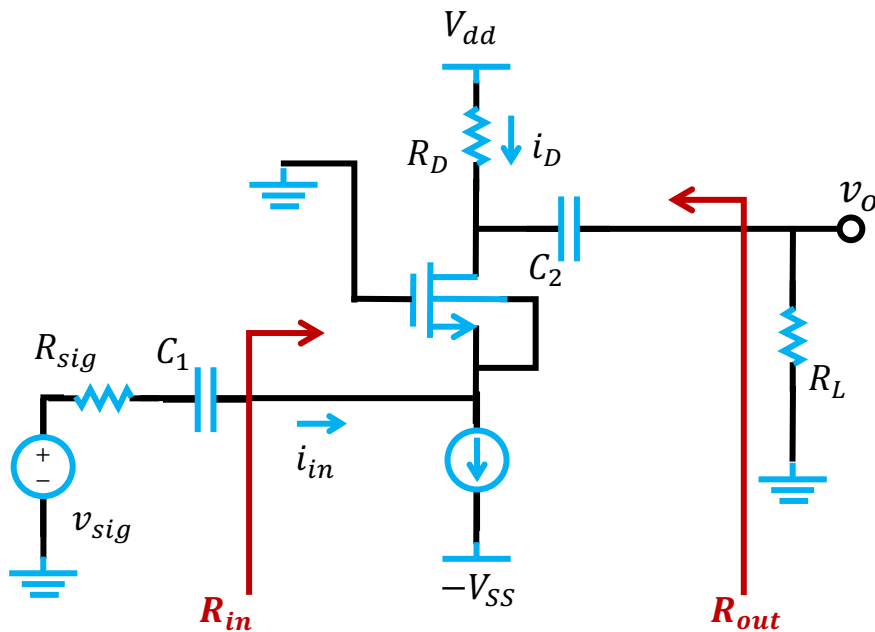
QUESTION: Find out the voltage gain, the input and output resistances with given v_{in} , R_{sig} , k_n , R_G , R_D , C_1 , C_2 , C_S , V_{dd} , V_{SS} , and V_{th} .



- **Step 1: perform DC analysis**
- **Step 2: perform AC analysis**
 - Step 2.1: turn off DC sources
 - Step 2.2: Calculate small-signal model parameter, g_m
 - Step 2.3: replace the transistor with the small-signal model
 - Step 2.4: Analyze the resulting circuit

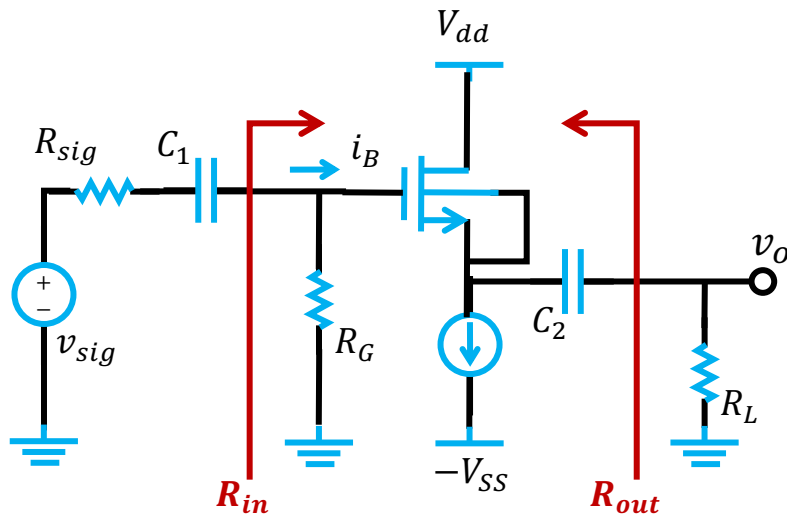
Example 4: CG Amplifier

QUESTION: Find out the voltage gain, the input and output resistances with given v_{in} , R_{sig} , k_n , R_D , C_1 , C_2 , V_{dd} , V_{SS} , and V_{th} . The body effect can not be neglected.

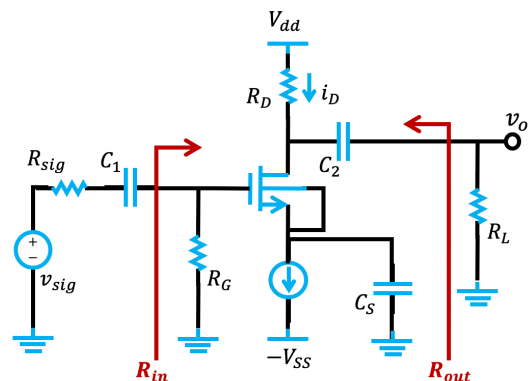


Example 5: CD Amplifier

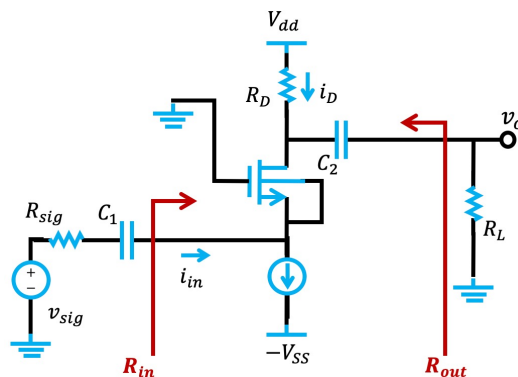
QUESTION: Find out the voltage gain, the input and output resistances with given v_{in} , R_{sig} , k_n , R_G , C_1 , C_2 , V_{dd} , V_{SS} , and V_{th} .



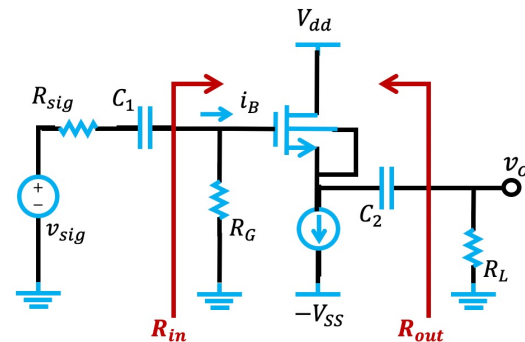
MOSFET Amp. Config.



CS Amplifier



CG Amplifier



CD Amplifier

R_{in}

R_G

$\frac{1}{g_m}$

R_G

R_{out}

$r_o || R_D$

R_D

$r_o || \frac{1}{g_m}$

G_v

$-g_m(R_D || R_L || r_o) \frac{R_G}{R_{sig} + R_G}$

$\frac{1}{R_{sig}g_m + 1} g_m(R_D || R_L)$

$\frac{R_G}{R_{sig} + R_G} \frac{g_m(R_L || r_o)}{g_m(R_L || r_o) + 1}$

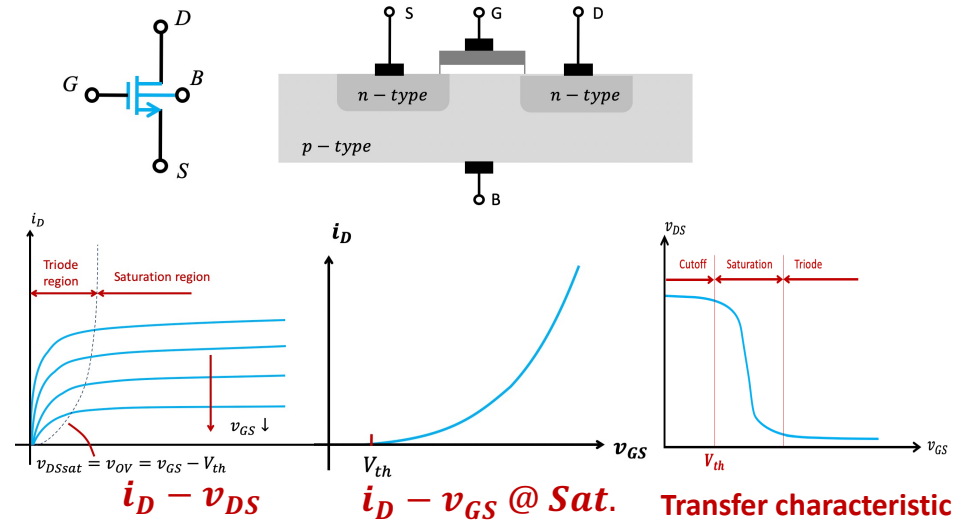
Outline

■ Introduction to MOSFET

- WHAT does it look like
- HOW does it work

■ The characteristic curves

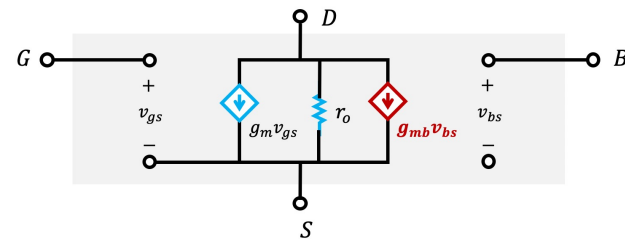
- $i_D - v_{DS}$ characteristics
- $i_D - v_{GS}$ characteristics
- The transfer characteristic



■ Circuit analysis techniques with MOSFET

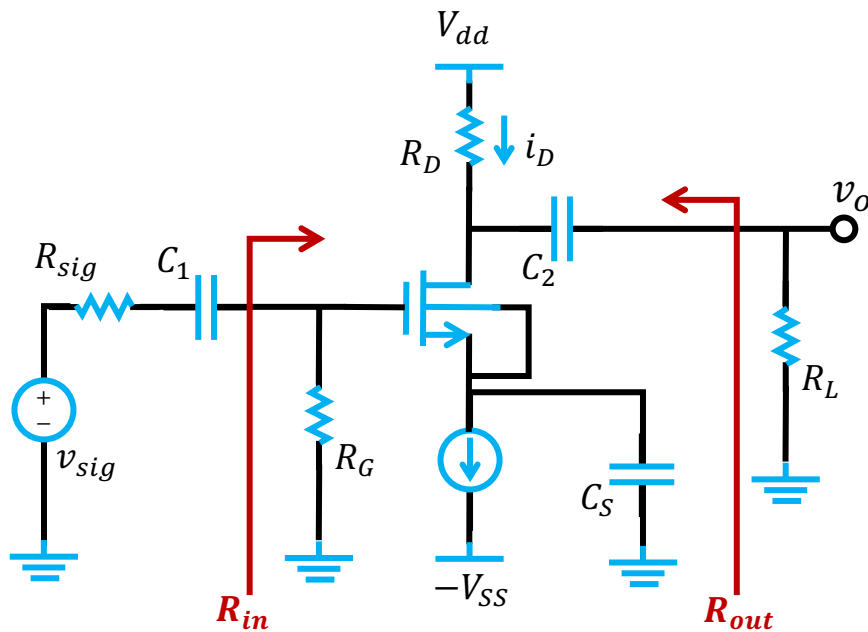
- DC analysis techniques
- AC analysis techniques
 - Locate the bias point
 - Small-signal operation & model
 - Basic BJT Amplifier Configurations

□ The frequency response



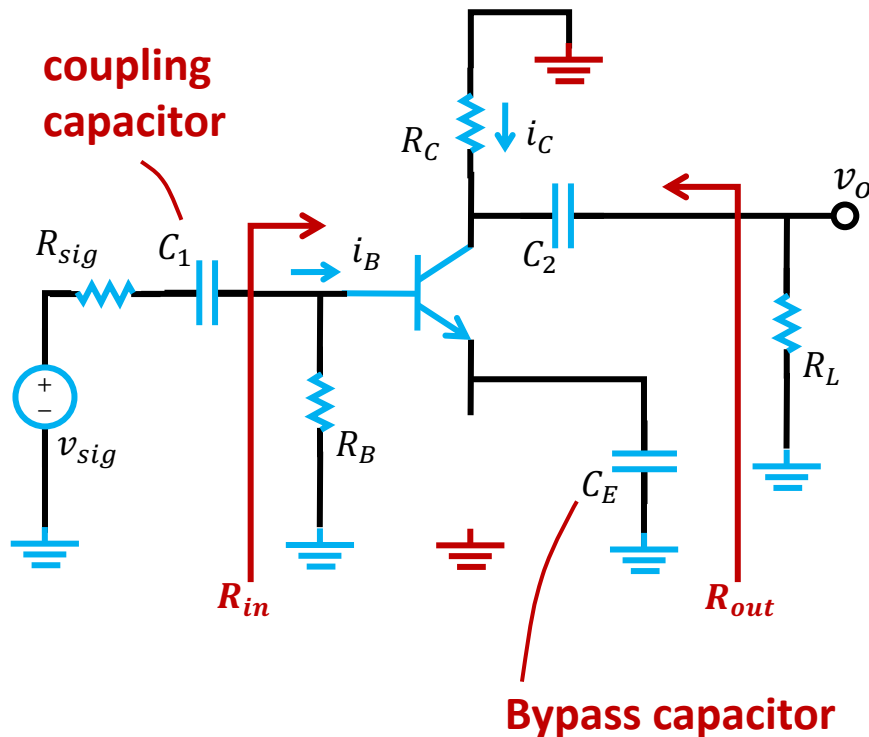
Example 6: CS Amp. w/ High Freq. Mod.

QUESTION: Find out the voltage gain, the input and output resistances with given v_{in} , R_{sig} , k_n , R_G , R_D , C_1 , C_2 , C_S , V_{dd} , V_{SS} , and V_{th} .



Recall: Example 5: BJT CE Amplifier

QUESTION: Find out the voltage gain, the input and output resistances with given v_{in} , R_{sig} , R_B , R_C , C_1 , C_2 , C_E , V_{dd} , V_{SS} , and β .



Step 2: perform AC analysis

- Step 2.1: turn off DC sources
 - SHORT all voltage sources
 - OPEN all current sources

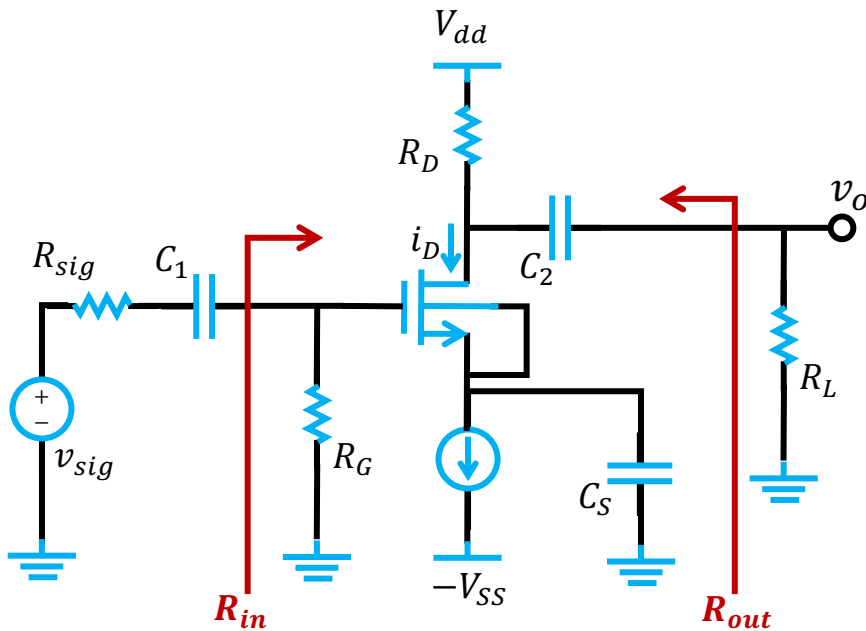
$$Z_{C_E} = \frac{1}{sC_E}$$

The **bypass capacitor** C_E provides a very **LOW** impedance to ground at all signal frequencies of interest

The **coupling capacitor** C_1 and C_2 act as **perfect short circuit** at all signal frequencies of interest

Example 6: CS Amp. w/ High Freq. Mod.

QUESTION: Find out the voltage gain, the input and output resistances with given v_{in} , R_{sig} , k_n , R_G , R_D , C_1 , C_2 , C_S , V_{dd} , V_{SS} , and V_{th} .

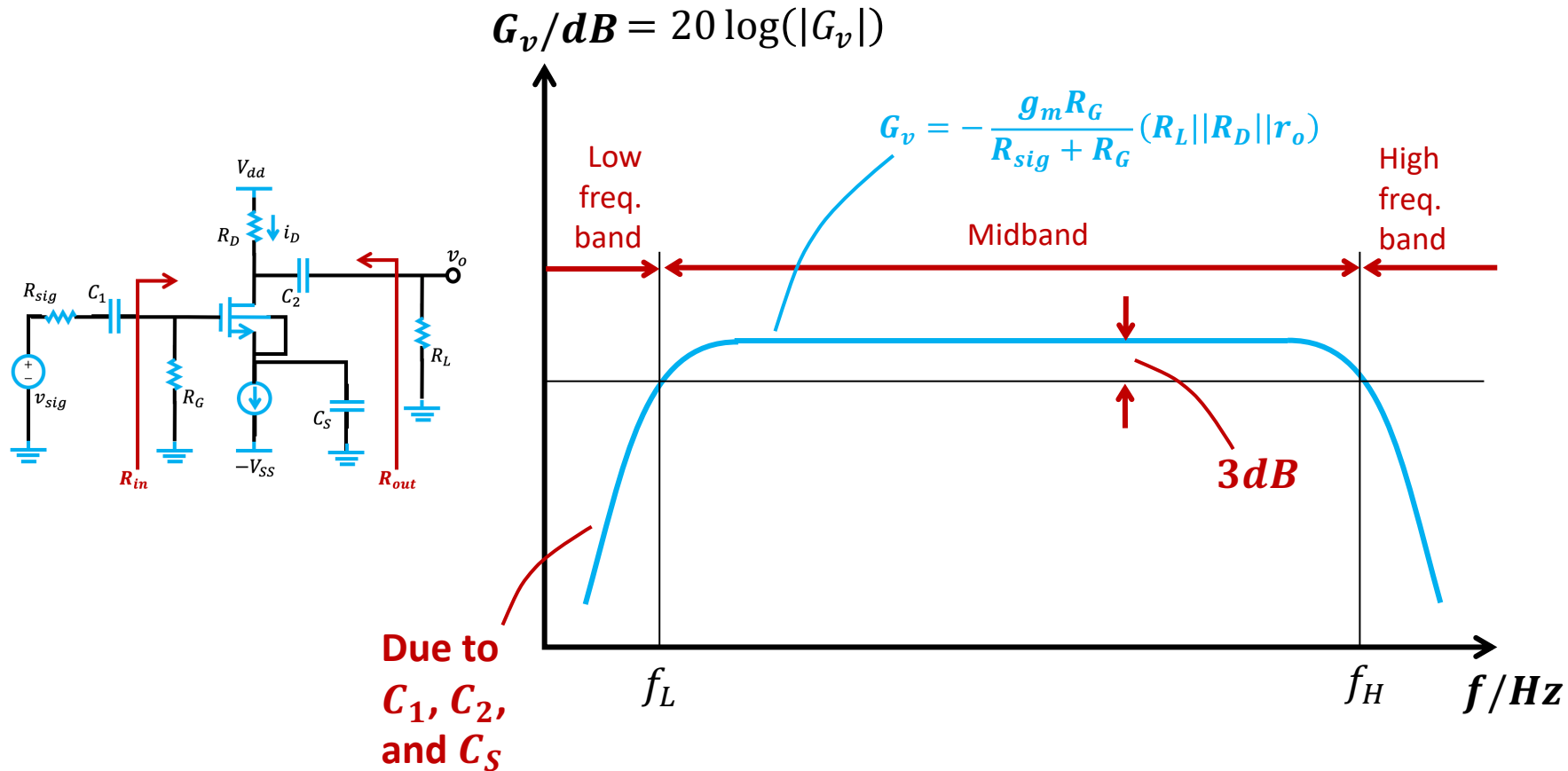


IDEALLY

- C_1 and C_2 are **coupling capacitors**
- C_S is **bypass capacitor**
- C_1 , C_2 , and C_S **IDEALLY** are short circuit @DC

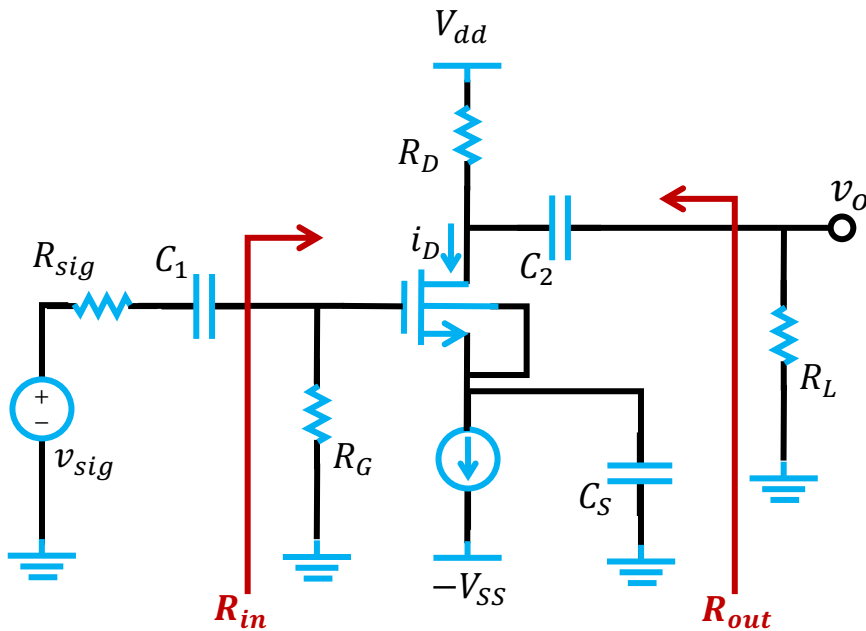
WHAT HAPPEN IN A REAL SCENARIO?

The Frequency Response



Example 6: CS Amp. w/ High Freq. Mod.

QUESTION: Find out the voltage gain, the input and output resistances with given v_{in} , R_{sig} , k_n , R_G , R_D , C_1 , C_2 , C_S , V_{dd} , V_{SS} , and V_{th} .

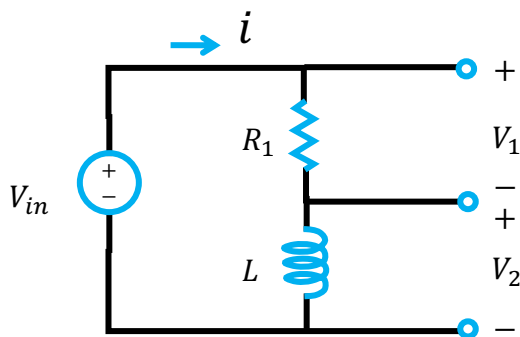


- Check the effect from C_1
- According to KVL

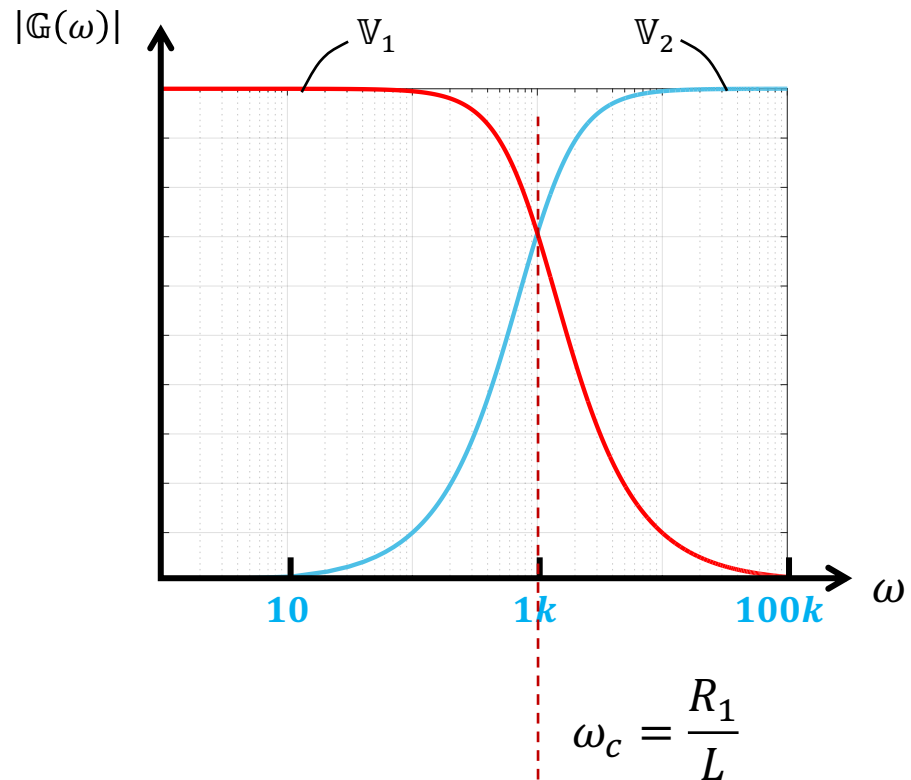
$$\begin{aligned} v_G &= \frac{R_G}{R_{sig} + \frac{1}{sC_1} + R_G} v_{sig} \\ &= v_{sig} \frac{R_G}{R_{sig} + R_G} \frac{1}{1 + \frac{1}{sC_1(R_{sig} + R_G)}} \end{aligned}$$

Recall: Transfer Function for HPF

QUESTION: design a stereo amplifier with two output channels to split the high and low frequencies.

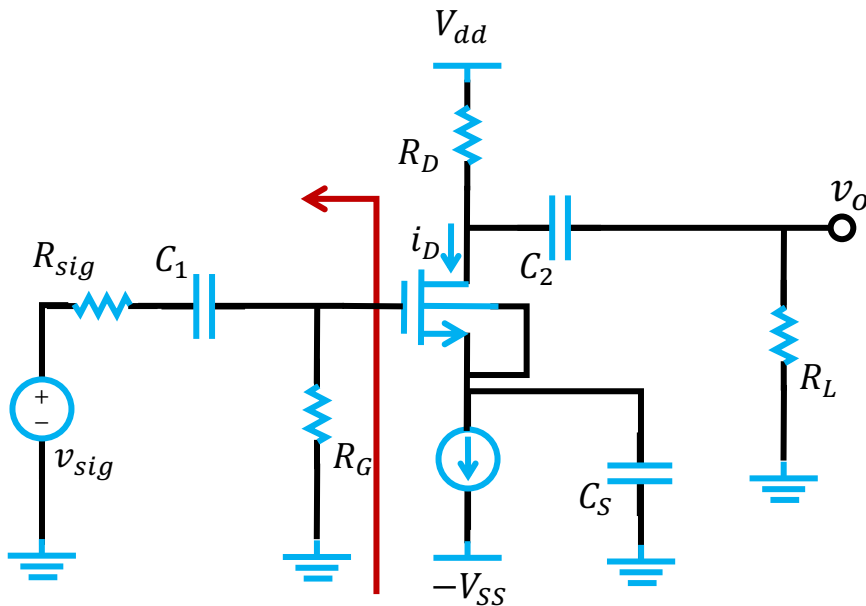


$$\left[\begin{aligned} \mathbb{G}_1(\omega) &= \frac{V_1}{V_{in}} = \frac{R_1}{R_1 + j\omega L} \\ \mathbb{G}_2(\omega) &= \frac{V_2}{V_{in}} = \frac{j\omega L}{R_1 + j\omega L} \end{aligned} \right]$$



Example 6: CS Amp. w/ High Freq. Mod.

QUESTION: Find out the voltage gain, the input and output resistances with given v_{in} , R_{sig} , k_n , R_G , R_D , C_1 , C_2 , C_S , V_{dd} , V_{SS} , and V_{th} .



- Check the effect from C_1
- According to KVL

$$v_G = \frac{R_G}{R_{sig} + \frac{1}{sC_1} + R_G} v_{sig}$$

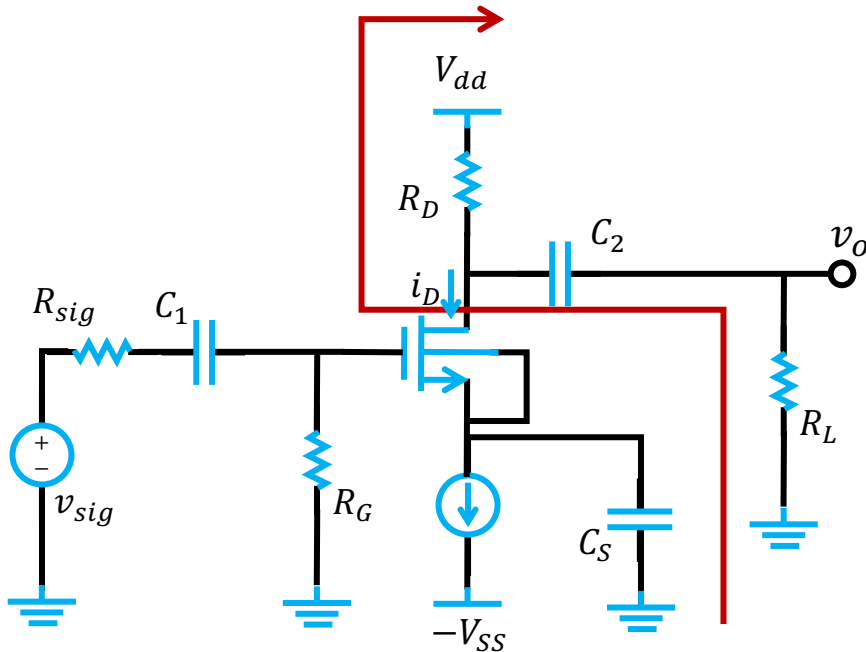
$$= v_{sig} \frac{R_G}{R_{sig} + R_G} \frac{s}{s + \frac{1}{C_1(R_{sig} + R_G)}}$$

This is a High Pass

$$\omega_{L1} = \frac{1}{C_1(R_{sig} + R_G)}$$

Example 6: CS Amp. w/ High Freq. Mod.

QUESTION: Find out the voltage gain, the input and output resistances with given v_{in} , R_{sig} , k_n , R_G , R_D , C_1 , C_2 , C_S , V_{dd} , V_{SS} , and V_{th} .



- Check the effect from C_2
- According to KVL

$$v_{out} = -i_d \frac{R_D R_L}{R_D + R_L + \frac{1}{sC_2}}$$

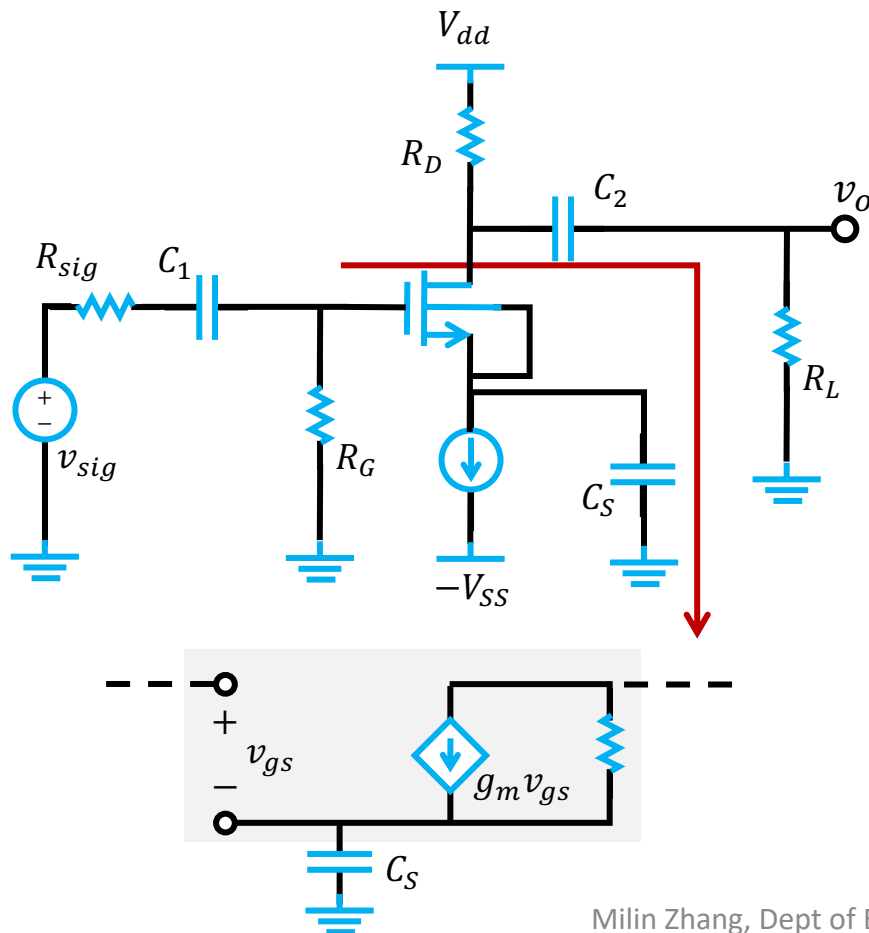
$$= -i_d R_L \frac{s}{s + \frac{1}{C_2(R_D + R_L)}}$$

This is a High Pass

$$\omega_{L2} = \frac{1}{C_2(R_D + R_L)}$$

Example 6: CS Amp. w/ High Freq. Mod.

QUESTION: Find out the voltage gain, the input and output resistances with given v_{in} , R_{sig} , k_n , R_G , R_D , C_1 , C_2 , C_S , V_{dd} , V_{SS} , and V_{th} .



- Check the effect from C_S
- According to KVL

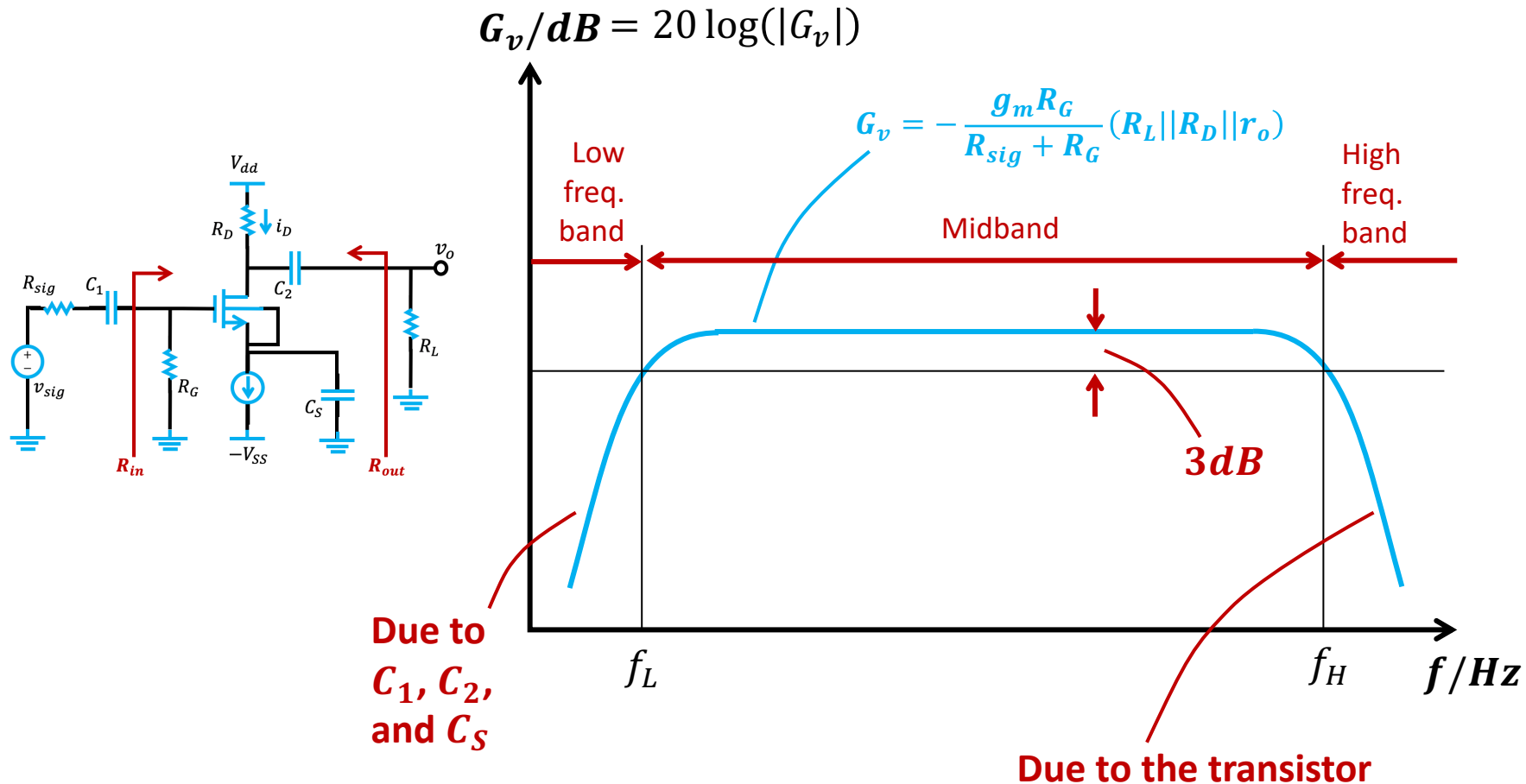
$$v_G - g_m v_{gs} \frac{1}{sC_S} = v_{gs}$$

$$v_{gs} = \frac{s}{s + \frac{g_m}{C_S}} v_G$$

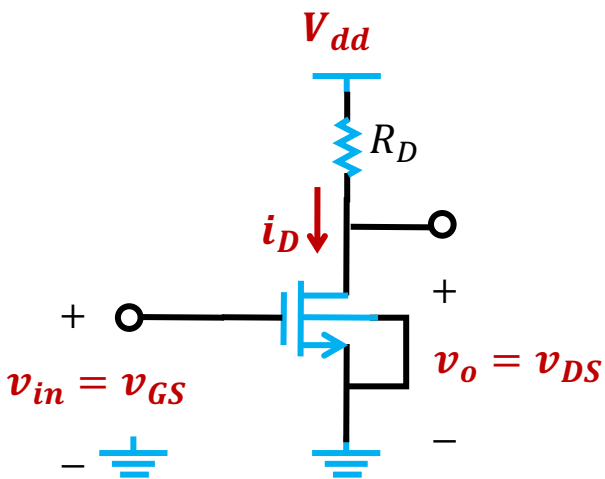
This is a High Pass

$$\omega_{L3} = \frac{g_m}{C_S}$$

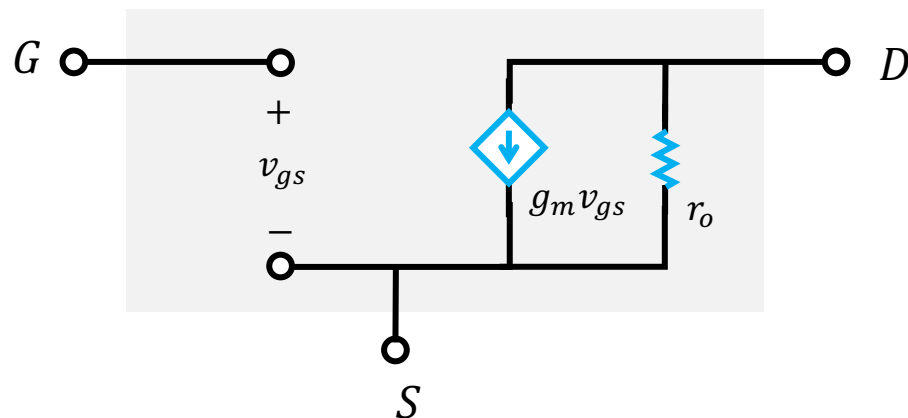
The Frequency Response



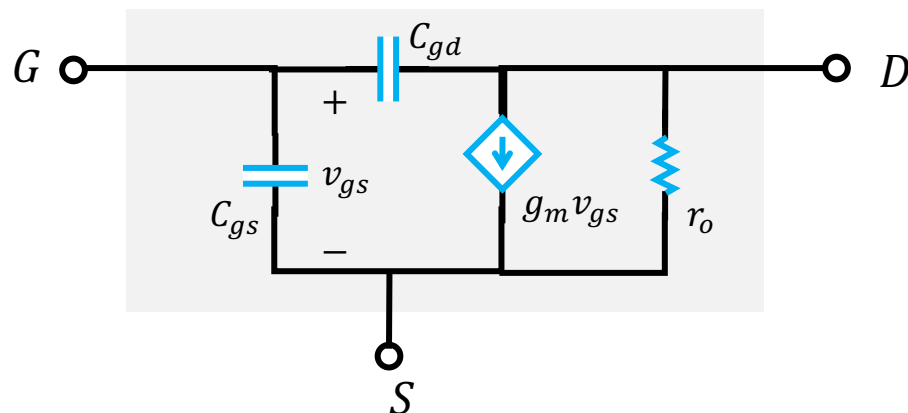
Small-Signal Model



SIMPLIFIED HYBRID- π MODEL

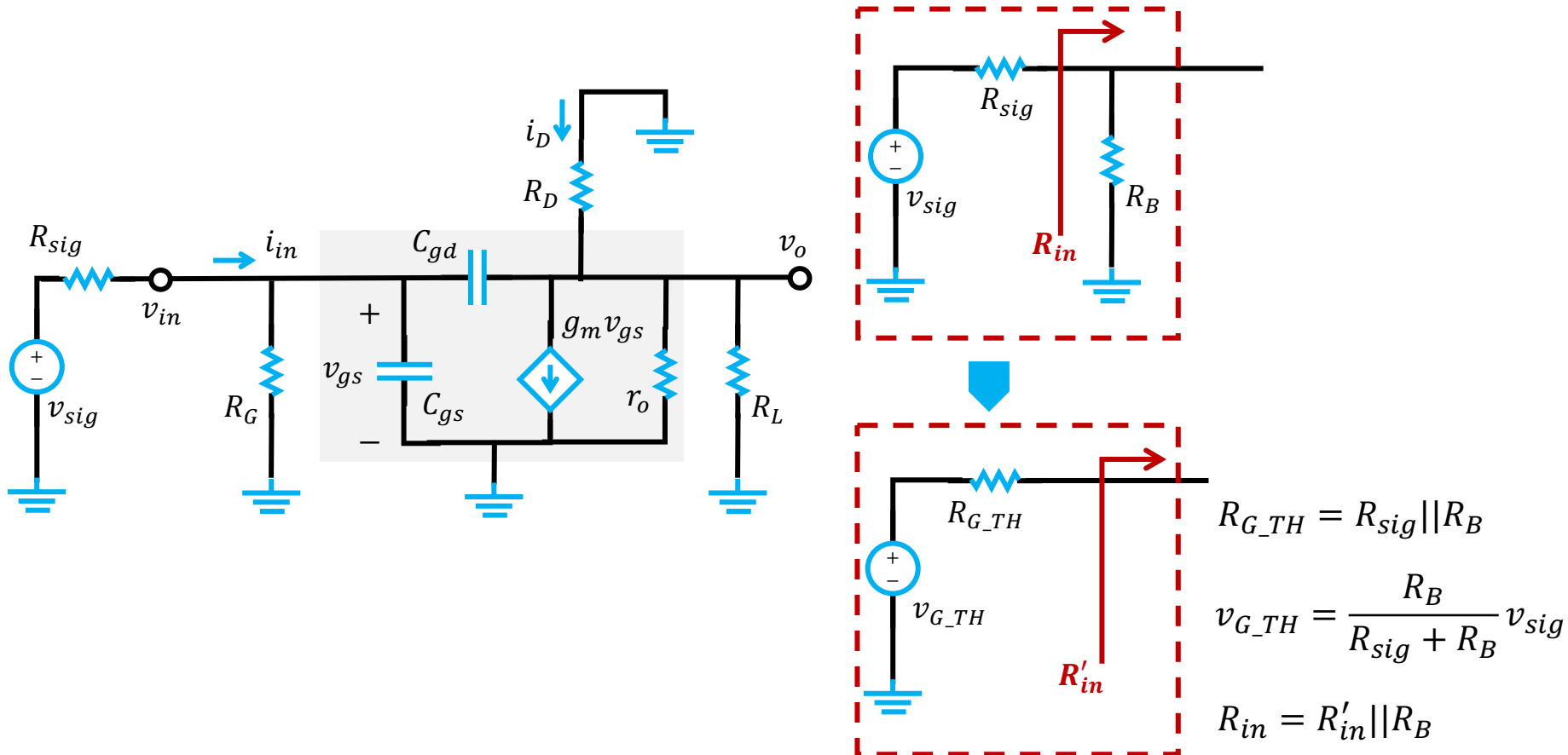


THE HIGH FREQUENCY MODEL



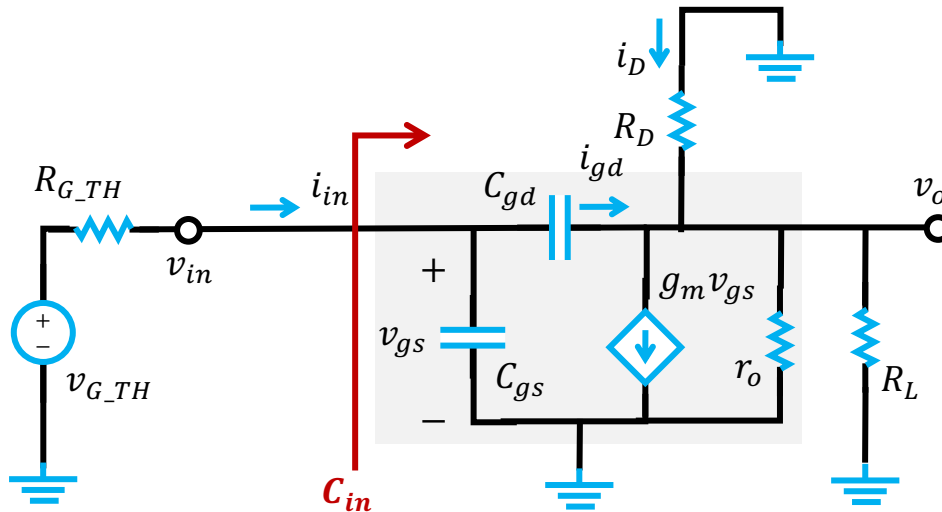
Example 6: CS Amp. w/ High Freq. Mod.

QUESTION: Find out the voltage gain, the input and output resistances with given v_{in} , R_{sig} , k_n , R_G , R_D , C_1 , C_2 , C_S , V_{dd} , V_{SS} , and V_{th} .



Example 6: CS Amp. w/ High Freq. Mod.

QUESTION: Find out the voltage gain, the input and output resistances with given v_{in} , R_{sig} , k_n , R_G , R_D , C_1 , C_2 , C_S , V_{dd} , V_{SS} , and V_{th} .



- According to KVL & KCL, and device characteristics

$$v_{out} = \frac{(-g_m v_{gs} + i_{gd})(R_L || R_D || r_o)}{\text{Define } R'_L}$$

$$\downarrow \quad g_m v_{gs} \gg i_{gd}$$

$$\approx -g_m v_{gs} R'_L$$

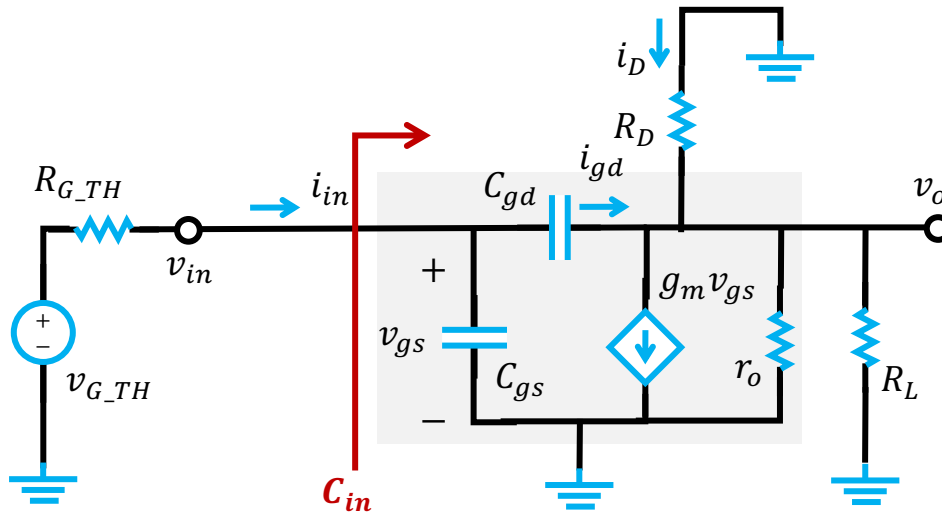
$$i_{gd} = (v_{gs} - v_{out})sC_{gd}$$

$$= (1 + g_m R'_L)sC_{gd}v_{gs}$$

$$i_{in} = i_{gs} + i_{gd} = \frac{(C_{gs} + C_{gd}(1 + g_m R'_L))s v_{gs}}{\text{Define } C_{in}}$$

Example 6: CS Amp. w/ High Freq. Mod.

QUESTION: Find out the voltage gain, the input and output resistances with given v_{in} , R_{sig} , k_n , R_G , R_D , C_1 , C_2 , C_S , V_{dd} , V_{SS} , and V_{th} .



$$R_{G_TH} = R_{sig} || R_B$$

$$v_{G_TH} = \frac{R_B}{R_{sig} + R_B} v_{sig}$$

$$C_{in} = C_{gs} + C_{gd}(1 + g_m R'_L)$$

- The overall gain

$$G_v = \frac{v_o}{v_{sig}} = \frac{-g_m v_{gs} R'_L}{v_{sig}}$$

$$= \frac{-g_m R'_L}{v_{sig}} \frac{v_{G_TH}}{1 + s R_{G_TH} C_{in}}$$

$$= \boxed{-\frac{g_m R'_L R_G}{R_{sig} + R_G}} \boxed{\frac{1}{1 + s R_{G_TH} C_{in}}}$$

**Midband
Gain**

Low Pass
 $\omega_{H1} = \frac{1}{R_{G_TH} C_{in}}$

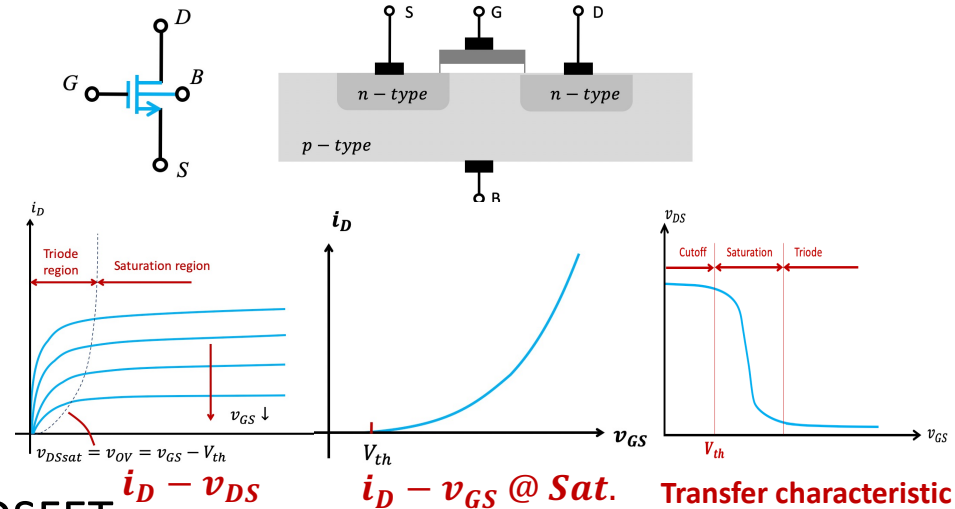
Outline

■ Introduction to MOSFET

- WHAT does it look like
- HOW does it work

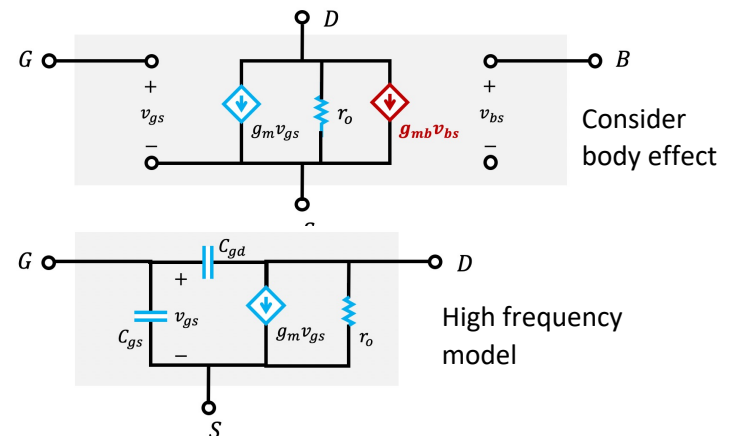
■ The characteristic curves

- $i_D - v_{DS}$ characteristics
- $i_D - v_{GS}$ characteristics
- The transfer characteristic



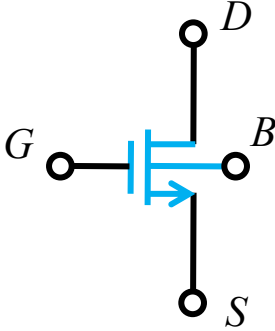
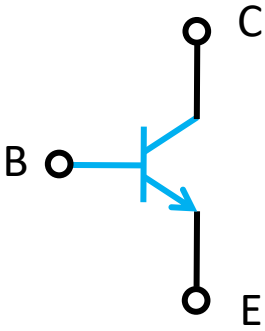
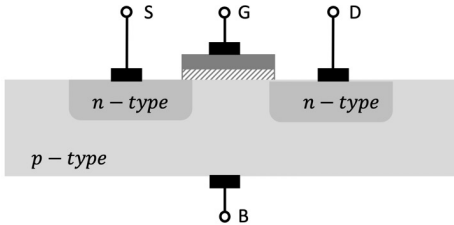
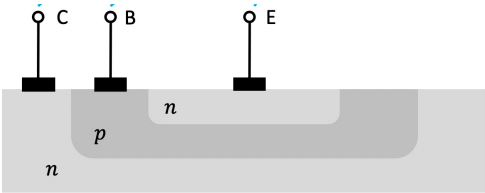
■ Circuit analysis techniques with MOSFET

- DC analysis techniques
- AC analysis techniques
 - Locate the bias point
 - Small-signal operation & model
 - Basic BJT Amplifier Configurations
- The frequency response



■ MOSFET v.s. BJT

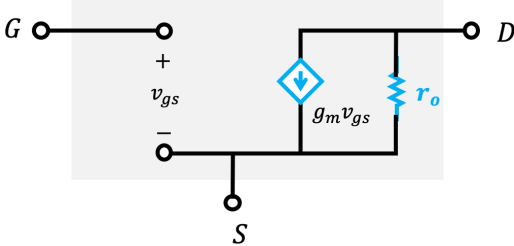
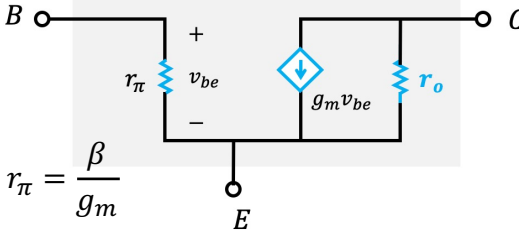
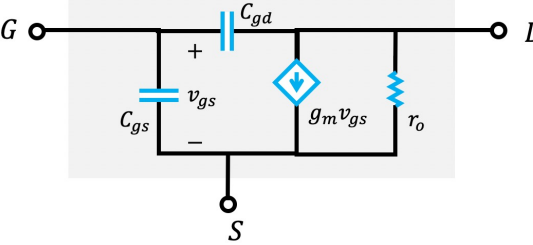
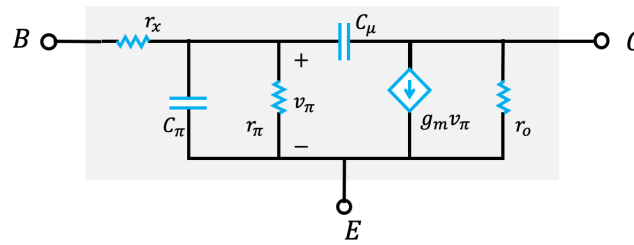
MOSFET v.s. BJT

	NMOS	<i>n</i> pn BJT
Circuit symbol		
Physical Structure		

MOSFET v.s. BJT

	NMOS	<i>n</i> pn BJT
To Operate in Active mode	biased in saturation region $v_{GS} > V_{th}$ └ To induce a channel $v_{GS} - v_{DS} < V_{th}$ └ To pinch-off channel @ drain	biased in active region $V_{BE} > V_{BEon}$ └ <i>EBJ</i> forward biased $V_{BC} < V_{BCon}$ └ <i>CBJ</i> reverse biased
Characteristics in active mode	$i_D = (\mu_n C_{ox}) \left(\frac{W}{L} \right) \left[(v_{GS} - V_{th}) v_{DS} - \frac{1}{2} v_{DS}^2 \right]$ $i_G = 0$	$i_C = I_S e^{\frac{v_{BE}}{V_T}} \left(1 + \frac{v_{CE}}{V_A} \right)$ $i_B = i_C / \beta$

MOSFET v.s. BJT

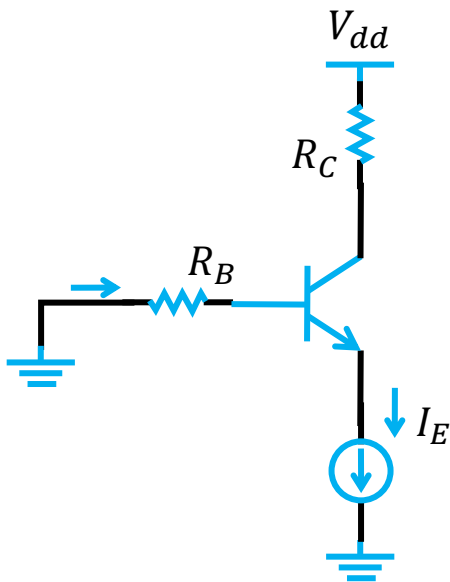
	NMOS	nnp BJT
The small signal model		
High frequency model		
Transconductance	$g_m = k_n (V_{GS} - V_{th})$	$g_m = \frac{I_C}{V_T}$
Input resistance w/ Source/Emitter grounded	$R_{in} = \infty$	$R_{in} = \frac{V_T}{I_B} = \frac{\beta}{g_m}$

Outline

- Introduction to MOSFET
- The characteristic curves
- Circuit analysis techniques with MOSFET
 - DC analysis techniques
 - AC analysis techniques
 - The frequency response
- MOSFET v.s. BJT
- More examples of useful modules
 - **Current sources**

Recall: HOW to Locate the Bias Point

- Method 3 – Using Constant-Current Source



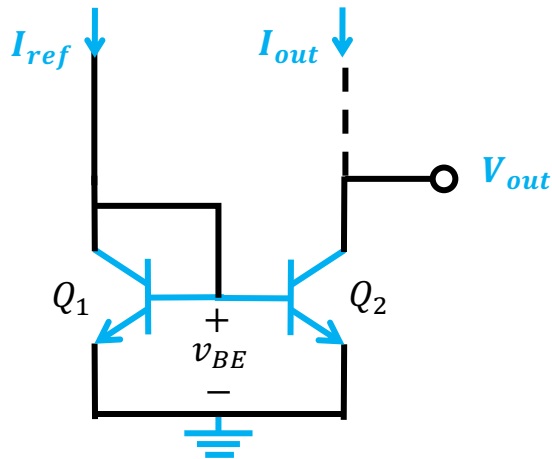
- I_E is independent of β and R_B
- How to realize a current source?



Current mirror explained in next chapter

Example 6: BJT current mirror

QUESTION: Find out the relationship between I_{ref} and I_{out} with Q_1 and Q_2 are matched.



- Since Q_1 and Q_2 are matched $\Rightarrow \beta_{Q_1} = \beta_{Q_2} = \beta$

- According to the BJT characteristic

$$I_{C1} = I_S e^{\frac{v_{BE1}}{V_T}} \quad \text{and} \quad I_{out} = I_{C2} = I_S e^{\frac{v_{BE2}}{V_T}}$$

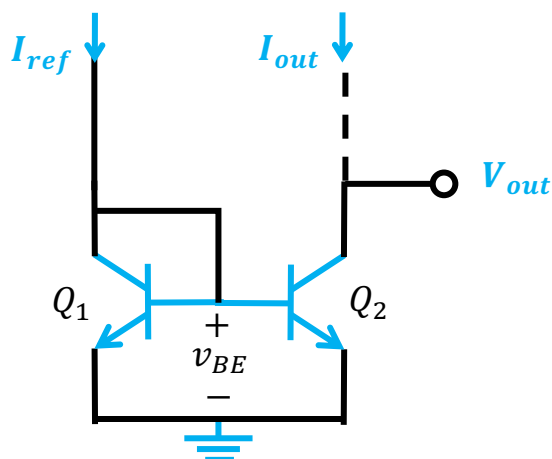
- According to KCL

$$I_{ref} = I_{C1} + I_{B1} + I_{B2} = I_{C1} + \frac{I_{C1}}{\beta} + \frac{I_{C2}}{\beta}$$

$$\Rightarrow \frac{I_{out}}{I_{ref}} = \frac{1}{1 + \frac{2}{\beta}} \xrightarrow{\beta \rightarrow \infty} 1$$

Example 6: BJT current mirror

QUESTION: Find out the relationship between I_{ref} and I_{out} with Q_1 and Q_2 are matched.



- What if the area of EBJ of Q_2 is m times of Q_1 ?

► $I_{S2} = mI_{S1}$

- According to the BJT characteristic

$$I_{C1} = I_{S1} e^{\frac{v_{BE1}}{V_T}} \quad \text{and} \quad I_{out} = I_{C2} = I_{S2} e^{\frac{v_{BE2}}{V_T}}$$

► $\frac{I_{out}}{I_{ref}} = \frac{m}{1 + \frac{1}{\beta} + \frac{m}{\beta}} \xrightarrow{\beta \rightarrow \infty} m$

collector current $i_C = I_S e^{\frac{v_{BE}}{V_T}}$

Saturation current $I_S = \frac{A_E q D_n n_i^2}{N_A W}$

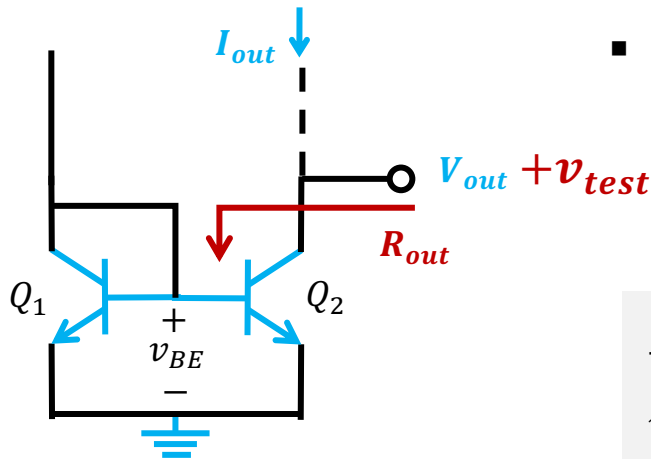
Labels for I_S equation:

- Cross-sectional area (A_E)
- Electron diffusivity in base (D_n)
- Carrier density (n_i^2)
- Base width (W)
- Doping concentration in the base (N_A)

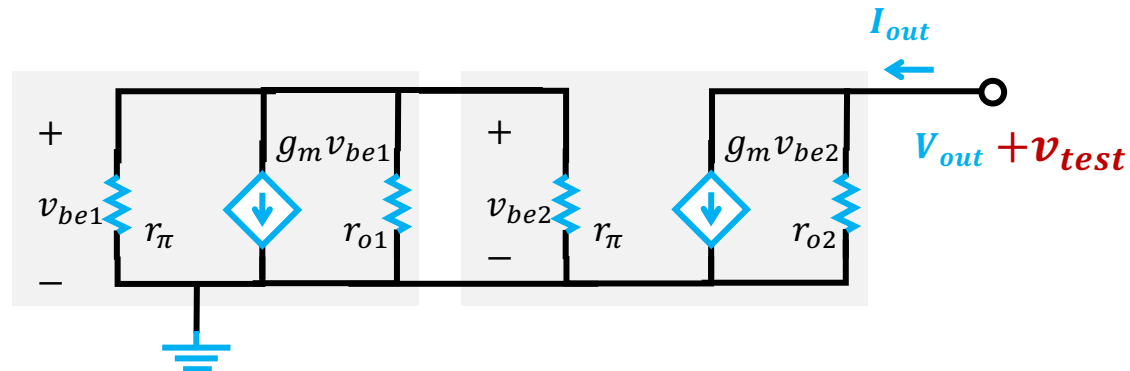
Example 6: BJT current mirror

QUESTION: Find out the relationship between I_{ref} and I_{out} with Q_1 and Q_2 are matched.

- Step 1: remove I_{ref}
- Step 2: Apply a **SMALL** testing voltage. Look for ΔI_{out}



$$R_o = r_{o2} = \frac{V_{A2}}{I_{out}}$$

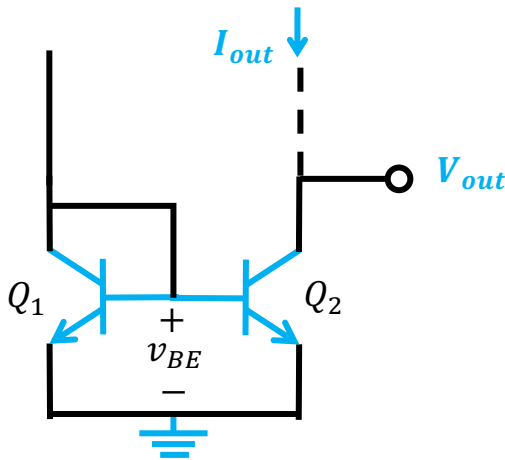


$$R_o = \infty \text{ if neglect the Early Effect}$$

Ideal current source

Example 6: BJT current mirror

QUESTION: Find out the relationship between I_{ref} and I_{out} with Q_1 and Q_2 are matched.



- If take the Early Effect into account
- According to the BJT characteristic

$$I_{C1} = I_S e^{\frac{v_{BE1}}{V_T}} \left(1 + \frac{V_{CE1}}{V_{A1}} \right)$$

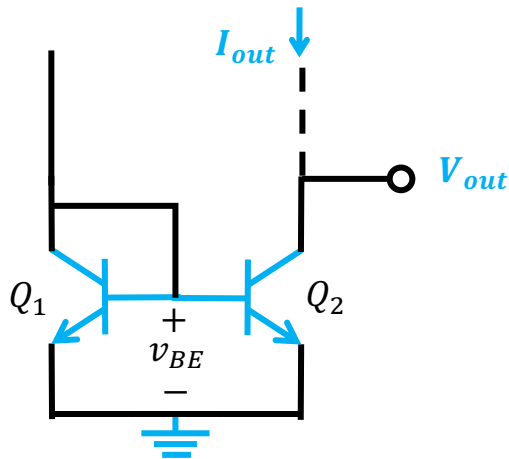
$$I_{out} = I_{C2} = m I_S e^{\frac{v_{BE2}}{V_T}} \left(1 + \frac{V_{CE2}}{V_{A2}} \right)$$

$$I_{B1} \approx \frac{1}{\beta} I_S e^{\frac{v_{BE1}}{V_T}}$$

$$I_{B2} \approx \frac{m}{\beta} I_S e^{\frac{v_{BE2}}{V_T}}$$

Example 6: BJT current mirror

QUESTION: Find out the relationship between I_{ref} and I_{out} with Q_1 and Q_2 are matched.



- According to KCL

$$I_{ref} = I_{C1} + I_{B1} + I_{B2} = \frac{m \left(1 + \frac{V_{CE2}}{V_{A2}}\right)}{\left(1 + \frac{V_{CE1}}{V_{A1}}\right) + \frac{1+m}{\beta}}$$

$\frac{1+m}{\beta}$ is very small

$$\approx \frac{m \left(1 + \frac{V_{CE2}}{V_{A2}}\right)}{\left(1 + \frac{V_{CE1}}{V_{A1}}\right) + \frac{1+m}{\beta} \left(1 + \frac{V_{CE1}}{V_{A1}}\right)} = \frac{m}{1 + \frac{1+m}{\beta} \left(1 + \frac{V_{CE1}}{V_{A1}}\right)} \left(1 + \frac{V_{CE2}}{V_{A2}}\right)$$

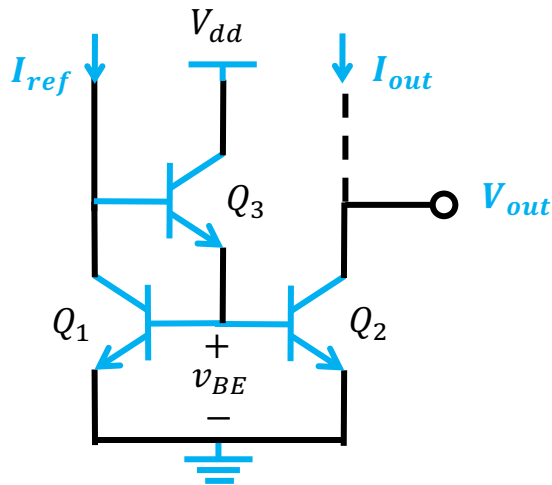
$$\approx \frac{m}{1 + \frac{1+m}{\beta}} \left(1 + \frac{V_{CE2}}{V_{A2}}\right) \left(1 - \frac{V_{CE1}}{V_{A1}}\right) \approx \frac{m}{1 + \frac{1+m}{\beta}} \left(1 + \frac{V_{CE2}}{V_{A2}} - \frac{V_{CE1}}{V_{A1}}\right)$$

$$= \frac{m}{1 + \frac{1+m}{\beta}} \left(1 + \frac{V_{out} - V_{BE}}{V_{A2}}\right)$$

The error term $\xrightarrow{V_{out}=V_{BE}} 0$

Example 7: BJT current mirror

QUESTION: Find out the relationship between I_{ref} and I_{out} with Q_1 and Q_2 are matched.



- Since Q_1 and Q_2 are matched $\Rightarrow \beta_{Q_1} = \beta_{Q_2} = \beta$

- According to the BJT characteristic

$$I_{C1} = I_S e^{\frac{v_{BE1}}{V_T}} \quad \text{and} \quad I_{out} = I_{C2} = I_S e^{\frac{v_{BE2}}{V_T}}$$

- According to KCL

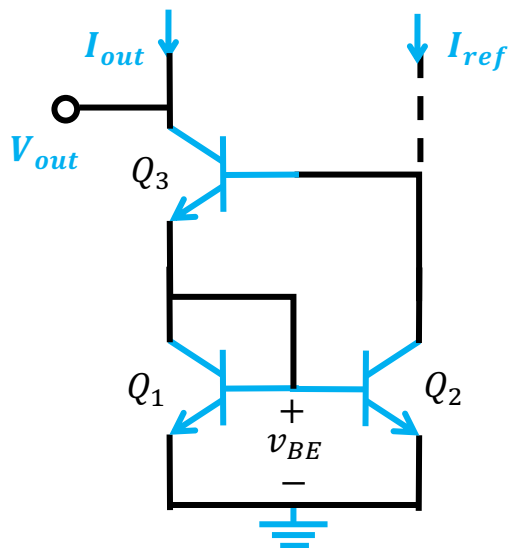
$$I_{ref} = I_{C1} + I_{B3} = I_{C1} + \frac{I_{E3}}{1 + \beta}$$

$$\Rightarrow \frac{I_{out}}{I_{ref}} = \frac{1}{1 + \frac{2}{\beta^2 + \beta}} \xrightarrow{\beta \rightarrow \infty} \frac{1}{1 + \frac{2}{\beta^2}}$$

- The output resistance $R_o = r_{o2} = \frac{V_{A2}}{I_{out}}$

Example 8: Wilson current mirror

QUESTION: Find out the relationship between I_{ref} and I_{out} with Q_1 and Q_2 are matched.



- Since Q_1 and Q_2 are matched $\Rightarrow \beta_{Q_1} = \beta_{Q_2} = \beta$

- According to the BJT characteristic

$$I_{C1} = I_{C2} = I_S e^{\frac{v_{BE}}{V_T}} \quad \text{and} \quad I_{out} = I_{C3} = \frac{\beta}{1 + \beta} I_{E3}$$

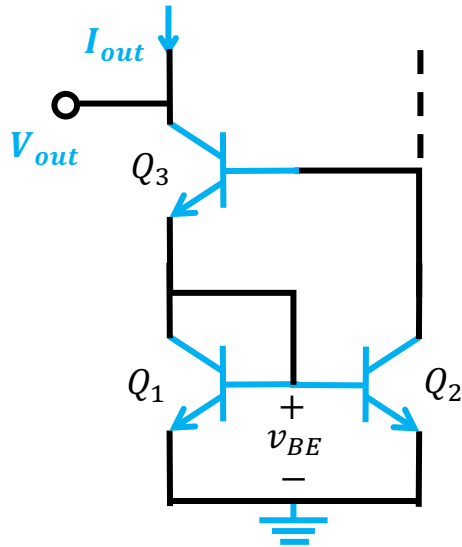
- According to KCL

$$\begin{cases} I_{ref} = I_{C2} + I_{B3} = \frac{1 + 2/\beta}{1 + \beta} I_{C1} + I_{C2} \\ I_{out} = \frac{\beta}{1 + \beta} (I_{C1} + I_{B1} + I_{B2}) \end{cases}$$

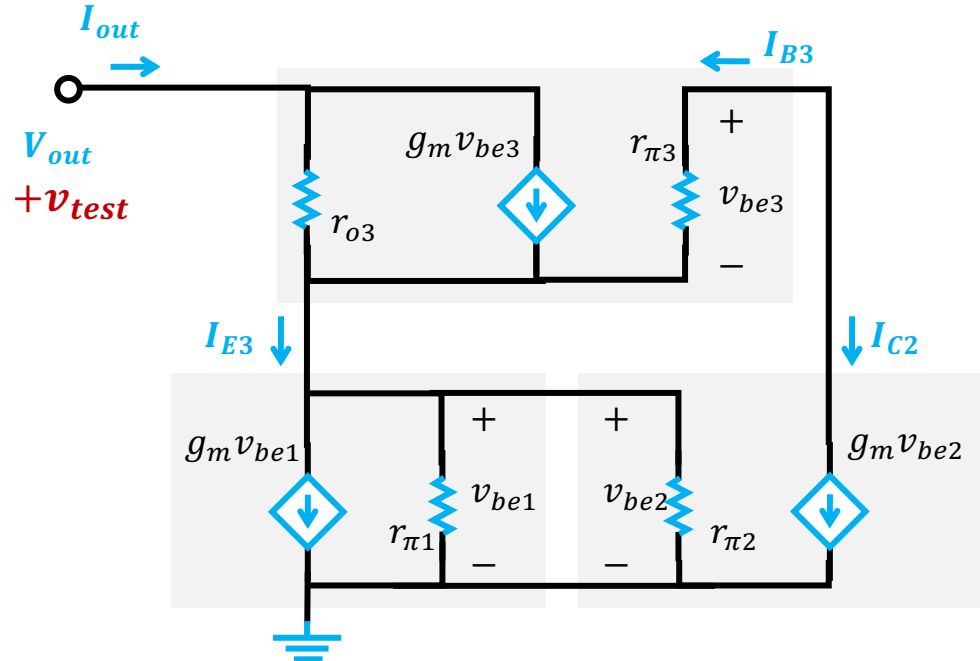
$$\Rightarrow \frac{I_{out}}{I_{ref}} = \frac{1}{1 + \frac{2}{\beta(\beta + 2)}} \xrightarrow{\beta \rightarrow \infty} \frac{1}{1 + \frac{2}{\beta^2}}$$

Example 8: Wilson current mirror

QUESTION: Find out the relationship between I_{ref} and I_{out} with Q_1 and Q_2 are matched.

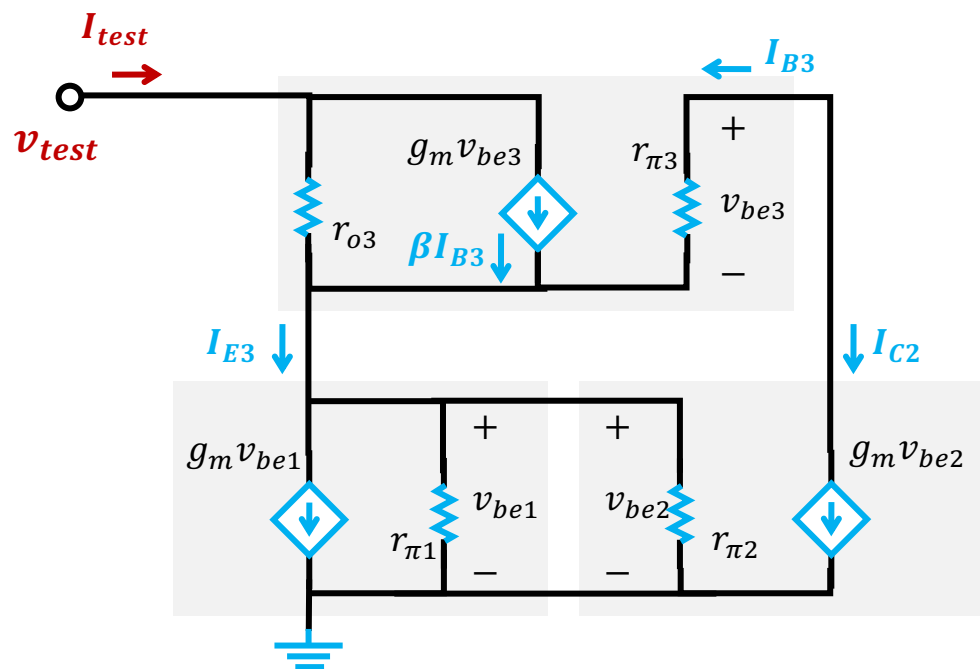


- Step 1: remove I_{ref}
- Step 2: Apply a **SMALL** testing voltage. Look for ΔI_{out}



Example 8: Wilson current mirror

QUESTION: Find out the relationship between I_{ref} and I_{out} with Q_1 and Q_2 are matched.



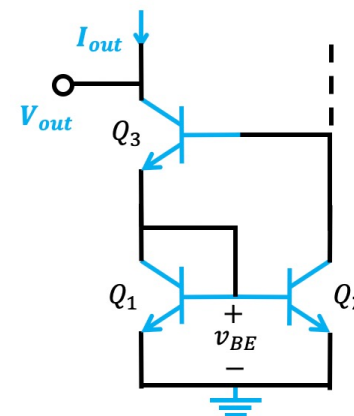
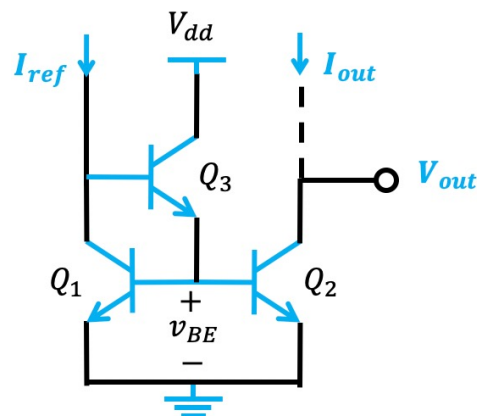
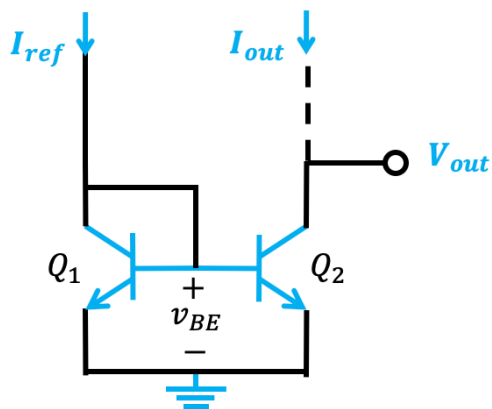
- According to KCL

$$\begin{cases} I_{E3} = \frac{I_{C1}}{\beta} + \frac{I_{C2}}{\beta} + I_{C1} \\ I_{E3} = I_{r_{o3}} + \beta I_{B3} + I_{B3} \\ I_{C1} = I_{C2} = -I_{B3} \end{cases}$$

- The input resistance

$$R_o = \frac{v_{test}}{I_{test}} \approx \frac{r_{o3} I_{r_{o3}}}{I_{r_{o3}} + \beta I_{B3}} = \frac{\left(\frac{2}{\beta} + 2 + \beta\right) I_{C1} r_{o3} + \frac{1}{\beta} I_{C1} r_{\pi 1}}{\left(\frac{2}{\beta} + 1\right) I_{C1}} \approx \frac{\beta}{2} r_{o3}$$

Summary: BJT current mirror



$$\frac{I_{out}}{I_{ref}}$$

$$\frac{1}{1 + \frac{2}{\beta}}$$

$$\frac{1}{1 + \frac{2}{\beta^2}}$$



$$\frac{1}{1 + \frac{2}{\beta^2}}$$



$$R_o$$

$$r_{o2}$$

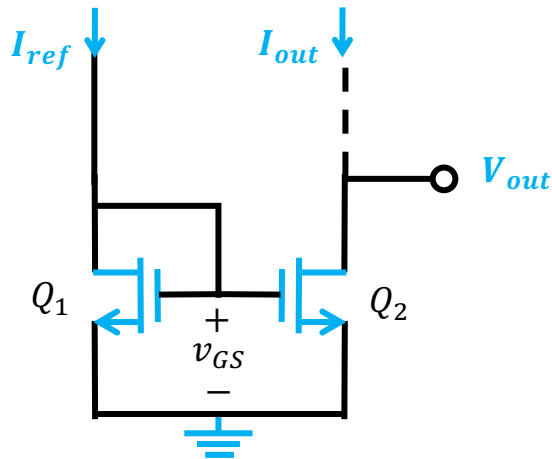
$$r_{o2}$$

$$\frac{\beta}{2} r_{o3}$$



Example 9: MOSFET current mirror

QUESTION: Find out the relationship between I_{ref} and I_{out} with Q_1 and Q_2 are matched.



- Since Q_1 and Q_2 are matched $\Rightarrow k_{n,Q_1} = k_{n,Q_2} = k_n$

- According to the MOSFET characteristic

$$\begin{cases} v_{GS} > V_{th} \\ v_{GS} - v_{DS} = 0 < V_{th} \end{cases}$$

$\Rightarrow$ Saturation region

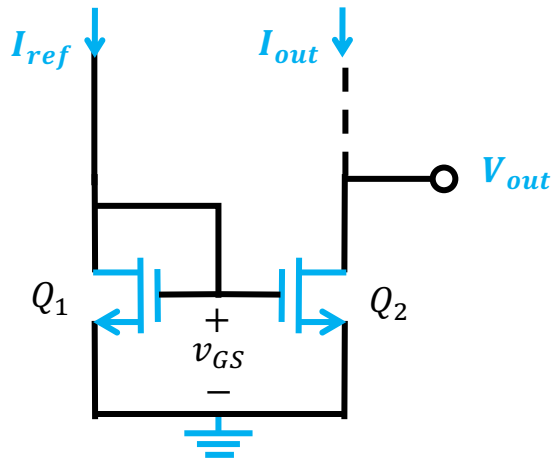
$$I_{ref} = \frac{1}{2} k_{n,Q_1} (V_{GS1} - V_{th})^2$$

$$I_{out} = \frac{1}{2} k_{n,Q_2} (V_{GS1} - V_{th})^2$$

$$\Rightarrow \frac{I_{out}}{I_{ref}} = 1$$

Example 9: MOSFET current mirror

QUESTION: Find out the relationship between I_{ref} and I_{out} with Q_1 and Q_2 are matched.



- What if $\left(\frac{W}{L}\right)_2 \neq \left(\frac{W}{L}\right)_1$
- According to the MOSFET characteristic

$$I_{ref} = \frac{1}{2} \mu_n C_{ox} \left(\frac{W}{L}\right)_1 (V_{GS1} - V_{th})^2$$

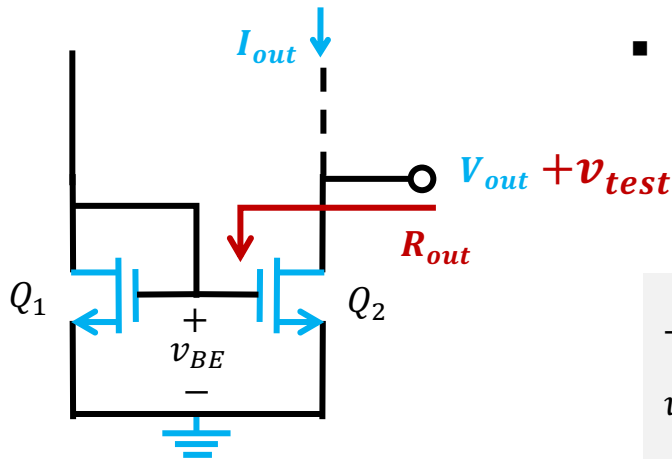
$$I_{out} = \frac{1}{2} \mu_n C_{ox} \left(\frac{W}{L}\right)_2 (V_{GS1} - V_{th})^2$$

$$\Rightarrow \frac{I_{out}}{I_{ref}} = \frac{(W/L)_2}{(W/L)_1}$$

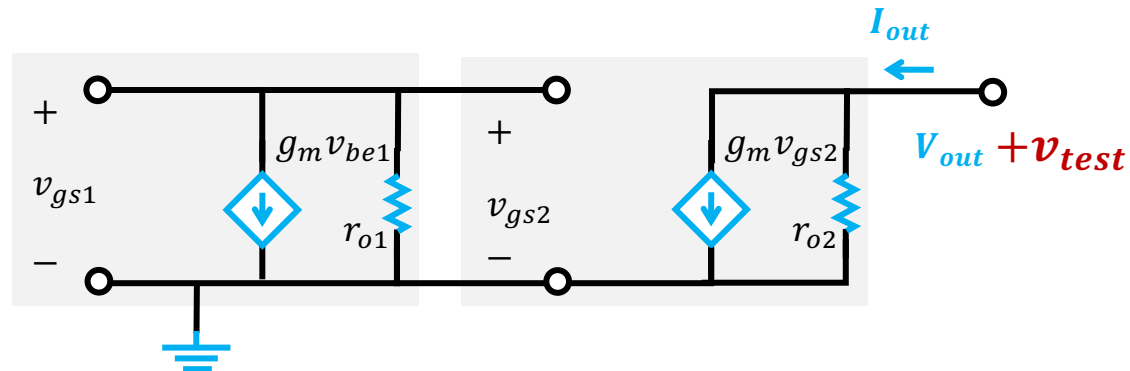
Example 9: MOSFET current mirror

QUESTION: Find out the relationship between I_{ref} and I_{out} with Q_1 and Q_2 are matched.

- Step 1: remove I_{ref}
- Step 2: Apply a **SMALL** testing voltage. Look for ΔI_{out}

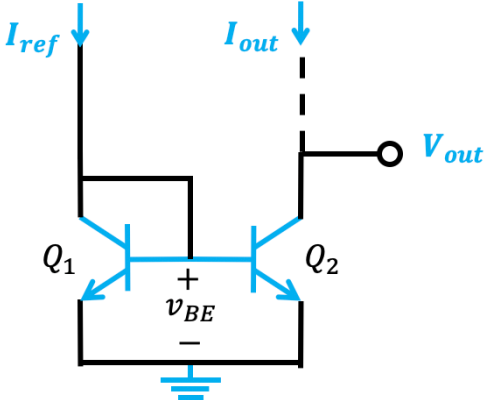
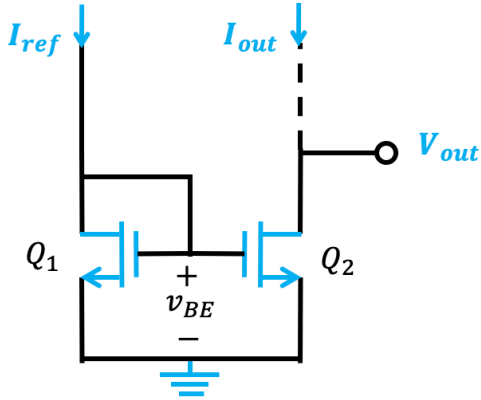


► $R_o = r_{o2} = \frac{V_{A2}}{I_{out}}$



► $R_o = \infty$ if neglect the Early Effect

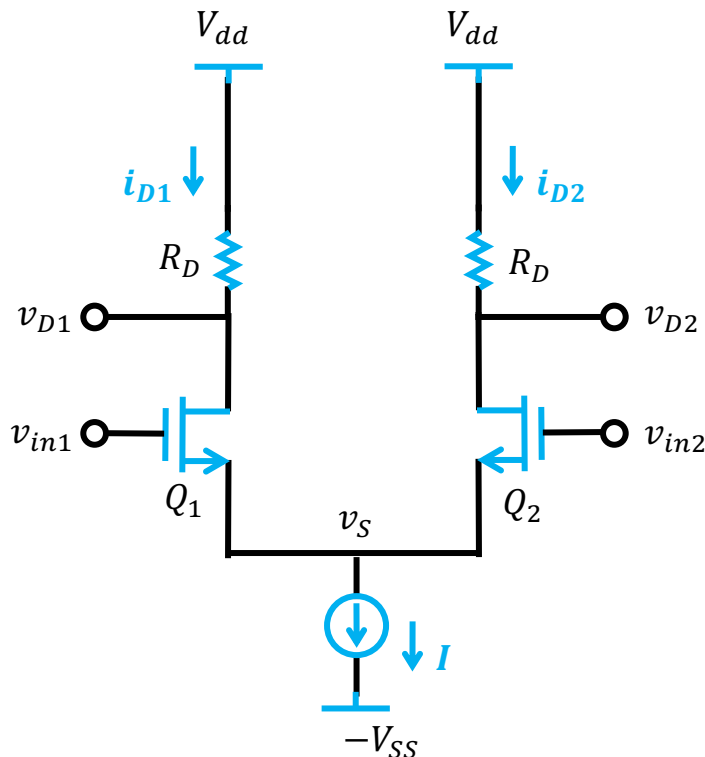
MOSFET v.s. BJT current mirror

		
$\frac{I_{out}}{I_{ref}}$	$\frac{1}{1 + \frac{2}{\beta}}$	1
R_o	r_{o2}	r_{o2}

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 - Current sources
 - **The MOS differential pair**

What is MOS Differential Pair



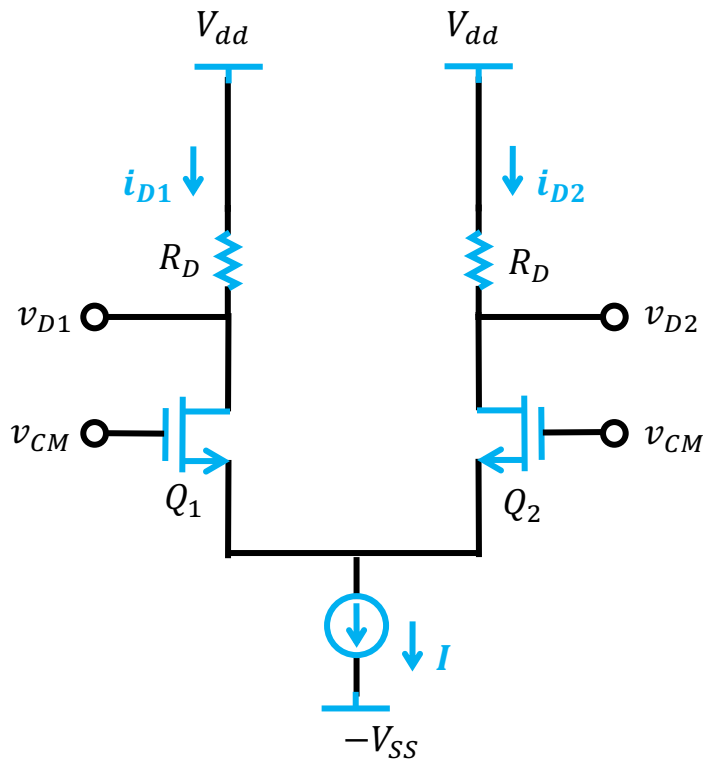
Features of MOS differential pair

- Two **matched** transistors

$$\left(\mu_n C_{ox} \frac{W}{L}\right)_1 = \left(\mu_n C_{ox} \frac{W}{L}\right)_2 = \mu_n C_{ox} \frac{W}{L}$$

- The two **Sources** are joined together
- A constant **current source** for biasing connected to the sources
- The transistors are typically biased in **saturation region**

Operation w/ Common-Mode Input



- According to KCL

$$i_{D1} = i_{D2} = \frac{1}{2}I$$

- Assume the trans. are biased in sat. region

$$\begin{cases} i_{D1} = \frac{1}{2}\mu_n C_{ox} \frac{W}{L} (V_{GS1} - V_{th})^2 \\ i_{D2} = \frac{1}{2}\mu_n C_{ox} \frac{W}{L} (V_{GS2} - V_{th})^2 \end{cases}$$

$$\blacktriangleright V_{GS1} = V_{GS2} = V_{th} + \sqrt{\frac{I}{\mu_n C_{ox} \frac{W}{L}}}$$

- According to KVL

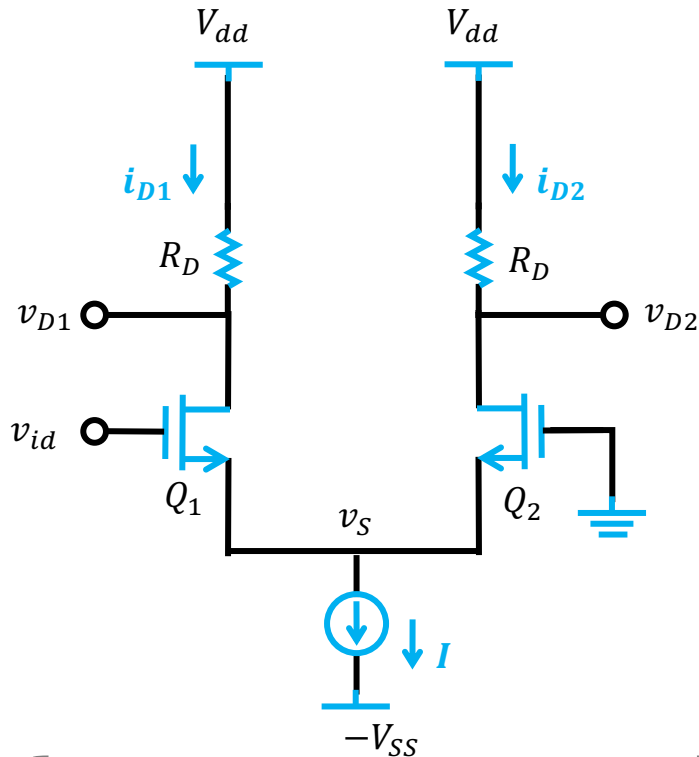
$$v_{D1} = v_{D2} = V_{dd} - i_{D1}R_D$$

- There must be $\begin{cases} v_{GS} > V_{th} \\ v_{GD} < V_{th} \end{cases}$ to fit the assumption

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Operation w/ Differential-Mode Input



$$\left[\begin{array}{l} V_{GS1} = V_{GS2} = V_{th} + \sqrt{\frac{I}{\mu_n C_{ox} \frac{W}{L}}} \\ v_{D1} = v_{D2} = V_{dd} - i_{D1} R_D \end{array} \right]$$

- According to KVL

$$v_{id} - v_{GS1} = v_{in2} - v_{GS2}$$

► $v_{id} = v_{GS1} - v_{GS2}$

- Define differential output $v_{od} = v_{D2} - v_{D1}$
- Assume the trans. are biased in sat. region

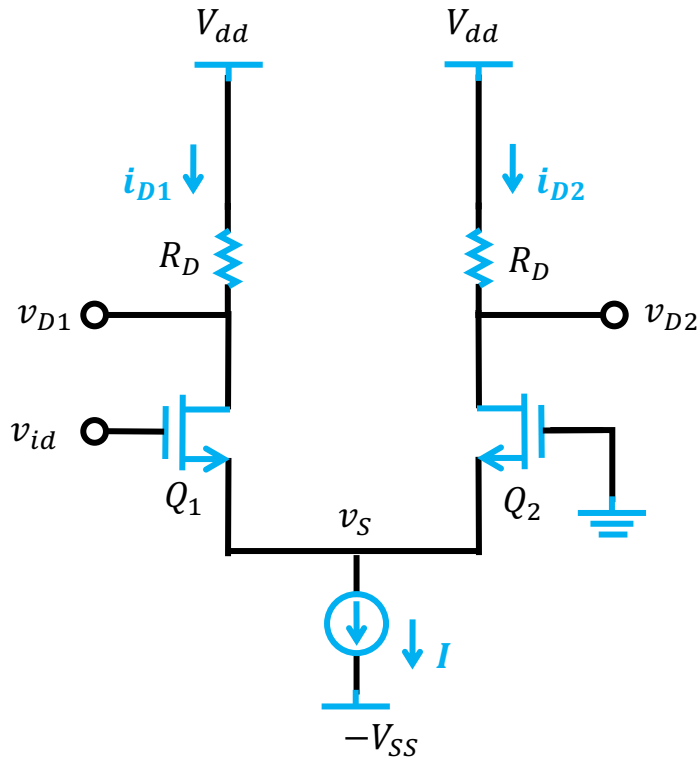
$$i_{D1} > i_{D2} \quad \text{if } v_{id} > 0$$

► $v_{od} = v_{D2} - v_{D1} > 0$

$$i_{D1} < i_{D2} \quad \text{if } v_{id} < 0$$

► $v_{od} = v_{D2} - v_{D1} < 0$

Operation w/ Differential-Mode Input



- Let's find the range of v_{id}
- A highest value of v_{id} is achieved when

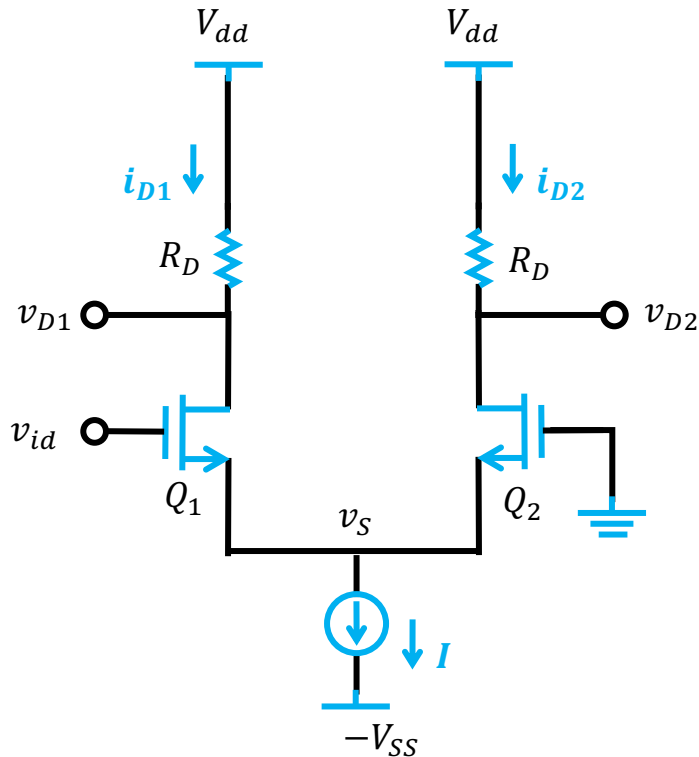
$$\begin{cases} i_{D1} = I \\ i_{D2} = 0 \quad \text{with } v_{GS2} = V_{th} \end{cases}$$

► $v_{GS1} = v_{id,max} + V_{th}$
- Assume the trans. are biased in sat. region

$$I = \frac{1}{2} \mu_n C_{ox} \frac{W}{L} (V_{GS1} - V_{th})^2$$

►
$$v_{id,max} = \sqrt{\frac{2I}{k_n}}$$

Operation w/ Differential-Mode Input



- Let's find the range of v_{id}
- A highest value of v_{id} is $v_{id,max} = \sqrt{\frac{2I}{k_n}}$

$$@ \begin{cases} i_{D1} = I \\ i_{D2} = 0 \end{cases} \text{ with } v_{GS2} = V_{th}$$

- Similarly, the lowest value of v_{id} is

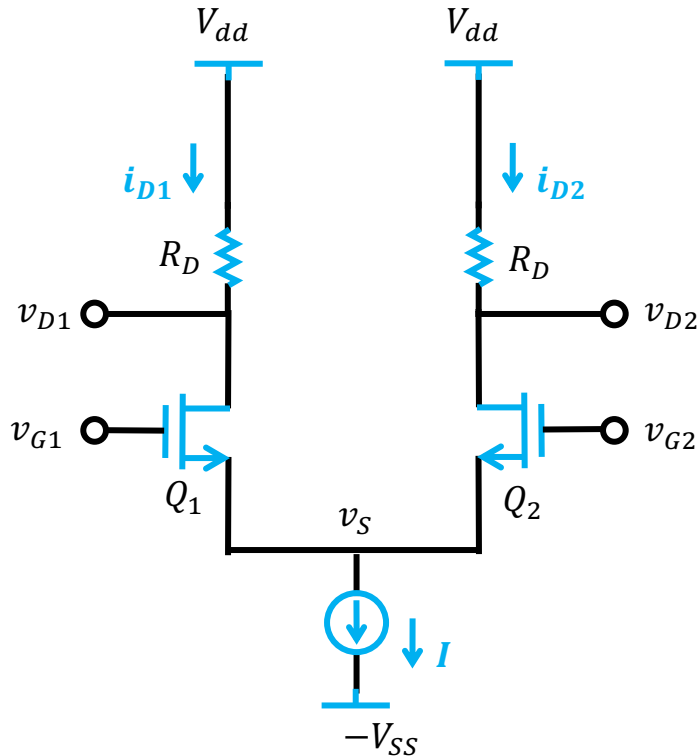
$$v_{id,min} = -\sqrt{\frac{2I}{k_n}}$$

NOTE: both of the transistors are biased in the saturation region

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 - **Large-signal operation**

Large-Signal Operation



DEFINE **DIFFERENTIAL INPUT** $v_{id} = v_{G1} - v_{G2}$

ASSUME Q_1 & Q_2 are biased in saturation region

- According to characteristic

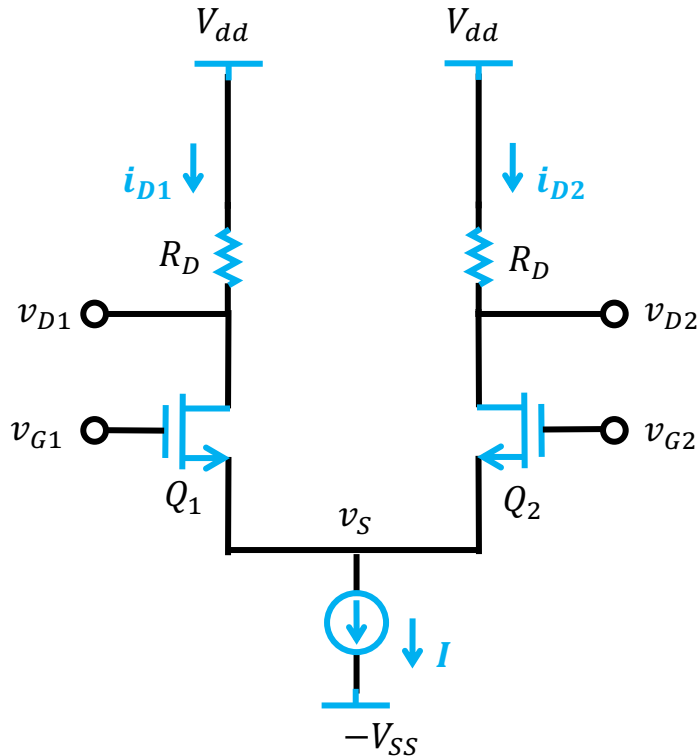
$$\begin{cases} i_{D1} = \frac{1}{2}k_n(V_{GS1} - V_{th})^2 \\ i_{D2} = \frac{1}{2}k_n(V_{GS2} - V_{th})^2 \end{cases}$$

$$\Rightarrow \sqrt{i_{D1}} - \sqrt{i_{D2}} = \sqrt{\frac{1}{2}k_n}(v_{GS1} - v_{GS2})$$

- According to KCL

$$i_{D1} + i_{D2} = I$$

Large-Signal Operation

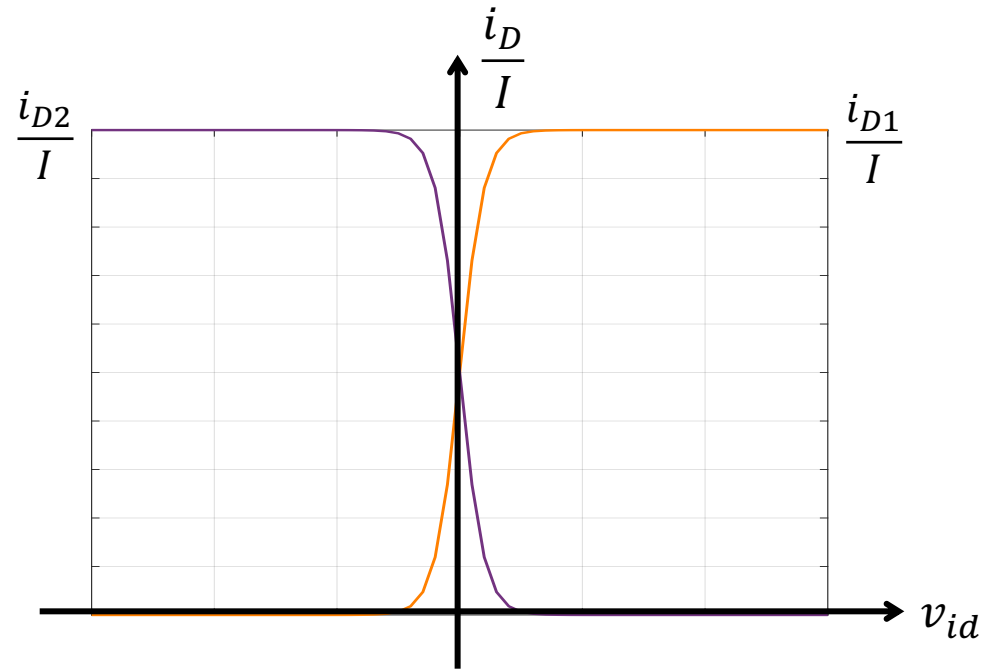
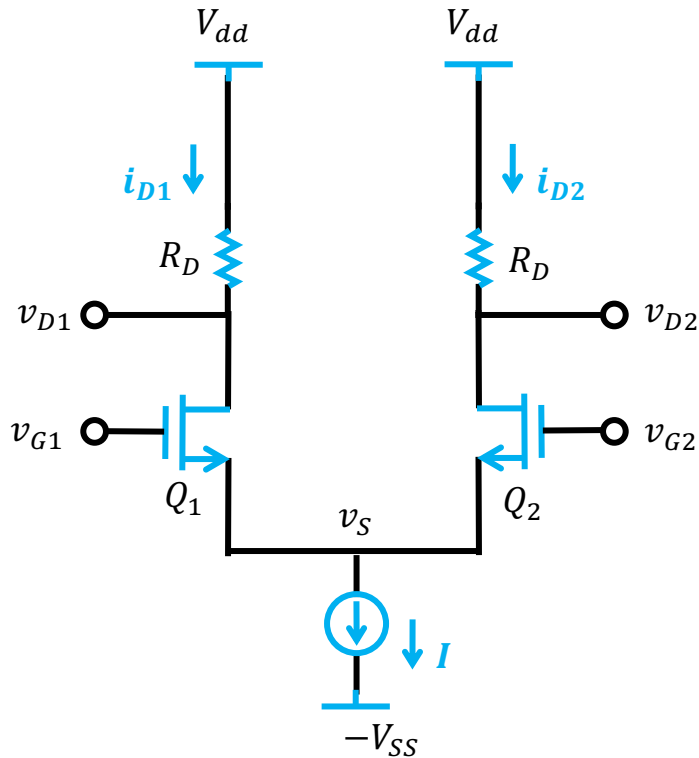


$$\begin{cases} \sqrt{i_{D1}} - \sqrt{i_{D2}} = \sqrt{\frac{1}{2}k_n}(v_{GS1} - v_{GS2}) \\ i_{D1} + i_{D2} = I \end{cases}$$

$$\begin{cases} i_{D1} = \frac{I}{2} + \frac{v_{id}}{2} \sqrt{k_n I \left(1 - \frac{v_{id}^2 k_n}{4I}\right)} \\ i_{D2} = \frac{I}{2} - \frac{v_{id}}{2} \sqrt{k_n I \left(1 - \frac{v_{id}^2 k_n}{4I}\right)} \end{cases}$$

▪ If $v_{id} = 0$ $\Rightarrow i_{D1} = i_{D2} = \frac{I}{2}$

Large-Signal Operation

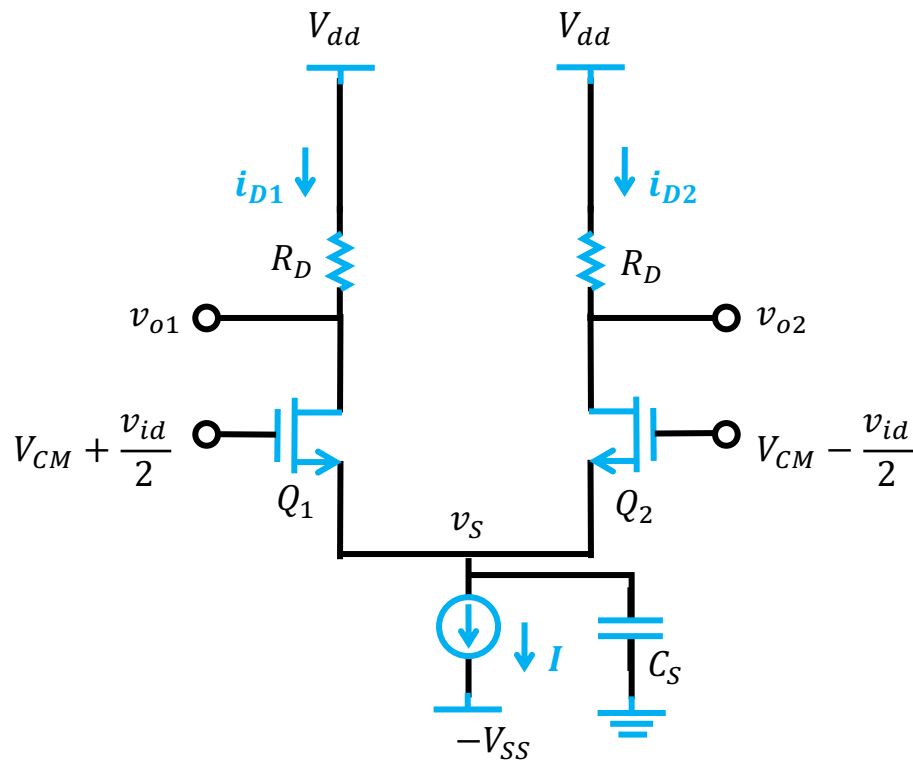


$$\begin{cases} i_{D1} = \frac{I}{2} + \frac{v_{id}}{2} \sqrt{k_n I \left(1 - \frac{v_{id}^2 k_n}{4I} \right)} \\ i_{D2} = \frac{I}{2} - \frac{v_{id}}{2} \sqrt{k_n I \left(1 - \frac{v_{id}^2 k_n}{4I} \right)} \end{cases}$$

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 - **Small-signal operation**

Small-Signal Operation



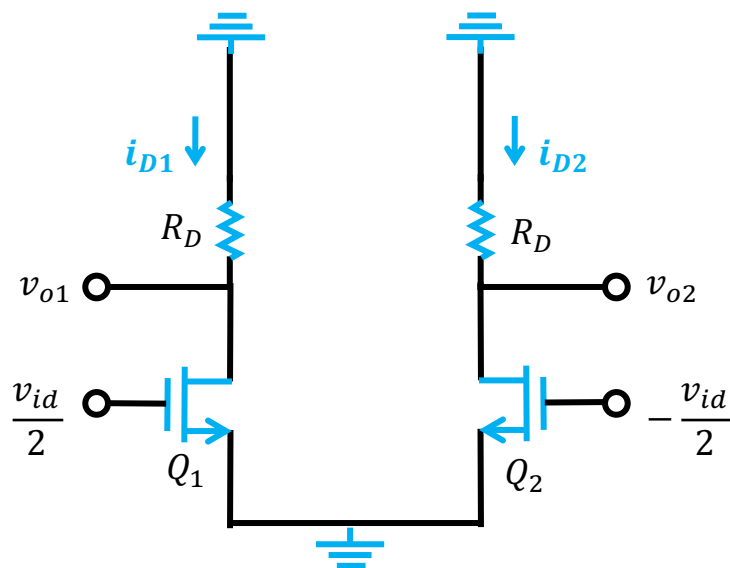
V_{CM} is a DC input

$\frac{v_{id}}{2}$ is a very small AC signal

- **Step 1: perform DC analysis**

$$I_{D1} = I_{D2} = \frac{I}{2}$$

Small-Signal Operation



V_{CM} is a DC input

$\frac{v_{id}}{2}$ is a very small AC signal

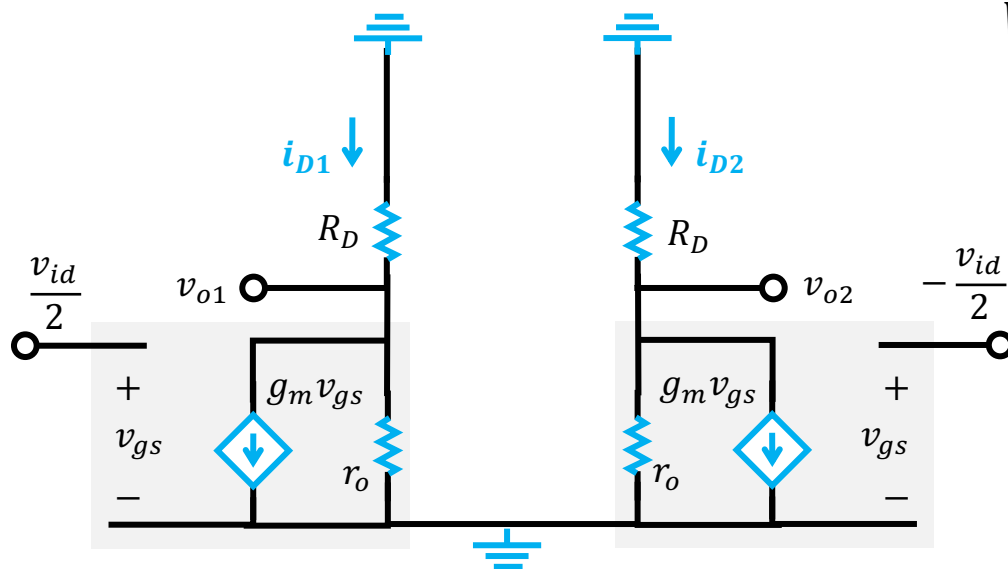
- **Step 1: perform DC analysis**

$$I_{D1} = I_{D2} = \frac{I}{2}$$

- **Step 2: perform AC analysis**

- Step 2.1: turn off DC sources
- Step 2.2: Calculate small-signal model parameters, g_m

Small-Signal Operation



V_{CM} is a DC input

$\frac{v_{id}}{2}$ is a very small AC signal

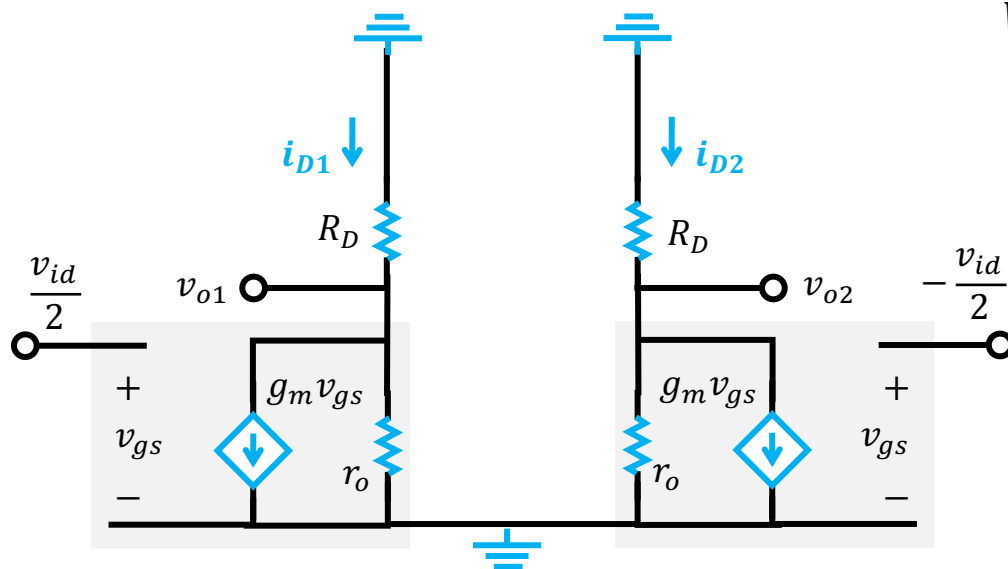
- **Step 1: perform DC analysis**

$$I_{D1} = I_{D2} = \frac{I}{2}$$

- **Step 2: perform AC analysis**

- Step 2.1: turn off DC sources
- Step 2.2: Calculate small-signal model parameters, g_m
- Step 2.3: replace the transistor with the small-signal model
- Step 2.4: Analyze the resulting circuit

Small-Signal Operation



V_{CM} is a DC input

$\frac{v_{id}}{2}$ is a very small AC signal

- Step 1: perform DC analysis

$$I_{D1} = I_{D2} = \frac{I}{2}$$

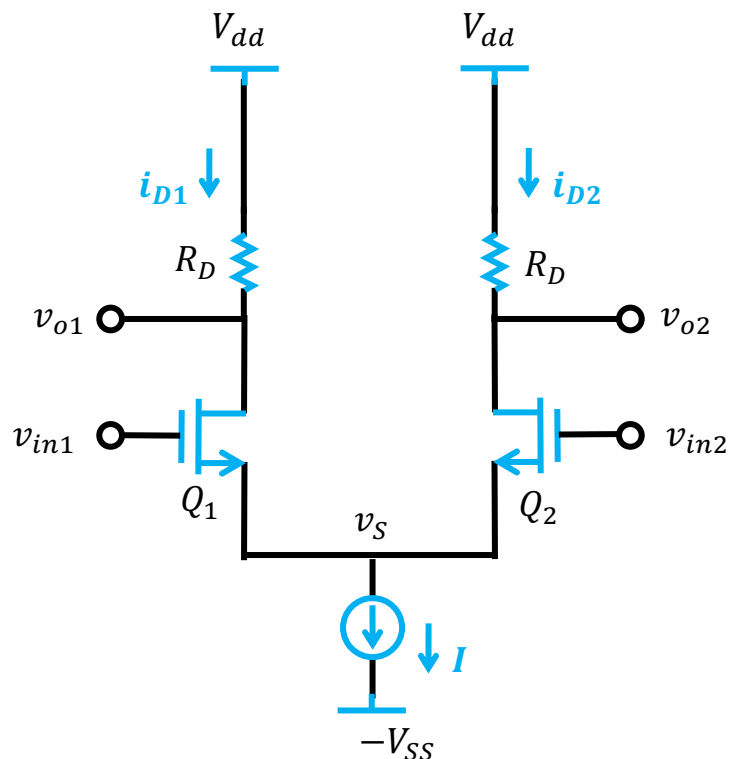
- Step 2: perform AC analysis

$$v_{o1} = -g_m \frac{v_{id}}{2} (R_D || r_o)$$

$$v_{o2} = g_m \frac{v_{id}}{2} (R_D || r_o)$$

► $v_o = v_{o2} - v_{o1} = -g_m v_{id} (R_D || r_o)$

Summary: Differential Pair



DEFINE **COMMON MODE VOLTAGE**

$$V_{iCM} \equiv \frac{1}{2}(v_{in1} + v_{in2})$$

DEFINE **DIFFERENTIAL MODE VOLTAGE**

$$v_{id} \equiv v_{in1} - v_{in2}$$

The voltage gain

$$|A_{CM}| = \frac{v_{o2} - v_{o1}}{v_{iCM}} = 0$$

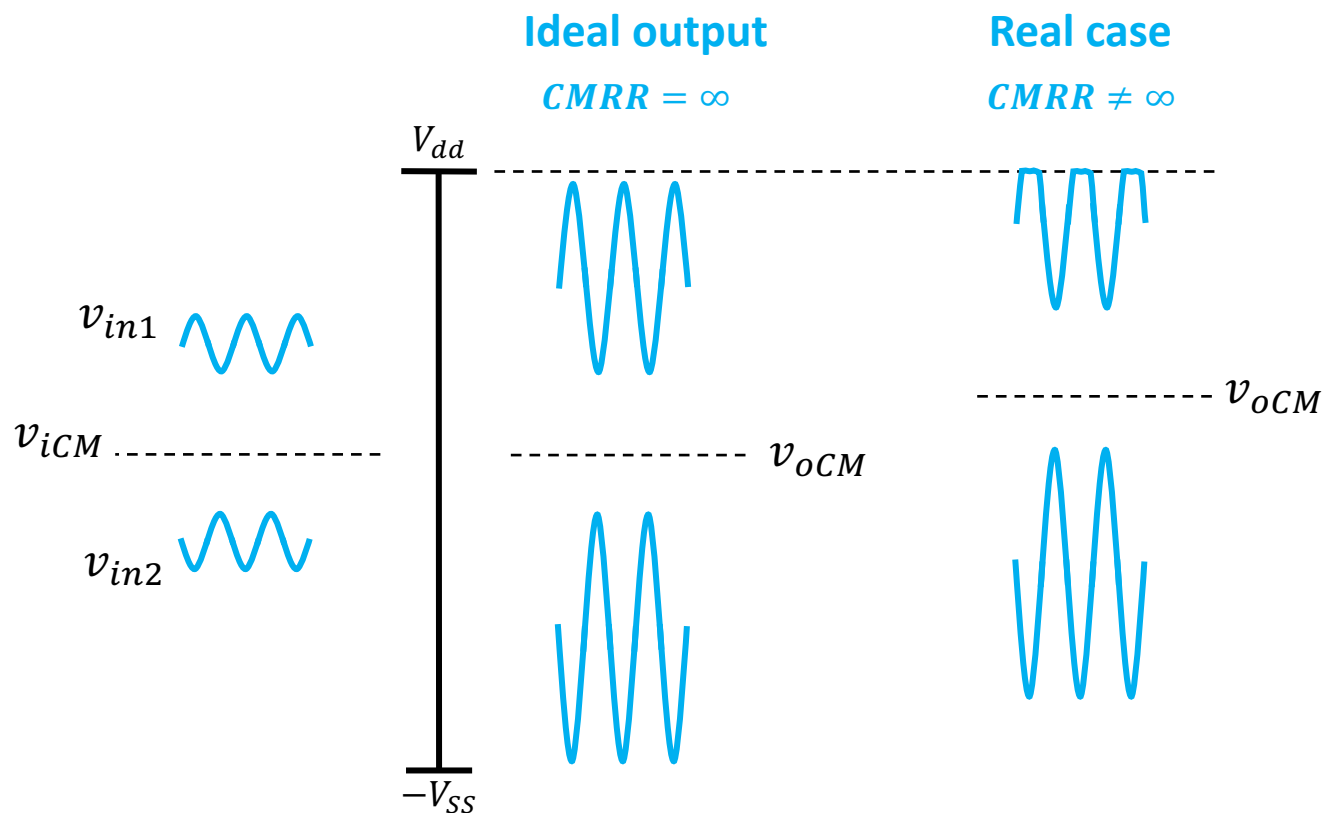
$$|A_d| = \frac{v_{o2} - v_{o1}}{v_{id}} = g_m(R_D || r_o)$$

DEFINE **COMMON MODE REJECTION RATIO**

$$CMRR \equiv \left| \frac{A_d}{A_{CM}} \right| \xrightarrow{\text{Ideally}} \infty$$

Summary: Differential Pair

Why we need CMRR? $CMRR \equiv \left| \frac{A_d}{A_{CM}} \right| \xrightarrow{\text{Ideally}} \infty$



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Reading tasks & learning goals

- Reading tasks

- Microelectronic Circuits, 6th edition
 - Chapter 5, 7.4.2–7.5, 8.1–8.2.2, 9.1–9.3

- Learning goals

- Know the **structure** of a MOSFET and **how it works**
- Well understand the **characteristic** of MOSFET
- Well understand how to locate the **bias point** of a MOSFET in circuit
- Well understand how to analyze a circuit with **MOSFET**
- Well understand the similarity and difference between BJT and MOSFET
- Well understand the MOSFET amplification circuit **configuration** and **analysis**
- Know the fundamental analysis on **current source** and **differential pairs**